

STAT 210B: Theoretical Statistics

Lecture Notes

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These notes accompany the graduate course **STAT 210B: Theoretical Statistics** at UC Berkeley. Parts of the material overlap with standard references such as [3, 4] and related lecture notes [2]. Comments and corrections are welcome.

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1 Introduction

1.1 Why non-asymptotic?

Non-asymptotic bounds keep the sample size fixed and aim to track the effect of all influential parameters. This point of view is often more informative than taking limits, especially in modern high-dimensional settings where several parameters (dimension, noise level, model complexity) may grow simultaneously.

A simple example is mean estimation. Suppose we observe independent and identically distributed random variables $X_1, \dots, X_n$ with mean μ and (for simplicity) known variance σ^2 . A classical confidence interval is

$$\left[\frac{1}{n} \sum_{i=1}^n X_i - \frac{2\sigma}{\sqrt{n}}, \frac{1}{n} \sum_{i=1}^n X_i + \frac{2\sigma}{\sqrt{n}} \right].$$

By the Central Limit Theorem (CLT), this interval asymptotically contains μ with probability $1 - 2\Phi(-2) \approx 0.9545$, where Φ is the standard normal cumulative distribution function. Indeed, if $Z \sim \mathcal{N}(0, 1)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma \sqrt{n}} \right| \geq t \right) = \mathbb{P}(|Z| \geq t) = 2\Phi(-t).$$

The point is not that the CLT is inaccurate; rather, it does not tell us how large n needs to be for a given distribution. For some distributions, the Gaussian approximation is excellent even for moderate n , while for others it can require much larger samples.

A second illustration comes from rare events. If $X_{n,1}, \dots, X_{n,n}$ are independent Bernoulli random variables with success probability p_n and $np_n \rightarrow \lambda$, then the sum behaves asymptotically as a Poisson random variable.

Theorem 1.1 (Poisson limit theorem). *Let $X_{n,i} \sim \text{Ber}(p_n)$ be independent. Assume that $np_n \rightarrow \lambda > 0$. Then*

$$\sum_{i=1}^n X_{n,i} \xrightarrow{d} \text{Poi}(\lambda) \quad \text{as } n \rightarrow \infty.$$

For instance, with $n = 50$ and $p = 1/50$, a Poisson approximation is typically more accurate than a Gaussian approximation. Non-asymptotic tools allow us to make such comparisons quantitatively at a finite sample size.

1.2 A motivating example: fixed-design Gaussian regression

A basic model where non-asymptotic bounds arise naturally is fixed-design Gaussian regression. We observe data $\{(Y_i, x_i)\}_{i=1}^n$ satisfying

$$Y_i = f(x_i) + \xi_i, \quad i = 1, \dots, n, \quad (1)$$

where $f : \mathcal{X} \rightarrow \mathbf{R}$ is unknown, the design points $x_i \in \mathcal{X}$ are fixed (nonrandom), and the noise variables ξ_i are independent $\mathcal{N}(0, \sigma^2)$.

Given an estimator $\hat{f}$, we measure error using the empirical loss

$$\|\hat{f} - f\|_n^2 = \frac{1}{n} \sum_{i=1}^n (\hat{f}(x_i) - f(x_i))^2, \quad \text{and the risk } \mathbb{E}\|\hat{f} - f\|_n^2.$$

A standard special case is linear regression: $x_i \in \mathbf{R}^d$ and $f(x_i) = \langle x_i, \theta^* \rangle$ for an unknown $\theta^* \in \mathbf{R}^d$. The least-squares estimator is

$$\hat{\theta} \in \arg \min_{\theta \in \mathbf{R}^d} \frac{1}{n} \sum_{i=1}^n (\langle x_i, \theta \rangle - Y_i)^2.$$

Assume that $\text{rank}(X) = d$, where $X \in \mathbf{R}^{n \times d}$ is the design matrix with rows $x_i^\top$. Then $\hat{\theta}$ is unique and one can show that

$$\mathbb{E}\|\hat{f} - f\|_n^2 = \frac{\sigma^2 d}{n}, \quad \text{where } \hat{f}(x_i) = \langle x_i, \hat{\theta} \rangle.$$

This is a typical non-asymptotic statement: it holds for every n and d (with $d \leq n$ and $\text{rank}(X) = d$), and it makes the dependence on n, d, σ^2 fully explicit. Here d, n , and σ^2 each play a specific role, and they can vary simultaneously in an arbitrary manner. In contrast, classical asymptotic analyses usually focus on specific limiting regimes, such as $n \rightarrow \infty$ with d fixed, or $n, d \rightarrow \infty$ with a prescribed relationship (for example, $d/n \rightarrow \gamma$). Even in this simple example, these regimes cover only a fraction of the possible interactions between d, σ , and n .

To put this viewpoint in context, we quote Pascal Massart [1]:

“Through our works, we have promoted a nonasymptotic approach in statistics which consists in taking the number of observations as it is and trying to evaluate the effect of all the influential parameters. At this very starting point, it seems to me that it is important to provide a first answer to the following question: why should we be interested by a nonasymptotic view for model selection at all? In my opinion, the motivation should neither be a strange interest for “small” sets of data nor a special taste for constants and inequalities rather than for limit theorems (although since mathematics is also a matter of taste, it is a possible way for getting involved in it. . .). On the contrary, the nonasymptotic point of view may turn to be especially relevant when the number of observations is large. It is indeed to fit large complex sets of data that one needs to deal with possibly huge collections of models at different scales. The nonasymptotic approach for model selection precisely allows the collection of models together with their dimensions to vary freely, letting the dimensions be possibly of the same order of magnitude as the number of observations.”

2 Concentration and sub-Gaussian tools

2.1 From Markov to Chernoff

Most concentration bounds begin with a refined application of Markov inequality. We start with two classical inequalities and then explain how exponential moments lead to much sharper bounds.

Proposition 2.1 (Markov inequality). *Let X be a non-negative random variable. Then for any $t > 0$,*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}X}{t}.$$

A first consequence is Chebyshev inequality.

Proposition 2.2 (Chebyshev inequality). *Let X be a random variable with finite variance. Then for any $t > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

Proof. Apply Markov inequality to $(X - \mathbb{E}X)^2$. □

For light-tailed distributions, Chebyshev inequality is often far from optimal. For instance, Gaussian tails decay at an exponential rate in t^2 .

Proposition 2.3 (Gaussian tail bounds). *Let $Z \sim \mathcal{N}(0, 1)$. For any $t > 0$,*

$$\left(\frac{1}{t} - \frac{1}{t^3}\right) \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \leq \mathbb{P}(Z \geq t) \leq \frac{1}{t} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

Proof. We have

$$\mathbb{P}(Z \geq t) = \frac{1}{\sqrt{2\pi}} \int_t^\infty e^{-x^2/2} dx.$$

For the upper bound, substitute $x = t + y$:

$$\mathbb{P}(Z \geq t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \int_0^\infty e^{-ty} e^{-y^2/2} dy \leq \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \int_0^\infty e^{-ty} dy = \frac{1}{t} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

For the lower bound, define $f(x) = \left(\frac{1}{x} - \frac{1}{x^3}\right) e^{-x^2/2}$ for $x > 0$. A direct computation gives $f'(x) = -(1 - 3x^{-4})e^{-x^2/2}$, and hence

$$\int_t^\infty (1 - 3x^{-4}) e^{-x^2/2} dx = f(t) = \left(\frac{1}{t} - \frac{1}{t^3}\right) e^{-t^2/2}.$$

Since $(1 - 3x^{-4}) \leq 1$, we have

$$\int_t^\infty e^{-x^2/2} dx \geq \int_t^\infty (1 - 3x^{-4}) e^{-x^2/2} dx = \left(\frac{1}{t} - \frac{1}{t^3}\right) e^{-t^2/2},$$

which yields the desired lower bound after multiplying by $1/\sqrt{2\pi}$. □

The proof above is a good warm-up: for a Gaussian variable we can compute tails directly. For a general random variable, the analog of “completing the square” is to work with exponential moments.

For a real random variable Z , define its moment generating function (MGF) by

$$M_Z(\lambda) = \mathbb{E}e^{\lambda Z}, \quad \lambda \in \mathbf{R}.$$

The key feature is multiplicativity under independence: if $Z_1, \dots, Z_n$ are independent, then

$$M_{Z_1 + \dots + Z_n}(\lambda) = \prod_{i=1}^n M_{Z_i}(\lambda).$$

This simple identity is behind most concentration inequalities for sums.

For Gaussian random variables the MGF has an explicit closed form: if $Z \sim \mathcal{N}(0, \sigma^2)$, then

$$M_Z(\lambda) = \mathbb{E}e^{\lambda Z} = e^{\lambda^2 \sigma^2 / 2}, \quad \lambda \in \mathbf{R}.$$

2.2 Sub-Gaussian random variables

The Gaussian MGF suggests a natural class of random variables with comparable tail behavior.

Definition 2.4 (Sub-Gaussian random variable). A mean-zero random variable X is *sub-Gaussian with variance parameter σ^2* if

$$\mathbb{E}e^{\lambda X} \leq e^{\lambda^2 \sigma^2 / 2} \quad \text{for all } \lambda \in \mathbf{R}.$$

Sub-Gaussianity can be characterized in several equivalent ways. We record one standard collection of equivalences; see [3, Proposition 2.5.2] for a complete proof and sharp constants.

Proposition 2.5 (Equivalent definitions of sub-Gaussianity). *Let X be a random variable with $\mathbb{E}X = 0$. The following conditions are equivalent; moreover, the corresponding parameters are equivalent up to absolute constant factors.*

1. *There exists $k_1 > 0$ such that for all $\lambda \in \mathbf{R}$,*

$$\mathbb{E}e^{\lambda X} \leq e^{k_1^2 \lambda^2}.$$

2. *There exists $k_2 > 0$ such that for all $t \geq 0$,*

$$\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/k_2^2}.$$

3. *There exists $k_3 > 0$ such that for all $p \geq 1$,*

$$\|X\|_{L_p} = (\mathbb{E}|X|^p)^{1/p} \leq k_3 \sqrt{p}.$$

4. *There exists $k_5 > 0$ such that*

$$\mathbb{E}e^{X^2/k_5^2} \leq 2.$$

Proof. We sketch two implications; the remaining ones are in [3, Proposition 2.5.2].

(1) $\Rightarrow$ (2) (Chernoff bound). For any $\lambda > 0$,

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}e^{\lambda X} \leq \exp(k_1^2 \lambda^2 - \lambda t).$$

Optimizing over λ yields $\mathbb{P}(X \geq t) \leq \exp(-t^2/(4k_1^2))$. Applying the same argument to $-X$ and using the union bound gives (2).

(2) $\Rightarrow$ (3). Using $\mathbb{E}|X|^p = \int_0^\infty \mathbb{P}(|X|^p \geq s) ds$ and the substitution $s = u^p$ gives

$$\mathbb{E}|X|^p = \int_0^\infty \mathbb{P}(|X| \geq u) p u^{p-1} du.$$

Assume without loss of generality that $k_2 = 1$. Then

$$\mathbb{E}|X|^p \leq 2p \int_0^\infty u^{p-1} e^{-u^2} du = p \int_0^\infty w^{p/2-1} e^{-w} dw = p \Gamma(p/2).$$

Standard bounds for the gamma function imply $\Gamma(p/2)^{1/p} \lesssim \sqrt{p}$, hence $\|X\|_{L_p} \lesssim \sqrt{p}$. $\square$

It is useful to keep a small supply of examples in mind; we will return to more examples throughout the course.

Example 2.6 (Gaussian random variables). If $Z \sim \mathcal{N}(0, \sigma^2)$, then Z is sub-Gaussian with variance parameter σ^2 , since $\mathbb{E}e^{\lambda Z} = e^{\lambda^2 \sigma^2 / 2}$ for all $\lambda \in \mathbf{R}$.

Lemma 2.7 (Hoeffding lemma (short proof, non-optimal constant)). *Let X be a random variable such that $\mathbb{E}X = 0$ and $X \in [a, b]$ almost surely. Then for all $\lambda \in \mathbf{R}$,*

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2(b-a)^2}{2}\right).$$

Proof. Let X' be an independent copy of X . By Jensen inequality (conditioning on X and using convexity of the exponential),

$$\mathbb{E}e^{\lambda X} = \mathbb{E}\left[e^{\lambda(X - \mathbb{E}[X'|X])}\right] \leq \mathbb{E}\left[e^{\lambda(X - X')}\right].$$

Set $Y = X - X'$. The random variable Y is symmetric: Y and $-Y$ have the same distribution, since swapping X and X' changes Y to $-Y$ without changing the joint law. Therefore,

$$\mathbb{E}e^{\lambda Y} = \mathbb{E}\left[\frac{e^{\lambda Y} + e^{-\lambda Y}}{2}\right] = \mathbb{E}[\cosh(\lambda Y)].$$

We use the elementary bound $\cosh(u) \leq e^{u^2/2}$ for all $u \in \mathbf{R}$. Indeed, comparing Taylor series,

$$\cosh(u) = \sum_{k=0}^{\infty} \frac{u^{2k}}{(2k)!} \leq \sum_{k=0}^{\infty} \frac{u^{2k}}{2^k k!} = e^{u^2/2},$$

where we used $(2k)! \geq 2^k k!$. Since $|Y| \leq b - a$ almost surely, we conclude that

$$\mathbb{E}e^{\lambda X} \leq \mathbb{E}e^{\lambda Y} = \mathbb{E}\cosh(\lambda Y) \leq \mathbb{E}e^{\lambda^2 Y^2 / 2} \leq e^{\lambda^2(b-a)^2 / 2}.$$

□

Remark 2.8. The constant in Theorem 2.7 is not optimal. A sharper argument gives $\mathbb{E}e^{\lambda X} \leq \exp(\lambda^2(b-a)^2/8)$; see, for example, [3, Proposition 2.5.2]. In most applications we only need the qualitative fact that bounded random variables are sub-Gaussian.

2.3 Sums of independent sub-Gaussians

One reason MGFs are convenient is that they interact well with independence. In particular, they lead to concentration bounds for sums that mirror the familiar identity $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i)$ for independent mean-zero random variables.

Proposition 2.9. *Let $X_1, \dots, X_n$ be independent random variables with means μ_i such that $X_i - \mu_i$ are sub-Gaussian with variance parameters σ_i^2 in the sense of Theorem 2.4. Then for any $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right),$$

and

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right).$$

Proof. By independence and the sub-Gaussian MGF bound,

$$\mathbb{E} \exp \left(\lambda \sum_{i=1}^n (X_i - \mu_i) \right) = \prod_{i=1}^n \mathbb{E} e^{\lambda(X_i - \mu_i)} \leq \exp \left(\frac{\lambda^2}{2} \sum_{i=1}^n \sigma_i^2 \right).$$

Apply Markov inequality to $\exp(\lambda \sum_{i=1}^n (X_i - \mu_i))$ and optimize over $\lambda > 0$ to obtain the upper tail. The lower tail follows by applying the same argument to $-\sum_i (X_i - \mu_i)$ and then using the union bound. $\square$

As a clean corollary, bounded random variables satisfy Hoeffding-type concentration.

Proposition 2.10 (Hoeffding inequality (with a non-optimal constant)). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}X_i = \mu_i$ and $X_i \in [a_i, b_i]$ almost surely. Then for any $t \geq 0$,*

$$\mathbb{P} \left(\left| \sum_{i=1}^n (X_i - \mu_i) \right| \geq t \right) \leq 2 \exp \left(- \frac{t^2}{2 \sum_{i=1}^n (b_i - a_i)^2} \right).$$

Proof. By Theorem 2.7, each $X_i - \mu_i$ is sub-Gaussian with variance parameter $\sigma_i^2 = (b_i - a_i)^2$. Plug into Theorem 2.9. $\square$

Example 2.11 (Averages of random signs). Let $\varepsilon_1, \dots, \varepsilon_n$ be independent random signs taking values ± 1 with probability $1/2$. Since $\varepsilon_i \in [-1, 1]$ and $\mathbb{E}\varepsilon_i = 0$, Theorem 2.10 gives, for $t \geq 0$,

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \right| \geq t \right) \leq 2 \exp \left(- \frac{nt^2}{8} \right).$$

Equivalently, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \right| \leq \sqrt{\frac{8 \log(2/\delta)}{n}}.$$

For comparison, Chebyshev inequality only yields $\mathbb{P}(|\frac{1}{n} \sum_{i=1}^n \varepsilon_i| \geq t) \leq 1/(nt^2)$.

2.4 The sub-Gaussian norm

The MGF definition is convenient for sums, but it is sometimes useful to work with a norm that measures the “size” of tails. The standard choice is the Orlicz ψ_2 norm.

Definition 2.12 (Sub-Gaussian norm). For a random variable X (not necessarily centered), define its sub-Gaussian norm by

$$\|X\|_{\psi_2} = \inf \left\{ t > 0 : \mathbb{E} \exp \left(\frac{X^2}{t^2} \right) \leq 2 \right\}.$$

It is standard (and can be checked directly) that $X \mapsto \|X\|_{\psi_2}$ defines a norm on random variables modulo almost sure equality; see [3, Section 2.5].

Proposition 2.13 (Equivalent characterizations). *There exist absolute constants $c, C > 0$ such that for any random variable X :*

1. for all $t \geq 0$, $\mathbb{P}(|X| \geq t) \leq 2 \exp(-ct^2/\|X\|_{\psi_2}^2)$;
2. $\mathbb{E} \exp(X^2/\|X\|_{\psi_2}^2) \leq 2$;

3. for all $p \geq 1$, $\|X\|_{L_p} \leq C\|X\|_{\psi_2}\sqrt{p}$;
 4. if $\mathbb{E}X = 0$, then for all $\lambda \in \mathbf{R}$, $\mathbb{E}\exp(\lambda X) \leq \exp(C\lambda^2\|X\|_{\psi_2}^2)$.

Remark 2.14. Items (1)–(3) do not require $\mathbb{E}X = 0$. Proofs and sharper constants can be found in [3, Proposition 2.5.2].

Centering does not significantly change the ψ_2 size. Indeed, if $\|X\|_{\psi_2} < \infty$, then $\|X - \mathbb{E}X\|_{\psi_2} \leq 2\|X\|_{\psi_2}$. Indeed, for any $t > \|X\|_{\psi_2}$ we have $\mathbb{E}\exp(X^2/t^2) \leq 2$, and Jensen gives $\exp((\mathbb{E}X)^2/t^2) \leq \mathbb{E}\exp(X^2/t^2) \leq 2$, hence $\|\mathbb{E}X\|_{\psi_2} = |\mathbb{E}X|/\sqrt{\log 2} \leq t$. Therefore $\|X - \mathbb{E}X\|_{\psi_2} \leq \|X\|_{\psi_2} + \|\mathbb{E}X\|_{\psi_2} \leq \|X\|_{\psi_2} + t$, and letting $t \rightarrow \|X\|_{\psi_2}$ yields the claim.

Example 2.15 (Bounded random variables). If $|X| \leq M$ almost surely, then $\|X\|_{\psi_2} \leq M/\sqrt{\log 2}$. Indeed, $\exp(X^2/t^2) \leq \exp(M^2/t^2)$ almost surely, so choosing $t = M/\sqrt{\log 2}$ makes $\mathbb{E}\exp(X^2/t^2) \leq 2$.

The next proposition is the ψ_2 -norm analogue of the variance additivity identity. It says that independent sub-Gaussian variables combine in an “ ℓ_2 ” fashion.

Proposition 2.16 (Sums of independent sub-Gaussians). *There exists an absolute constant $C > 0$ such that if $X_1, \dots, X_n$ are independent mean-zero random variables, then*

$$\left\| \sum_{i=1}^n X_i \right\|_{\psi_2}^2 \leq C \sum_{i=1}^n \|X_i\|_{\psi_2}^2.$$

Proof. By Theorem 2.13(4), for each i and all $\lambda \in \mathbf{R}$,

$$\mathbb{E}e^{\lambda X_i} \leq \exp(C_0\lambda^2\|X_i\|_{\psi_2}^2)$$

for an absolute constant C_0 . Using independence and multiplying MGFs,

$$\mathbb{E}\exp\left(\lambda \sum_{i=1}^n X_i\right) = \prod_{i=1}^n \mathbb{E}e^{\lambda X_i} \leq \exp\left(C_0\lambda^2 \sum_{i=1}^n \|X_i\|_{\psi_2}^2\right).$$

Apply Theorem 2.5 ((1) to (4)) and use the definition of $\|\cdot\|_{\psi_2}$ to conclude the proof. $\square$

As a corollary, we have the following form of Hoeffding’s inequality.

Corollary 2.17 (Hoeffding’s inequality in terms of ψ_2 norms). *Let $X_1, \dots, X_n$ be independent random variables with $\max_i \|X_i\|_{\psi_2} < \infty$. Then for any $t \geq 0$,*

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right| \geq t\right) \leq 2\exp\left(-\frac{ct^2}{\sum_{i=1}^n \|X_i\|_{\psi_2}^2}\right),$$

where $c > 0$ is an absolute constant.

2.5 Khintchine-type bounds

The sub-Gaussian norm controls moments, and this leads to a useful inequality for linear combinations. In particular, for independent sub-Gaussian summands the natural geometry is Euclidean: the size of $\sum_i a_i X_i$ is governed by $\|a\|_2$.

Proposition 2.18 (Khinchine inequality for sub-Gaussian variables). *Let $X_1, \dots, X_n$ be independent mean-zero sub-Gaussian random variables such that $\text{Var}(X_i) = 1$ for all i . Let $a = (a_1, \dots, a_n) \in \mathbf{R}^n$ and $p \geq 2$. Then*

$$\left(\sum_{i=1}^n a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^n a_i X_i \right\|_{L_p} \leq C K \sqrt{p} \left(\sum_{i=1}^n a_i^2 \right)^{1/2},$$

where $K = \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}$ and $C > 0$ is an absolute constant.

Proof. For the lower bound, $\|\cdot\|_{L_p} \geq \|\cdot\|_{L_2}$ for $p \geq 2$, hence

$$\begin{aligned} \left\| \sum_{i=1}^n a_i X_i \right\|_{L_p} &\geq \left(\mathbb{E} \left| \sum_{i=1}^n a_i X_i \right|^2 \right)^{1/2} = \sqrt{\text{Var} \left(\sum_{i=1}^n a_i X_i \right)} \\ &= \left(\sum_{i=1}^n a_i^2 \text{Var}(X_i) \right)^{1/2} = \left(\sum_{i=1}^n a_i^2 \right)^{1/2}. \end{aligned}$$

For the upper bound, Theorem 2.13(3) gives $\|Y\|_{L_p} \leq C_1 \sqrt{p} \|Y\|_{\psi_2}$ for an absolute constant C_1 (see also [3, Proposition 2.6.1]). Apply this to $Y = \sum_{i=1}^n a_i X_i$ and use Theorem 2.16:

$$\begin{aligned} \left\| \sum_{i=1}^n a_i X_i \right\|_{L_p} &\leq C_1 \sqrt{p} \left\| \sum_{i=1}^n a_i X_i \right\|_{\psi_2} \\ &\leq C_2 \sqrt{p} \left(\sum_{i=1}^n \|a_i X_i\|_{\psi_2}^2 \right)^{1/2} \\ &= C_2 \sqrt{p} \left(\sum_{i=1}^n a_i^2 \|X_i\|_{\psi_2}^2 \right)^{1/2} \leq C_2 K \sqrt{p} \left(\sum_{i=1}^n a_i^2 \right)^{1/2}. \end{aligned}$$

□

Example 2.19 (Random signs). Let $\varepsilon_1, \dots, \varepsilon_n$ be independent random signs. By Theorem 2.15, $\|\varepsilon_i\|_{\psi_2} \leq 1/\sqrt{\log 2}$. Thus Theorem 2.18 yields, for $p \geq 2$,

$$\left\| \sum_{i=1}^n a_i \varepsilon_i \right\|_{L_p} \leq C \sqrt{p} \left(\sum_{i=1}^n a_i^2 \right)^{1/2}.$$

In particular, taking $p = 2$ and expanding the square gives the exact identity $\mathbb{E}(\sum_i a_i \varepsilon_i)^2 = \sum_i a_i^2$.

3 Sub-exponential random variables

Sub-Gaussian tails are not the whole story. A common and very important class of random variables has tails closer to $\exp(-ct)$, which is still fast but weaker than $\exp(-ct^2)$. This class naturally appears when we look at quadratic functionals (squares, energies) and products.

Example 3.1 (A centered χ^2 variable is not sub-Gaussian). Let $Z \sim \mathcal{N}(0, 1)$ and define $X = Z^2 - 1$ so that $\mathbb{E}X = 0$. Then, for $\lambda < 1/2$,

$$\mathbb{E}e^{\lambda X} = \mathbb{E}e^{\lambda(Z^2-1)} = e^{-\lambda} \mathbb{E}e^{\lambda Z^2} = e^{-\lambda} \frac{1}{\sqrt{1-2\lambda}},$$

and $\mathbb{E}e^{\lambda X} = +\infty$ for $\lambda \geq 1/2$. Thus X cannot satisfy a sub-Gaussian MGF bound for all $\lambda \in \mathbf{R}$.

3.1 The sub-exponential norm

Definition 3.2 (Sub-exponential norm). For a random variable X , define its sub-exponential norm by

$$\|X\|_{\psi_1} = \inf \left\{ t > 0 : \mathbb{E} \exp\left(\frac{|X|}{t}\right) \leq 2 \right\}.$$

Proposition 3.3 (Equivalent characterizations). *There exist absolute constants $c, C > 0$ such that for any random variable X :*

1. for all $t \geq 0$, $\mathbb{P}(|X| \geq t) \leq 2 \exp(-ct/\|X\|_{\psi_1})$;
2. $\mathbb{E} \exp(|X|/\|X\|_{\psi_1}) \leq 2$;
3. for all $p \geq 1$, $\|X\|_{L_p} \leq Cp\|X\|_{\psi_1}$;
4. if $\mathbb{E}X = 0$, then for all λ with $|\lambda| \leq c/\|X\|_{\psi_1}$,

$$\mathbb{E} e^{\lambda X} \leq \exp(C\lambda^2\|X\|_{\psi_1}^2).$$

The sub-exponential class contains the sub-Gaussian class.

Proposition 3.4. *Any sub-Gaussian random variable is sub-exponential, and $\|X\|_{\psi_1} \lesssim \|X\|_{\psi_2}$.*

It is often convenient to remember that taking squares converts ψ_2 into ψ_1 .

Proposition 3.5. *For any random variable X ,*

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2.$$

Proof. By definition,

$$\|X^2\|_{\psi_1} = \inf \left\{ t > 0 : \mathbb{E} \exp\left(\frac{X^2}{t}\right) \leq 2 \right\}.$$

Writing $t = s^2$ shows that the condition $\mathbb{E} \exp(X^2/t) \leq 2$ is equivalent to $\mathbb{E} \exp(X^2/s^2) \leq 2$. $\square$

Proposition 3.6. *If X and Y are random variables (not necessarily independent), then*

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}.$$

Proof. Let $a = \|X\|_{\psi_2}$ and $b = \|Y\|_{\psi_2}$. By Young inequality, $|uv| \leq \frac{u^2}{2} + \frac{v^2}{2}$ for all real u, v . With $u = X/a$ and $v = Y/b$, we obtain

$$\frac{|XY|}{ab} \leq \frac{X^2}{2a^2} + \frac{Y^2}{2b^2}.$$

Therefore,

$$\mathbb{E} \exp\left(\frac{|XY|}{ab}\right) \leq \mathbb{E} \left[\exp\left(\frac{X^2}{2a^2}\right) \exp\left(\frac{Y^2}{2b^2}\right) \right].$$

By Cauchy-Schwarz and the definition of $\|\cdot\|_{\psi_2}$,

$$\mathbb{E} \exp\left(\frac{|XY|}{ab}\right) \leq \left(\mathbb{E} \exp\left(\frac{X^2}{a^2}\right) \right)^{1/2} \left(\mathbb{E} \exp\left(\frac{Y^2}{b^2}\right) \right)^{1/2} \leq \sqrt{2} \cdot \sqrt{2} = 2.$$

Thus $\|XY\|_{\psi_1} \leq ab$ by Theorem 3.2. $\square$

3.2 Examples

It is useful to keep a few canonical sub-exponential distributions in mind. The following examples can be verified directly from Theorem 3.2.

Example 3.7 (Exponential random variable). Let $Y \sim \text{Exp}(\lambda)$ with rate $\lambda > 0$, i.e. $\mathbb{P}(Y \geq y) = e^{-\lambda y}$ for $y \geq 0$. Then Y is sub-exponential and

$$\|Y\|_{\psi_1} = \frac{2}{\lambda}.$$

Indeed, since $Y \geq 0$,

$$\mathbb{E} \exp\left(\frac{|Y|}{t}\right) = \mathbb{E} \exp\left(\frac{Y}{t}\right) = \frac{\lambda}{\lambda - 1/t}, \quad t > \frac{1}{\lambda}.$$

The condition $\mathbb{E} \exp(Y/t) \leq 2$ is equivalent to $t \geq 2/\lambda$, which proves the claim.

Example 3.8 (Laplace distribution). Let X have the Laplace (double exponential) distribution with density $\frac{1}{2b} e^{-|x|/b}$, $b > 0$. Then $|X|$ is exponential with rate $1/b$, and therefore

$$\|X\|_{\psi_1} = \| |X| \|_{\psi_1} = 2b.$$

Example 3.9 (Squares and products). If $Z \sim \mathcal{N}(0, 1)$, then Z is sub-Gaussian and hence Z^2 is sub-exponential by Theorem 3.5. More generally, if X and Y are sub-Gaussian, then the product XY is sub-exponential by Theorem 3.6. These two operations (squaring and taking products) are a common way sub-exponential variables enter high-dimensional probability.

3.3 Bernstein inequality

We now turn to Bernstein-type inequalities. The ψ_1 form in Theorem 3.10 is convenient because it only assumes sub-exponential tails and it automatically interpolates between a *quadratic* regime (typical of Gaussian behavior) and a *linear* regime (typical of purely exponential tails). In many applications, however, it is also helpful to have a version that makes the role of the *variance* explicit. We record both viewpoints and then explain how to translate tail bounds into high-probability statements.

Theorem 3.10 (Bernstein's inequality (ψ_1 form)). *Let $X_1, \dots, X_n$ be independent mean-zero sub-exponential random variables. Then for all $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-c \min\left(\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_{1 \leq i \leq n} \|X_i\|_{\psi_1}}\right)\right),$$

and therefore, by symmetry and the union bound,

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-c \min\left(\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_{1 \leq i \leq n} \|X_i\|_{\psi_1}}\right)\right),$$

where $c > 0$ is an absolute constant.

Proof. This is [3, Theorem 2.8.1]. The proof applies Chernoff's method to $\exp(\lambda \sum_{i=1}^n X_i)$, bounds the MGF using sub-exponentiality, and optimizes over λ . $\square$

A variance-aware bound for Bernoulli. For two-point distributions one can sometimes obtain MGF bounds that adapt sharply to the parameter. A useful example is the following inequality.

Lemma 3.11 (Kearns–Saul inequality). *Let $p \in (0, 1)$ and $\lambda \in \mathbf{R}$. Then*

$$p \exp(\lambda(1-p)) + (1-p) \exp(-\lambda p) \leq \exp\left(\lambda^2 \frac{1-2p}{4 \log\left(\frac{1-p}{p}\right)}\right).$$

(The right-hand side is interpreted by continuity at $p = \frac{1}{2}$.)

Example 3.12 (Centered Bernoulli). Let $Y \sim \text{Ber}(p)$ and set $X = Y - p$. Then $\mathbb{E}X = 0$ and

$$\text{Var}(X) = p(1-p).$$

By Theorem 3.11,

$$\mathbb{E}e^{\lambda X} = p e^{\lambda(1-p)} + (1-p) e^{-\lambda p} \leq \exp\left(\lambda^2 \frac{1-2p}{4 \log\left(\frac{1-p}{p}\right)}\right).$$

When p is close to 0 or 1, the variance $p(1-p)$ is small and the bound becomes correspondingly tighter; at $p = \frac{1}{2}$ it matches the familiar proxy variance $1/4$.

3.4 A variance form of sub-exponentiality

Besides the norm-based definition of sub-exponentiality, it is often convenient to work with a two-parameter MGF bound. This separates a *variance scale* and a *tail scale*.

Definition 3.13 ((σ^2, b) -sub-exponential). A mean-zero random variable X is sub-exponential with parameters (σ^2, b) if for all λ with $|\lambda| < 1/b$,

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}\right).$$

A simple sufficient condition for Theorem 3.13 is a moment growth assumption of Bernstein type.

Definition 3.14 (Bernstein moment condition). Let X be a random variable with mean $\mu = \mathbb{E}X$ and variance $\sigma^2 = \text{Var}(X)$. We say that X satisfies Bernstein’s moment condition with parameter b if for every integer $k \geq 2$,

$$|\mathbb{E}[(X - \mu)^k]| \leq \frac{1}{2} k! \sigma^2 b^{k-2}.$$

Boundedness is a clean way to ensure Theorem 3.14, though it is far from necessary.

Proposition 3.15. *Assume that $|X - \mu| \leq B$ almost surely. Then X satisfies Bernstein’s moment condition with parameter $b = B/3$.*

Proof. For $k \geq 2$,

$$|\mathbb{E}[(X - \mu)^k]| \leq \mathbb{E}[(X - \mu)^2 |X - \mu|^{k-2}] \leq \mathbb{E}[(X - \mu)^2] B^{k-2} = \sigma^2 B^{k-2}.$$

For $k \geq 3$, note that $k! \geq 2 \cdot 3^{k-2}$, hence

$$\sigma^2 B^{k-2} \leq \frac{1}{2} k! \sigma^2 (B/3)^{k-2}.$$

This yields the claim (and $k = 2$ is immediate). □

Lemma 3.16. *Let X be mean-zero with variance σ^2 . If X satisfies Bernstein's moment condition with parameter b , then X is (σ^2, b) -sub-exponential in the sense of Theorem 3.13.*

Proof. Fix $|\lambda| < 1/b$ and expand the MGF into a Taylor series:

$$\mathbb{E}e^{\lambda X} = 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{\lambda^k \mathbb{E}[X^k]}{k!}.$$

Using Theorem 3.14 (with $\mu = 0$) and taking absolute values,

$$\mathbb{E}e^{\lambda X} \leq 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{|\lambda|^k}{k!} \cdot \frac{1}{2} k! \sigma^2 b^{k-2} = 1 + \frac{\lambda^2 \sigma^2}{2} \left(1 + \sum_{k=3}^{\infty} (b|\lambda|)^{k-2} \right).$$

The geometric series sums to $\sum_{k=3}^{\infty} (b|\lambda|)^{k-2} = \frac{b|\lambda|}{1-b|\lambda|}$, so

$$\mathbb{E}e^{\lambda X} \leq 1 + \frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}.$$

Finally, $1 + u \leq e^u$ for all $u \in \mathbf{R}$, hence

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}\right),$$

which is exactly Theorem 3.13. □

We now state the corresponding variance form of Bernstein's inequality.

Theorem 3.17 (Bernstein's inequality (variance form)). *Let $X_1, \dots, X_n$ be independent random variables with means $\mu_i = \mathbb{E}X_i$ and variances $\sigma_i^2 = \text{Var}(X_i)$. Assume that each centered variable $X_i - \mu_i$ satisfies Bernstein's moment condition with parameter b . Then for all $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{t^2}{2(\sum_{i=1}^n \sigma_i^2 + bt)}\right),$$

and hence, by symmetry and the union bound,

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2(\sum_{i=1}^n \sigma_i^2 + bt)}\right).$$

Proof. This follows from Chernoff's method combined with Theorem 3.16; see, e.g., [4, Proposition 2.14] for details. □

3.5 From tail bounds to high-probability bounds

A tail bound becomes a high-probability statement by solving for the deviation level. For example, if we know that

$$\mathbb{P}(S \geq t) \leq e^{-\psi(t)} \quad (t \geq 0)$$

for some increasing function ψ , then setting $\delta = e^{-\psi(t)}$ yields the equivalent form: with probability at least $1 - \delta$, one has $S < \psi^{-1}(\log(1/\delta))$. We will use this translation repeatedly.

3.6 Comparing Hoeffding and Bernstein on a centered Bernoulli

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p) - p$ and set $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Since $-p \leq X_i \leq 1 - p$, Hoeffding's inequality gives, for all $t \geq 0$,

$$\mathbb{P}(|\bar{X}_n| \geq t) \leq 2 \exp(-2nt^2). \quad (2)$$

Equivalently, with probability at least $1 - \delta$,

$$|\bar{X}_n| \leq \sqrt{\frac{\log(2/\delta)}{2n}}. \quad (3)$$

This bound depends on n and δ , but not on p .

Bernstein's inequality, on the other hand, incorporates the variance $\text{Var}(X_i) = p(1 - p)$. Moreover, since $|X_i| \leq 1$, Theorem 3.15 yields Bernstein's moment condition with $b = 1/3$. Applying Theorem 3.17 gives

$$\mathbb{P}(|\bar{X}_n| \geq t) \leq 2 \exp\left(-\frac{nt^2}{2p(1-p) + 2t/3}\right). \quad (4)$$

Solving for t in terms of δ yields: with probability at least $1 - \delta$,

$$|\bar{X}_n| \leq \sqrt{\left(\frac{\log(2/\delta)}{3n}\right)^2 + \frac{2p(1-p)\log(2/\delta)}{n}} + \frac{\log(2/\delta)}{3n}. \quad (5)$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, one can further simplify this to

$$|\bar{X}_n| \leq \sqrt{\frac{2p(1-p)\log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}.$$

The comparison is intuitive: when p is close to $1/2$, the variance is large and Hoeffding can be tighter; when p is close to 0 or 1, the variance is small and Bernstein becomes preferable.

3.7 Another application: relative deviations for a binomial

It is sometimes more natural to control deviations relative to the mean. Let $X \sim \text{Bin}(n, p)$, so $X = \sum_{i=1}^n Y_i$ with $Y_i \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p)$ and $\mathbb{E}X = np$. Applying Theorem 3.17 to $Y_i - p$ (where $|Y_i - p| \leq 1$ and $\text{Var}(Y_i - p) = p(1 - p)$), we obtain: for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|X - \mathbb{E}X| \leq \sqrt{2np(1-p)\log(2/\delta)} + \frac{2}{3}\log(2/\delta).$$

To relate this to $\mathbb{E}X$, use $\sqrt{ab} \leq \frac{1}{2}(\gamma a + b/\gamma)$ for $a, b \geq 0$ and $\gamma > 0$. With $a = 2\mathbb{E}X$ and $b = (1-p)\log(2/\delta)$, we get

$$\sqrt{2np(1-p)\log(2/\delta)} \leq \gamma \mathbb{E}X + \frac{(1-p)\log(2/\delta)}{2\gamma}.$$

Hence, with probability at least $1 - \delta$,

$$|X - \mathbb{E}X| \leq \gamma \mathbb{E}X + \log(2/\delta) \left(\frac{1-p}{2\gamma} + \frac{2}{3} \right).$$

If the rightmost term is at most $\gamma \mathbb{E}X$, then $|X - \mathbb{E}X| \leq 2\gamma \mathbb{E}X$, i.e.

$$(1 - 2\gamma)\mathbb{E}X \leq X \leq (1 + 2\gamma)\mathbb{E}X.$$

This is a typical route from an additive Bernstein bound to a multiplicative Chernoff-type statement. There is, however, a classical bound tailored specifically to sums of Bernoulli random variables, usually called a *multiplicative Chernoff bound*, with slightly sharper constants.

Theorem 3.18 (Multiplicative Chernoff bounds). *Let $Y_1, \dots, Y_n$ be independent Bernoulli random variables with $\mathbb{P}(Y_i = 1) = p_i$. Set $X = \sum_{i=1}^n Y_i$ and $\mu = \mathbb{E}X = \sum_{i=1}^n p_i$. Then, it holds that*

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(-\frac{\varepsilon^2}{2 + \varepsilon} \mu\right), \quad 0 \leq \varepsilon,$$

and

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) \leq \exp\left(-\frac{\varepsilon^2}{2} \mu\right), \quad 0 \leq \varepsilon \leq 1.$$

Proof. For the upper tail, fix $\lambda > 0$ and apply Chernoff's method:

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) = \mathbb{P}\left(e^{\lambda X} \geq e^{\lambda(1+\varepsilon)\mu}\right) \leq \exp(-\lambda(1 + \varepsilon)\mu) \mathbb{E}e^{\lambda X}.$$

Since $Y_i \in \{0, 1\}$,

$$\mathbb{E}e^{\lambda Y_i} = (1 - p_i) + p_i e^\lambda = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1)),$$

and therefore, by independence,

$$\mathbb{E}e^{\lambda X} = \prod_{i=1}^n \mathbb{E}e^{\lambda Y_i} \leq \exp\left(\sum_{i=1}^n p_i(e^\lambda - 1)\right) = \exp(\mu(e^\lambda - 1)).$$

Hence

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(\mu(e^\lambda - 1 - \lambda(1 + \varepsilon))\right).$$

Choosing $\lambda = \log(1 + \varepsilon)$ gives

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(-\mu((1 + \varepsilon) \log(1 + \varepsilon) - \varepsilon)\right).$$

It remains to use the standard inequality, valid for all $\varepsilon \geq 0$,

$$(1 + \varepsilon) \log(1 + \varepsilon) - \varepsilon \geq \frac{\varepsilon^2}{2 + \varepsilon}.$$

One quick way to check it is to consider $h(u) = (1 + u) \log(1 + u) - u - \frac{u^2}{2+u}$, note that $h(0) = 0$, and compute $h'(u) = \log(1 + u) - \frac{u(4+u)}{(2+u)^2} \geq 0$ using $\log(1 + u) \geq \frac{2u}{2+u}$ for $u \geq 0$. This proves the first bound.

For the lower tail, apply the same argument to $e^{-\lambda X}$ with $\lambda > 0$:

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) = \mathbb{P}\left(e^{-\lambda X} \geq e^{-\lambda(1-\varepsilon)\mu}\right) \leq \exp(\lambda(1 - \varepsilon)\mu) \mathbb{E}e^{-\lambda X}.$$

As before, $\mathbb{E}e^{-\lambda Y_i} = 1 + p_i(e^{-\lambda} - 1) \leq \exp(p_i(e^{-\lambda} - 1))$, hence $\mathbb{E}e^{-\lambda X} \leq \exp(\mu(e^{-\lambda} - 1))$ and therefore

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) \leq \exp\left(\mu(e^{-\lambda} - 1 + \lambda(1 - \varepsilon))\right).$$

Choosing $\lambda = -\log(1 - \varepsilon)$ (valid for $0 \leq \varepsilon < 1$) yields

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) \leq \exp\left(-\mu((1 - \varepsilon) \log(1 - \varepsilon) + \varepsilon)\right).$$

Finally, for $0 \leq \varepsilon \leq 1$ one has

$$(1 - \varepsilon) \log(1 - \varepsilon) + \varepsilon \geq \frac{\varepsilon^2}{2},$$

which follows by considering $g(u) = (1 - u) \log(1 - u) + u - \frac{u^2}{2}$ on $[0, 1]$, noting $g(0) = 0$ and $g'(u) = -\log(1 - u) - u \geq 0$ because $-\log(1 - u) \geq u$. This gives the second bound. $\square$

Remark 3.19. Compared to the Bernstein-based relative deviation bound above, Theorem 3.18 is specialized to Bernoulli sums and typically gives sharper constants in the multiplicative regime.

3.8 Application: Statistical learning theory

This section illustrates how the concentration tools developed so far translate into simple learning guarantees. We begin with the classical finite-class setting, and then record a few useful inequalities for maxima of random variables. We conclude with McDiarmid's inequality, which provides concentration for functions of independent samples under a bounded-differences condition.

3.9 A finite hypothesis class in the realizable case

We consider a basic binary classification model. Let $\mathcal{X}$ be an instance space and let $\mathcal{F}$ be a finite class of functions $f : \mathcal{X} \rightarrow \{0, 1\}$, with $M = |\mathcal{F}|$. Assume there is an unknown “true” classifier $f^* \in \mathcal{F}$. We observe the labeled sample

$$(X_1, f^*(X_1)), \dots, (X_n, f^*(X_n)),$$

where $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{P}$ for an unknown distribution $\mathcal{P}$ on $\mathcal{X}$.

Definition 3.20 (Risk). The *risk* of $f \in \mathcal{F}$ is the probability of misclassification on a fresh draw from $\mathcal{P}$:

$$R(f) = \mathbb{P}_{X \sim \mathcal{P}}(f(X) \neq f^*(X)).$$

Definition 3.21 (Empirical risk). The *empirical risk* of $f \in \mathcal{F}$ is the fraction of mistakes on the observed sample:

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(f(X_i) \neq f^*(X_i)).$$

For a fixed f , the indicators $\mathbf{1}(f(X_i) \neq f^*(X_i))$ are i.i.d. Bernoulli random variables with mean $R(f)$. Thus $R_n(f)$ is a rescaled binomial variable, and Bernstein's inequality yields a variance-sensitive deviation bound.

Proposition 3.22 (A Bernstein bound for the empirical risk). *Fix $f \in \mathcal{F}$ and let $\delta \in (0, 1)$. With probability at least $1 - \delta$,*

$$R(f) - R_n(f) \leq \sqrt{\frac{2R(f)(1 - R(f)) \log(1/\delta)}{n}} + \frac{2 \log(1/\delta)}{3n} \leq \sqrt{\frac{2R(f) \log(1/\delta)}{n}} + \frac{2 \log(1/\delta)}{3n}. \quad (6)$$

Proof. Let $U_i = \mathbf{1}(f(X_i) \neq f^*(X_i))$, so $\mathbb{E}U_i = R(f)$ and $\text{Var}(U_i) = R(f)(1 - R(f))$. Applying Theorem 3.17 to the centered variables $U_i - \mathbb{E}U_i$ (noting $|U_i - \mathbb{E}U_i| \leq 1$) gives, for all $t \geq 0$,

$$\mathbb{P}(R(f) - R_n(f) \geq t) = \mathbb{P}\left(\sum_{i=1}^n (R(f) - U_i) \geq nt\right) \leq \exp\left(-\frac{nt^2}{2R(f)(1 - R(f)) + \frac{2}{3}t}\right).$$

Solving the inequality $\frac{nt^2}{2R(f)(1 - R(f)) + \frac{2}{3}t} \geq \log(1/\delta)$ for t yields the first bound in (6). The second bound follows from $1 - R(f) \leq 1$. $\square$

A standard way to make (6) uniform over the class is to apply the union bound.

Corollary 3.23 (Uniform control over a finite class). *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$, simultaneously for all $f \in \mathcal{F}$,*

$$R(f) - R_n(f) \leq \sqrt{\frac{2R(f) \log(M/\delta)}{n}} + \frac{2 \log(M/\delta)}{3n}.$$

Proof. Apply Theorem 3.22 with δ/M and take the union bound over the M functions in $\mathcal{F}$. $\square$

Now define the empirical risk minimizer

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} R_n(f).$$

In the realizable setting, $f^* \in \mathcal{F}$ satisfies $R_n(f^*) = 0$, so we may choose $\hat{f}$ so that $R_n(\hat{f}) = 0$. Applying Theorem 3.23 to $\hat{f}$ yields, with probability at least $1 - \delta$,

$$R(\hat{f}) \leq \sqrt{\frac{2R(\hat{f}) \log(M/\delta)}{n}} + \frac{2 \log(M/\delta)}{3n}.$$

This inequality is self-bounding and implies $R(\hat{f}) \lesssim \frac{\log(M/\delta)}{n}$. A simple (loose but convenient) way to see this is the following: set $a = \log(M/\delta)/n$. If $R(\hat{f}) > 10a$, then

$$\sqrt{2R(\hat{f})a} < \sqrt{\frac{2}{10}} R(\hat{f}) \quad \text{and} \quad \frac{2a}{3} < \frac{1}{15} R(\hat{f}),$$

so the right-hand side is strictly smaller than $R(\hat{f})$, a contradiction. Hence $R(\hat{f}) \leq 10a$.

There is also a very direct argument that gives the same dependence (and often better constants).

Proposition 3.24 (A direct counting argument). *Fix $\delta \in (0, 1)$ and let $\hat{f} \in \mathcal{F}$ satisfy $R_n(\hat{f}) = 0$. Then, with probability at least $1 - \delta$,*

$$R(\hat{f}) \leq \frac{\log M + \log(1/\delta)}{n}.$$

Proof. Fix $\varepsilon \in (0, 1)$. By the union bound,

$$\mathbb{P}(\exists f \in \mathcal{F} : R(f) \geq \varepsilon, R_n(f) = 0) \leq \sum_{f \in \mathcal{F}} \mathbb{P}(R(f) \geq \varepsilon, R_n(f) = 0).$$

For a fixed f with $R(f) \geq \varepsilon$, each draw X_i is correctly classified with probability at most $1 - \varepsilon$, and the events are independent across i , so

$$\mathbb{P}(R_n(f) = 0) \leq (1 - \varepsilon)^n \leq e^{-\varepsilon n}.$$

Thus

$$\mathbb{P}(\exists f \in \mathcal{F} : R(f) \geq \varepsilon, R_n(f) = 0) \leq M e^{-\varepsilon n}.$$

Choose $\varepsilon = (\log M + \log(1/\delta))/n$, so that $M e^{-\varepsilon n} = \delta$. On the complementary event (of probability at least $1 - \delta$), every f with $R_n(f) = 0$ must satisfy $R(f) \leq \varepsilon$. In particular, $R(\hat{f}) \leq \varepsilon$. $\square$

3.10 Maximal inequalities

In many problems we need to control not only a single random quantity, but the maximum over a collection. The next bounds are elementary, but surprisingly useful.

Theorem 3.25 (Maximum of sub-Gaussian variables). *Let $X_1, \dots, X_n$ be mean-zero (not necessarily independent) random variables such that*

$$\mathbb{E} e^{\lambda X_i} \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \quad \text{for all } \lambda \in \mathbf{R} \text{ and all } i.$$

Then

$$\mathbb{E} \left[\max_{1 \leq i \leq n} X_i \right] \leq \sqrt{2\sigma^2 \log n}.$$

Proof. Fix $\lambda > 0$. Since $\log$ is concave, Jensen's inequality yields

$$\mathbb{E}\left[\max_i X_i\right] = \frac{1}{\lambda} \mathbb{E}\left[\log\left(e^{\lambda \max_i X_i}\right)\right] \leq \frac{1}{\lambda} \log \mathbb{E}\left[e^{\lambda \max_i X_i}\right] = \frac{1}{\lambda} \log \mathbb{E}\left[\max_i e^{\lambda X_i}\right].$$

Using $\max_i a_i \leq \sum_i a_i$ for $a_i \geq 0$ and linearity of expectation,

$$\mathbb{E}\left[\max_i e^{\lambda X_i}\right] \leq \mathbb{E}\left[\sum_{i=1}^n e^{\lambda X_i}\right] = \sum_{i=1}^n \mathbb{E}e^{\lambda X_i} \leq n \exp\left(\frac{\lambda^2 \sigma^2}{2}\right).$$

Therefore,

$$\mathbb{E}\left[\max_i X_i\right] \leq \frac{\log n}{\lambda} + \frac{\lambda \sigma^2}{2}.$$

Optimizing over $\lambda > 0$ gives the claim. $\square$

Theorem 3.26 (Maximum of sub-exponential variables). *Let $X_1, \dots, X_n$ be mean-zero (not necessarily independent) random variables such that for some $\sigma^2, b \geq 0$,*

$$\mathbb{E}e^{\lambda X_i} \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b\lambda}\right) \quad \text{for all } 0 < \lambda < 1/b \text{ and all } i,$$

with the convention that $1/0 = \infty$. Then there exists an absolute constant $C > 0$ such that

$$\mathbb{E}\left[\max_{1 \leq i \leq n} X_i\right] \leq C(\sigma \sqrt{\log n} + b \log n).$$

In particular, for an absolute constant $C' > 0$,

$$\mathbb{E}\left[\max_{1 \leq i \leq n} X_i\right] \leq C' \left(\max_{1 \leq i \leq n} \|X_i\|_{\psi_1} \right) \log n.$$

Proof. As in the proof of Theorem 3.25, for any $0 < \lambda < 1/b$ we have

$$\mathbb{E}\left[\max_i X_i\right] \leq \frac{1}{\lambda} \log \left(\sum_{i=1}^n \mathbb{E}e^{\lambda X_i} \right) \leq \frac{\log n}{\lambda} + \frac{\lambda \sigma^2}{2(1 - b\lambda)}.$$

Choose

$$\lambda = \min \left\{ \frac{\sqrt{2 \log n}}{\sigma}, \frac{1}{2b} \right\},$$

with the conventions $1/0 = \infty$ and $\sqrt{2 \log n}/0 = \infty$. Then $b\lambda \leq 1/2$, so $(1 - b\lambda)^{-1} \leq 2$. Substituting this choice of λ into the previous display gives the stated bound up to an absolute constant.

For the final statement, recall that $\|X\|_{\psi_1}$ controls sub-exponential MGF bounds up to absolute constants: indeed, if $\|X\|_{\psi_1} \leq K$, then one may take $\sigma^2 \asymp K^2$ and $b \asymp K$ in the hypothesis above. $\square$

Proposition 3.27 (A crude L_p bound for maxima). *For any random variables $X_1, \dots, X_n$ and any $p \geq 1$,*

$$\mathbb{E}\left[\max_{1 \leq i \leq n} |X_i|\right] \leq n^{1/p} \max_{1 \leq i \leq n} \|X_i\|_{L_p}.$$

Proof. Since $x \mapsto x^{1/p}$ is concave on $\mathbf{R}_+$, Jensen's inequality gives

$$\mathbb{E}\left[\max_i |X_i|\right] = \mathbb{E}\left[\left(\max_i |X_i|^p\right)^{1/p}\right] \leq \left(\mathbb{E}\left[\max_i |X_i|^p\right]\right)^{1/p}.$$

Moreover, $\max_i |X_i|^p \leq \sum_{i=1}^n |X_i|^p$, so

$$\mathbb{E} \left[\max_i |X_i|^p \right] \leq \sum_{i=1}^n \mathbb{E} |X_i|^p \leq n \max_i \mathbb{E} |X_i|^p.$$

Taking p -th roots yields the claim. $\square$

The next proposition complements Theorem 3.27 by showing that maxima inherit sub-Gaussian or sub-exponential behavior, with the familiar $\sqrt{\log n}$ or $\log n$ price.

Proposition 3.28 (ψ -norms of a maximum). *Let $X_1, \dots, X_n$ be random variables and set*

$$M = \max_{1 \leq i \leq n} |X_i|.$$

Then

$$\|M\|_{\psi_2} \leq C \sqrt{\log(2n)} \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}, \quad \|M\|_{\psi_1} \leq C \log(2n) \max_{1 \leq i \leq n} \|X_i\|_{\psi_1},$$

where $C > 0$ is an absolute constant.

Proof. We prove the ψ_2 statement; the ψ_1 case is analogous. Let $K = \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}$ and let $p \geq 1$. By the standard moment growth of sub-Gaussian random variables, for every i ,

$$\|X_i\|_{L_p} \leq C_1 \sqrt{p} K$$

for an absolute constant $C_1 > 0$.

Next, observe that the same argument used in the proof of Theorem 3.27 gives the stronger inequality

$$\|M\|_{L_p} = \left(\mathbb{E} \left[\max_i |X_i|^p \right] \right)^{1/p} \leq \left(\sum_{i=1}^n \mathbb{E} |X_i|^p \right)^{1/p} \leq n^{1/p} \max_{1 \leq i \leq n} \|X_i\|_{L_p}.$$

Combining with the moment bound above yields

$$\|M\|_{L_p} \leq C_1 n^{1/p} \sqrt{p} K.$$

Now set $p_0 = \lceil \log(2n) \rceil$. If $1 \leq p \leq p_0$, then by monotonicity of L_p norms,

$$\|M\|_{L_p} \leq \|M\|_{L_{p_0}} \leq C_1 n^{1/p_0} \sqrt{p_0} K \leq C_2 \sqrt{\log(2n)} K$$

for an absolute constant $C_2 > 0$, since $n^{1/p_0} \leq e$ and $p_0 \asymp \log(2n)$. If $p \geq p_0$, then still $n^{1/p} \leq n^{1/p_0} \leq e$, and hence

$$\|M\|_{L_p} \leq C_3 \sqrt{p} K$$

for an absolute constant $C_3 > 0$.

Finally, using the standard equivalence between ψ_2 norms and moment growth,

$$\|M\|_{\psi_2} \asymp \sup_{p \geq 1} p^{-1/2} \|M\|_{L_p},$$

we conclude that $\|M\|_{\psi_2} \leq C \sqrt{\log(2n)} K$. The ψ_1 bound follows similarly, using $\|X_i\|_{L_p} \leq C' p \|X_i\|_{\psi_1}$ and the equivalence $\|M\|_{\psi_1} \asymp \sup_{p \geq 1} p^{-1} \|M\|_{L_p}$. $\square$

Sometimes we would like to keep a “variance term” while still using a soft (non-deterministic) notion of boundedness. A convenient way to do this is to truncate the summands and pay separately for the event that at least one summand is unusually large.

Proposition 3.29 (Bernstein with a ψ_1 -random bound). *Let $Y_1, \dots, Y_m$ be i.i.d. random variables with $\mathbb{E}Y_1 = 0$, and set*

$$\sigma^2 = \mathbb{E}Y_1^2, \quad M = \max_{1 \leq i \leq m} |Y_i|, \quad K = \|M\|_{\psi_1}.$$

Then there exists an absolute constant $c > 0$ such that for every $u > 0$,

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m Y_i \geq u\right) \leq 2 \exp\left(-cm \left(\frac{u^2}{\sigma^2} \wedge \frac{u}{K}\right)\right).$$

Proof. If $\sigma^2 = 0$, then $Y_1 = 0$ almost surely and the claim is immediate. If $\sigma^2 = +\infty$ or $K = +\infty$, then the right-hand side equals 2 and the claim is again trivial. Assume henceforth that $0 < \sigma^2 < \infty$ and $0 < K < \infty$.

Let $t = mu$ and write $S = \sum_{i=1}^m Y_i$, so that

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m Y_i \geq u\right) = \mathbb{P}(S \geq t).$$

Fix a truncation level

$$\rho = 4K,$$

and define

$$X_i = Y_i \mathbf{1}\{|Y_i| \leq \rho\}, \quad Z_i = Y_i - X_i = Y_i \mathbf{1}\{|Y_i| > \rho\}.$$

Since $Y_i = X_i + Z_i$ and $\mathbb{E}Y_i = 0$, we have

$$S = \sum_{i=1}^m (X_i - \mathbb{E}X_i) + \sum_{i=1}^m (Z_i - \mathbb{E}Z_i).$$

Therefore,

$$\mathbb{P}(S \geq t) \leq \mathbb{P}\left(\sum_{i=1}^m (X_i - \mathbb{E}X_i) \geq \frac{t}{2}\right) + \mathbb{P}\left(\sum_{i=1}^m (Z_i - \mathbb{E}Z_i) \geq \frac{t}{2}\right). \quad (7)$$

For the truncated sum, $|X_i| \leq \rho$ almost surely, hence $|X_i - \mathbb{E}X_i| \leq 2\rho$. Moreover,

$$\text{Var}(X_i) \leq \mathbb{E}[X_i^2] \leq \mathbb{E}[Y_i^2] = \sigma^2, \quad \text{so} \quad \sum_{i=1}^m \text{Var}(X_i) \leq m\sigma^2.$$

Bernstein’s inequality for bounded, independent, mean-zero summands gives

$$\mathbb{P}\left(\sum_{i=1}^m (X_i - \mathbb{E}X_i) \geq \frac{t}{2}\right) \leq \exp\left(-c_1 \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{\rho}\right)\right)$$

for an absolute constant $c_1 > 0$. It remains to control the remainder sum. Define

$$q(s) = \mathbb{P}(M > s), \quad p(s) = \mathbb{P}(|Y_1| > s).$$

Since the Y_i are i.i.d. and $M = \max_{1 \leq i \leq m} |Y_i|$, we have

$$q(s) = 1 - (1 - p(s))^m, \quad s \geq 0.$$

By the definition of $K = \|M\|_{\psi_1}$, one has $\mathbb{E} \exp(M/K) \leq 2$, hence

$$q(s) = \mathbb{P}(M > s) \leq 2e^{-s/K}, \quad s \geq 0. \quad (8)$$

In particular, $q(\rho) \leq 2e^{-4} < \frac{1}{2}$, and since q is decreasing, $q(s) \leq \frac{1}{2}$ for all $s \geq \rho$. For such s ,

$$1 - q(s) = (1 - p(s))^m \leq e^{-mp(s)},$$

so

$$p(s) \leq -\frac{1}{m} \log(1 - q(s)) \leq \frac{q(s)}{m(1 - q(s))} \leq \frac{2}{m} q(s).$$

Combining with (8) yields

$$p(s) \leq \frac{4}{m} e^{-s/K}, \quad s \geq \rho. \quad (9)$$

Fix $\lambda \in (0, \frac{1}{2K}]$. Since $Z_1 = Y_1 \mathbf{1}\{|Y_1| > \rho\}$,

$$\mathbb{E} e^{\lambda Z_1} = 1 + \mathbb{E} \left[(e^{\lambda Y_1} - 1) \mathbf{1}\{|Y_1| > \rho\} \right] \leq 1 + \mathbb{E} \left[(e^{\lambda Y_1} - 1) \mathbf{1}\{Y_1 > \rho\} \right],$$

because $e^{\lambda y} - 1 \leq 0$ when $y \leq 0$. Using the tail integration formula for the increasing function $s \mapsto e^{\lambda s} - 1$ on $[\rho, \infty)$,

$$\mathbb{E} \left[(e^{\lambda Y_1} - 1) \mathbf{1}\{Y_1 > \rho\} \right] = (e^{\lambda \rho} - 1) \mathbb{P}(Y_1 > \rho) + \int_{\rho}^{\infty} \lambda e^{\lambda s} \mathbb{P}(Y_1 > s) ds.$$

Since $\mathbb{P}(Y_1 > s) \leq p(s)$ and (9) holds for $s \geq \rho$,

$$\mathbb{E} \left[(e^{\lambda Y_1} - 1) \mathbf{1}\{Y_1 > \rho\} \right] \leq \frac{4}{m} \left[(e^{\lambda \rho} - 1) e^{-\rho/K} + \int_{\rho}^{\infty} \lambda e^{\lambda s} e^{-s/K} ds \right].$$

Using $e^x - 1 \leq xe^x$ and $\lambda \leq 1/(2K)$,

$$(e^{\lambda \rho} - 1) e^{-\rho/K} \leq \lambda \rho e^{-(1/K - \lambda)\rho} \leq \lambda \rho,$$

and

$$\int_{\rho}^{\infty} \lambda e^{\lambda s} e^{-s/K} ds = \frac{\lambda}{1/K - \lambda} e^{-(1/K - \lambda)\rho} \leq 2\lambda K.$$

Therefore,

$$\mathbb{E} e^{\lambda Z_1} \leq 1 + \frac{C}{m} \lambda (\rho + K) \leq 1 + \frac{C'}{m} \lambda K \leq \exp \left(\frac{C'}{m} \lambda K \right), \quad (10)$$

for absolute constants $C, C' > 0$, where we used $\rho = 4K$.

For the centering term, set $V = |Y_1|$. Then $|Z_1| = V \mathbf{1}\{V > \rho\}$ and the identity

$$\mathbb{E} [V \mathbf{1}\{V > \rho\}] = \rho \mathbb{P}(V > \rho) + \int_{\rho}^{\infty} \mathbb{P}(V > s) ds$$

gives

$$\mathbb{E} |Z_1| = \rho p(\rho) + \int_{\rho}^{\infty} p(s) ds \leq \frac{C''}{m} K$$

by (9) and $\rho = 4K$, for an absolute constant $C'' > 0$. Hence, for $\lambda \leq 1/(2K)$,

$$\mathbb{E}e^{\lambda(Z_1 - \mathbb{E}Z_1)} = e^{-\lambda\mathbb{E}Z_1} \mathbb{E}e^{\lambda Z_1} \leq \exp(\lambda|\mathbb{E}Z_1|) \mathbb{E}e^{\lambda Z_1} \leq \exp\left(\frac{C_0}{m}\lambda K\right)$$

for an absolute constant $C_0 > 0$, using (10). By independence,

$$\mathbb{E} \exp\left(\lambda \sum_{i=1}^m (Z_i - \mathbb{E}Z_i)\right) = \left(\mathbb{E}e^{\lambda(Z_1 - \mathbb{E}Z_1)}\right)^m \leq \exp(C_0\lambda K).$$

Choose $\lambda = \frac{\log 2}{C_0 K}$ (which is at most $1/(2K)$ after increasing C_0 if needed). Then the last bound is ≤ 2 , and Markov's inequality yields

$$\mathbb{P}\left(\sum_{i=1}^m (Z_i - \mathbb{E}Z_i) \geq \frac{t}{2}\right) \leq e^{-\lambda t/2} \mathbb{E} \exp\left(\lambda \sum_{i=1}^m (Z_i - \mathbb{E}Z_i)\right) \leq 2 \exp\left(-c_2 \frac{t}{K}\right)$$

for an absolute constant $c_2 > 0$.

Insert the bounds into (7) and use $\rho = 4K$ to obtain

$$\mathbb{P}(S \geq t) \leq \exp\left(-c \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right) + 2 \exp\left(-c \frac{t}{K}\right) \leq 3 \exp\left(-c \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right)$$

for an absolute constant $c > 0$. Since $\mathbb{P}(S \geq t) \leq 1$, we may write the previous bound as

$$\mathbb{P}(S \geq t) \leq \min\left\{1, 3 \exp\left(-c \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right)\right\},$$

and this is dominated by

$$2 \exp\left(-c' \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right)$$

after possibly replacing c by a smaller absolute constant $c' > 0$. Finally, substituting $t = mu$ yields

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m Y_i \geq u\right) \leq 2 \exp\left(-c' m \left(\frac{u^2}{\sigma^2} \wedge \frac{u}{K}\right)\right),$$

as claimed. □

3.11 Bounded differences and McDiarmid's inequality

We now record a concentration inequality for functions of independent samples under a simple stability condition.

Definition 3.30 (Bounded differences). Let $\mathcal{X}$ be a set and let $g : \mathcal{X}^n \rightarrow \mathbf{R}$. We say that g satisfies the bounded differences property with constants $C_1, \dots, C_n$ if for every $i \in [n]$,

$$\sup_{\substack{x_1, \dots, x_n \in \mathcal{X} \\ x'_i \in \mathcal{X}}} \left| g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n) \right| \leq C_i.$$

Theorem 3.31 (McDiarmid's inequality). Let $X_1, \dots, X_n$ be independent random variables taking values in $\mathcal{X}$, and let $g : \mathcal{X}^n \rightarrow \mathbf{R}$ satisfy the bounded differences property with constants $C_1, \dots, C_n$. Then for

every $t \geq 0$,

$$\mathbb{P}(g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n) \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n C_i^2}\right).$$

Proof. Let $g = g(X_1, \dots, X_n)$ and define the *Doob martingale differences*

$$Y_i = \mathbb{E}[g \mid X_1, \dots, X_i] - \mathbb{E}[g \mid X_1, \dots, X_{i-1}], \quad i = 1, \dots, n,$$

so that $g - \mathbb{E}g = \sum_{i=1}^n Y_i$ and $\mathbb{E}[Y_i \mid X_1, \dots, X_{i-1}] = 0$.

For each i , define

$$h_i(x_1, \dots, x_i) = \mathbb{E}[g \mid X_1 = x_1, \dots, X_i = x_i].$$

By the bounded differences property,

$$\sup_{x_i, x'_i \in \mathcal{X}} |h_i(X_1, \dots, X_{i-1}, x_i) - h_i(X_1, \dots, X_{i-1}, x'_i)| \leq C_i,$$

and therefore Y_i , conditional on $X_1, \dots, X_{i-1}$, is supported on an interval of length at most C_i .

Hoeffding's lemma (in its standard conditional form) implies that for every $\lambda \in \mathbf{R}$,

$$\mathbb{E}[e^{\lambda Y_i} \mid X_1, \dots, X_{i-1}] \leq \exp\left(\frac{\lambda^2 C_i^2}{8}\right).$$

Iterating conditional expectations then gives

$$\mathbb{E}e^{\lambda(g - \mathbb{E}g)} = \mathbb{E}e^{\lambda \sum_{i=1}^n Y_i} \leq \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^n C_i^2\right).$$

Finally, Chernoff's method yields, for any $\lambda > 0$,

$$\mathbb{P}(g - \mathbb{E}g \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n C_i^2\right),$$

and optimizing over λ gives the stated bound. $\square$

Remark 3.32. As a sanity check, if $g(X_1, \dots, X_n) = \sum_{i=1}^n X_i$ with $X_i \in [0, 1]$, then $C_i = 1$ and Theorem 3.31 recovers Hoeffding's inequality.

3.12 Application of bounded differences: kernel density estimation

We briefly illustrate how bounded differences can be used to control the *random fluctuations* of an estimator around its mean. Throughout this section we focus on the one-dimensional setting for clarity.

Let $X_1, \dots, X_n$ be i.i.d. samples from an unknown density f on $\mathbf{R}$. Given a kernel $K : \mathbf{R} \rightarrow [0, \infty)$ with $\int_{\mathbf{R}} K(u) du = 1$ and a bandwidth $h > 0$, define the kernel density estimator

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{j=1}^n K\left(\frac{x - X_j}{h}\right).$$

A natural way to measure accuracy is the L^1 error

$$\|\hat{f}_n - f\|_1 = \int_{\mathbf{R}} |\hat{f}_n(x) - f(x)| dx.$$

Since f is unknown, this quantity cannot be evaluated directly from the data; instead, we study its expectation $\mathbb{E}\|\hat{f}_n - f\|_1$ and ask how strongly $\|\hat{f}_n - f\|_1$ concentrates around this mean.

For a deterministic input $(x_1, \dots, x_n) \in \mathbf{R}^n$, define

$$g(x_1, \dots, x_n) = \|\hat{f}_n - f\|_1,$$

where $\hat{f}_n$ is computed from the sample $(x_1, \dots, x_n)$. To apply Theorem 3.31, we estimate how much g can change if a single coordinate is modified.

Fix $i \in [n]$ and let $x'_i \in \mathbf{R}$ be another value (keeping the other coordinates fixed). Write $\hat{f}'_n$ for the estimator constructed from $(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)$. Then, by the triangle inequality,

$$\begin{aligned} & \left| g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n) \right| \\ & \leq \int_{\mathbf{R}} |\hat{f}_n(x) - \hat{f}'_n(x)| \, dx \\ & = \frac{1}{nh} \int_{\mathbf{R}} \left| K\left(\frac{x - x_i}{h}\right) - K\left(\frac{x - x'_i}{h}\right) \right| \, dx \\ & \leq \frac{1}{nh} \int_{\mathbf{R}} K\left(\frac{x - x_i}{h}\right) \, dx + \frac{1}{nh} \int_{\mathbf{R}} K\left(\frac{x - x'_i}{h}\right) \, dx \\ & = \frac{2}{n}, \end{aligned}$$

where in the last step we used the change of variables $u = (x - x_i)/h$ and $\int_{\mathbf{R}} K(u) \, du = 1$ (and similarly for x'_i).

Therefore g satisfies the bounded differences property with constants $C_i = 2/n$. Applying Theorem 3.31 to g and to $-g$ and using the union bound, we obtain: for every $t \geq 0$,

$$\mathbb{P}\left(\left|\|\hat{f}_n - f\|_1 - \mathbb{E}\|\hat{f}_n - f\|_1\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (2/n)^2}\right) = 2 \exp\left(-\frac{nt^2}{2}\right).$$

In words: while $\mathbb{E}\|\hat{f}_n - f\|_1$ captures the bias–variance tradeoff through the choice of bandwidth h , the random fluctuations of $\|\hat{f}_n - f\|_1$ around its mean are, in this sense, of order $n^{-1/2}$.

4 Concentration of norms and random projections

We now return to a theme that appears repeatedly in high-dimensional statistics: for many “nice” random vectors $X \in \mathbf{R}^d$, the Euclidean norm $\|X\|_2$ is tightly concentrated. A basic example is a standard Gaussian vector, where $\|X\|_2^2$ is a sum of independent $\chi^2(1)$ variables.

4.1 Gaussian norm concentration via Bernstein

Let $X \sim \mathcal{N}(0, I_d)$, *i.e.* $X_1, \dots, X_d \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, and consider

$$\|X\|_2^2 = \sum_{i=1}^d X_i^2.$$

Since X_i is sub-Gaussian, X_i^2 is sub-exponential and $\|X_i^2\|_{\psi_1} = \|X_i\|_{\psi_2}^2 \lesssim 1$. By the triangle inequality for $\|\cdot\|_{\psi_1}$, this implies $\|X_i^2 - 1\|_{\psi_1} \lesssim 1$ as well. Applying Bernstein’s inequality (Theorem 3.10) to the

independent mean-zero variables $X_i^2 - 1$ yields

$$\mathbb{P}(|\|X\|_2^2 - d| \geq t) \leq 2 \exp\left(-c \min\left(\frac{t^2}{d}, t\right)\right), \quad t \geq 0,$$

where $c > 0$ is an absolute constant. Equivalently, normalizing by d and restricting to $t \in (0, 1)$,

$$\mathbb{P}\left(\left|\frac{\|X\|_2^2}{d} - 1\right| \geq t\right) \leq 2 \exp(-ct^2 d). \quad (11)$$

This is a first concrete instance of a general phenomenon: in high dimension, sums of many weakly dependent terms often exhibit concentration at a scale set by the dimension.

4.2 Johnson–Lindenstrauss via a Gaussian projection

The concentration bound (11) immediately leads to a simple random construction of a dimension-reduction map that approximately preserves distances.

Fix $u_1, \dots, u_n \in \mathbf{R}^d$ and let $m \ll d$. Let $\Gamma \in \mathbf{R}^{m \times d}$ have i.i.d. $\mathcal{N}(0, 1)$ entries and define the random linear map

$$\Pi(v) = \frac{1}{\sqrt{m}} \Gamma v, \quad v \in \mathbf{R}^d.$$

For a fixed $v \neq 0$, the vector Γv is Gaussian with covariance $\|v\|_2^2 I_m$, hence

$$\Pi(v) \sim \mathcal{N}\left(0, \frac{\|v\|_2^2}{m} I_m\right) \quad \text{and} \quad \frac{\|\Pi(v)\|_2^2}{\|v\|_2^2} = \frac{1}{m} \left\| \Gamma \left(\frac{v}{\|v\|_2} \right) \right\|_2^2.$$

Applying (11) in dimension m yields that for any $t \in (0, 1)$,

$$\mathbb{P}\left(\left|\frac{\|\Pi(v)\|_2^2}{\|v\|_2^2} - 1\right| \geq t\right) \leq 2 \exp(-ct^2 m).$$

Now apply the union bound over the $\binom{n}{2}$ difference vectors $u_i - u_j$. (If $u_i = u_j$, there is nothing to prove; the inequalities below hold trivially.) Thus

$$\mathbb{P}\left(\exists 1 \leq i < j \leq n \text{ with } u_i \neq u_j : \left|\frac{\|\Pi(u_i - u_j)\|_2^2}{\|u_i - u_j\|_2^2} - 1\right| \geq t\right) \leq 2 \binom{n}{2} \exp(-ct^2 m).$$

In particular, if

$$m \geq \frac{1}{ct^2} \log\left(\frac{2\binom{n}{2}}{\delta}\right),$$

then with probability at least $1 - \delta$ we have simultaneously for all $1 \leq i < j \leq n$,

$$(1 - t)\|u_i - u_j\|_2^2 \leq \|\Pi(u_i) - \Pi(u_j)\|_2^2 \leq (1 + t)\|u_i - u_j\|_2^2,$$

where we used the linearity $\Pi(u_i) - \Pi(u_j) = \Pi(u_i - u_j)$. This is a standard Gaussian proof of the Johnson–Lindenstrauss lemma; the striking feature is that the required projection dimension m depends only logarithmically on n and does not depend on the ambient dimension d .

4.3 Concentration of $\|X\|_2$ for independent sub-Gaussian coordinates

The previous subsection focused on $\|X\|_2^2$. Often we would like a direct concentration statement for $\|X\|_2$ itself.

Proposition 4.1 (Concentration of $\|X\|_2$ for independent coordinates). *Let $X = (X_1, \dots, X_d) \in \mathbf{R}^d$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$ for all i . Set $K = \max_{1 \leq i \leq d} \|X_i\|_{\psi_2}$. Then*

$$\|\|X\|_2 - \sqrt{d}\|_{\psi_2} \leq CK^2,$$

where $C > 0$ is an absolute constant.

Proof. First note that $K \geq 1$. Indeed, since $\mathbb{E}X_i^2 = 1$, Jensen's inequality gives $\mathbb{E} \exp(X_i^2/t^2) \geq \exp(1/t^2)$. Therefore $\mathbb{E} \exp(X_i^2/t^2) \leq 2$ forces $t^2 \geq 1/\log 2$, so $\|X_i\|_{\psi_2} \geq 1/\sqrt{\log 2}$ for every i .

By Theorem 3.5, $\|X_i^2\|_{\psi_1} = \|X_i\|_{\psi_2}^2 \leq K^2$. Also, $\|1\|_{\psi_1} = 1/\log 2$. Using the triangle inequality for $\|\cdot\|_{\psi_1}$,

$$\|X_i^2 - 1\|_{\psi_1} \leq \|X_i^2\|_{\psi_1} + \|1\|_{\psi_1} \leq K^2 + \frac{1}{\log 2} \leq C_0 K^2$$

for an absolute constant $C_0 > 0$, where we used $K \geq 1$ in the last step.

Applying Bernstein's inequality (Theorem 3.10) to the independent mean-zero variables $X_i^2 - 1$ yields: for all $u \geq 0$,

$$\mathbb{P}\left(\left|\frac{\|X\|_2^2}{d} - 1\right| \geq u\right) = \mathbb{P}\left(\left|\sum_{i=1}^d (X_i^2 - 1)\right| \geq ud\right) \leq 2 \exp\left(-cd \min\left(\frac{u^2}{K^4}, \frac{u}{K^2}\right)\right), \quad (12)$$

where $c > 0$ is an absolute constant.

Next, for any $a \geq 0$ and $u \geq 0$,

$$|a - 1| \geq u \implies |a^2 - 1| = |a - 1||a + 1| \geq u \max\{1, u\} = \max\{u, u^2\}.$$

Applying this with $a = \|X\|_2/\sqrt{d}$ yields, for every $u \geq 0$,

$$\mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{d}} - 1\right| \geq u\right) \leq \mathbb{P}\left(\left|\frac{\|X\|_2^2}{d} - 1\right| \geq \max\{u, u^2\}\right).$$

Combining this with (12), a simple case check shows that

$$d \min\left(\frac{\max\{u, u^2\}^2}{K^4}, \frac{\max\{u, u^2\}}{K^2}\right) \geq \frac{u^2 d}{K^4}, \quad u \geq 0,$$

and hence

$$\mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{d}} - 1\right| \geq u\right) \leq 2 \exp\left(-c \frac{u^2 d}{K^4}\right), \quad u \geq 0.$$

Finally, substituting $u = t/\sqrt{d}$ gives, for all $t \geq 0$,

$$\mathbb{P}\left(\|\|X\|_2 - \sqrt{d}\| \geq t\right) \leq 2 \exp\left(-c \frac{t^2}{K^4}\right).$$

By the standard tail characterization of ψ_2 norms (Theorem 2.13), this implies $\|\|X\|_2 - \sqrt{d}\|_{\psi_2} \leq CK^2$ for an absolute constant $C > 0$. $\square$

4.4 Beyond independence: isotropic and sub-Gaussian random vectors

Many random vectors in high-dimensional statistics do not have independent coordinates. A convenient baseline is the following normalization.

Definition 4.2 (Isotropic random vector). A random vector $X \in \mathbf{R}^d$ is *isotropic* if $\mathbb{E}[XX^\top] = I_d$.

If Y has mean μ and invertible covariance matrix Σ , then $X = \Sigma^{-1/2}(Y - \mu)$ is isotropic. Isotropy, however, does not imply independence. For instance, if X is uniform on the sphere $\sqrt{d}S^{d-1}$, then X is isotropic, but its coordinates are strongly dependent.

To discuss concentration without independence, it is useful to formalize what it means for a random vector to have sub-Gaussian one-dimensional marginals.

Definition 4.3 (ψ_2 size of a random vector). For a random vector $X \in \mathbf{R}^d$, define

$$\|X\|_{\psi_2} = \sup_{v \in S^{d-1}} \|\langle v, X \rangle\|_{\psi_2}.$$

We call X *sub-Gaussian* if $\|X\|_{\psi_2} < \infty$.

A more refined, variance-sensitive formulation uses a covariance proxy.

Definition 4.4 (Sub-Gaussian with covariance proxy). Let $X \in \mathbf{R}^d$ be mean-zero and let $\Sigma \succeq 0$. We say that X is sub-Gaussian with covariance proxy Σ if for all $\lambda \in \mathbf{R}$ and all $v \in S^{d-1}$,

$$\mathbb{E} \exp(\lambda \langle v, X \rangle) \leq \exp\left(\frac{\lambda^2}{2} v^\top \Sigma v\right).$$

In particular, Theorem 4.4 implies $\text{Var}(\langle v, X \rangle) \leq v^\top \Sigma v$ for every v , and hence

$$\mathbb{E}\|X\|_2^2 = \text{Tr}(\text{Cov}(X)) \leq \text{Tr}(\Sigma).$$

Proposition 4.5 (Norm tail bound under a covariance proxy). *Let X be mean-zero and sub-Gaussian with covariance proxy Σ in the sense of Theorem 4.4. Then for every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(\|X\|_2 \geq \sqrt{\text{Tr}(\Sigma)} + \sqrt{2 \lambda_{\max}(\Sigma) \log(1/\delta)}\right) \leq \delta,$$

where $\lambda_{\max}(\Sigma)$ denotes the largest eigenvalue.

Remark 4.6. We will return to the proof of this result later. It can be viewed as a dimension-free concentration statement for the 1-Lipschitz function $x \mapsto \|x\|_2$, with the scale governed by the operator norm $\lambda_{\max}(\Sigma)$ and the location controlled by $\text{Tr}(\Sigma)$.

5 A norm bound for sub-Gaussian random vectors

In Theorem 4.5 we stated a tail bound for $\|X\|_2$ under a covariance-proxy sub-Gaussian assumption. We now give a proof. The argument is pleasant for two reasons:

- it does *not* require independence of the coordinates of X ;
- it showcases a useful “change-of-measure” technique, expressed through a variational identity for log-moment generating functions.

5.1 Kullback–Leibler divergence

Let ρ and π be probability densities on a measurable set $\Theta \subseteq \mathbf{R}^d$. The *Kullback–Leibler divergence* (or *relative entropy*) is

$$\text{KL}(\rho\|\pi) = \int_{\Theta} \rho(\theta) \log\left(\frac{\rho(\theta)}{\pi(\theta)}\right) d\theta,$$

when ρ is absolutely continuous with respect to π ; otherwise $\text{KL}(\rho\|\pi) = +\infty$. As usual, the integrand is interpreted as 0 on the set where $\rho(\theta) = 0$.

Lemma 5.1 (Non-negativity of KL). *For any densities ρ, π as above, $\text{KL}(\rho\|\pi) \geq 0$, and $\text{KL}(\rho\|\pi) = 0$ if and only if $\rho = \pi$ almost everywhere.*

Proof. By Jensen’s inequality (since $\log$ is concave),

$$-\text{KL}(\rho\|\pi) = \int_{\Theta} \rho(\theta) \log\left(\frac{\pi(\theta)}{\rho(\theta)}\right) d\theta \leq \log\left(\int_{\Theta} \rho(\theta) \frac{\pi(\theta)}{\rho(\theta)} d\theta\right) = \log\left(\int_{\Theta} \pi(\theta) d\theta\right) = 0.$$

Thus $\text{KL}(\rho\|\pi) \geq 0$. Moreover, strict concavity of $\log$ implies equality in Jensen’s inequality only when $\pi(\theta)/\rho(\theta)$ is almost surely constant under ρ , which is equivalent to $\rho = \pi$ a.e. $\square$

5.2 Donsker–Varadhan variational formula

The next lemma is a convenient way to “move” expectations between measures.

Lemma 5.2 (Donsker–Varadhan variational formula). *Let π be a probability density on $\Theta \subseteq \mathbf{R}^d$ and let $h : \Theta \rightarrow \mathbf{R}$ be measurable such that $\mathbb{E}_{\theta \sim \pi} e^{h(\theta)} < \infty$. Then*

$$\log \mathbb{E}_{\theta \sim \pi} e^{h(\theta)} = \sup_{\rho} \left\{ \mathbb{E}_{\theta \sim \rho} h(\theta) - \text{KL}(\rho\|\pi) \right\},$$

where the supremum is over all probability densities ρ such that $\text{KL}(\rho\|\pi) < \infty$.

Proof. Define the tilted density

$$\pi'(\theta) = \frac{\pi(\theta) e^{h(\theta)}}{\mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)}}.$$

Then π' is a probability density on Θ . For any density ρ with $\text{KL}(\rho\|\pi) < \infty$,

$$\begin{aligned} 0 \leq \text{KL}(\rho\|\pi') &= \mathbb{E}_{\theta \sim \rho} \log\left(\frac{\rho(\theta)}{\pi'(\theta)}\right) \\ &= \mathbb{E}_{\theta \sim \rho} \log\left(\frac{\rho(\theta)}{\pi(\theta)} \cdot \frac{\mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)}}{e^{h(\theta)}}\right) \\ &= \text{KL}(\rho\|\pi) + \log \mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)} - \mathbb{E}_{\theta \sim \rho} h(\theta). \end{aligned}$$

Rearranging gives

$$\log \mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)} \geq \mathbb{E}_{\theta \sim \rho} h(\theta) - \text{KL}(\rho\|\pi).$$

Since this holds for every admissible ρ , it holds for the supremum. Finally, equality is achieved by taking $\rho = \pi'$, because then $\text{KL}(\pi'\|\pi') = 0$ by Theorem 5.1. $\square$

5.3 A uniform change-of-measure inequality

We now combine Theorem 5.2 with Markov's inequality to obtain a bound that holds simultaneously over all choices of ρ .

Lemma 5.3. *Let π be a probability density on $\Theta \subseteq \mathbf{R}^d$. Let X be a random variable (or random vector), and let $f(X, \theta)$ be a measurable function. Assume that the expectations below are finite. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ (over the draw of X), the following holds: for every probability density ρ such that $\text{KL}(\rho \parallel \pi) < \infty$,*

$$\mathbb{E}_{\theta \sim \rho} f(X, \theta) \leq \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{f(X, \theta)}] + \text{KL}(\rho \parallel \pi) + \log(1/\delta). \quad (13)$$

Proof. Define

$$h_X(\theta) = f(X, \theta) - \log \mathbb{E}[e^{f(X, \theta)}],$$

where the expectation is with respect to the randomness of X . Now set

$$Y_X = \sup_{\rho} \left\{ \mathbb{E}_{\theta \sim \rho} h_X(\theta) - \text{KL}(\rho \parallel \pi) \right\}.$$

By Theorem 5.2, we also have

$$Y_X = \log \mathbb{E}_{\theta \sim \pi} e^{h_X(\theta)}.$$

A direct computation shows $\mathbb{E}[e^{Y_X}] = 1$:

$$\begin{aligned} \mathbb{E}[e^{Y_X}] &= \mathbb{E} \left[\mathbb{E}_{\theta \sim \pi} e^{h_X(\theta)} \right] \\ &= \mathbb{E}_{\theta \sim \pi} \mathbb{E} \left[\exp(f(X, \theta) - \log \mathbb{E} e^{f(X, \theta)}) \right] \\ &= \mathbb{E}_{\theta \sim \pi} \left[\frac{\mathbb{E} e^{f(X, \theta)}}{\mathbb{E} e^{f(X, \theta)}} \right] = 1. \end{aligned}$$

By Markov's inequality, for every $t \geq 0$,

$$\mathbb{P}(Y_X \geq t) = \mathbb{P}(e^{Y_X} \geq e^t) \leq e^{-t} \mathbb{E}[e^{Y_X}] = e^{-t}.$$

Taking $t = \log(1/\delta)$ yields $\mathbb{P}(Y_X \geq \log(1/\delta)) \leq \delta$. Equivalently, with probability at least $1 - \delta$,

$$Y_X \leq \log(1/\delta).$$

Unpacking the definition of Y_X , this means that with probability at least $1 - \delta$, for every admissible ρ ,

$$\mathbb{E}_{\theta \sim \rho} h_X(\theta) - \text{KL}(\rho \parallel \pi) \leq \log(1/\delta).$$

Finally, substitute the definition of h_X and rearrange to obtain (13). $\square$

5.4 A few Gaussian computations

We record several elementary facts that we will use in the proof of Theorem 4.5.

Lemma 5.4 (Quadratic forms of a spherical Gaussian). *Let $Y \sim \mathcal{N}(\mu, \sigma^2 I_d)$ and let $A \in \mathbf{R}^{d \times d}$ be any matrix. Then*

$$\mathbb{E}[Y^\top A Y] = \sigma^2 \text{Tr}(A) + \mu^\top A \mu.$$

Proof. Write $Y = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, I_d)$. Expanding and using $\mathbb{E}Z = 0$ gives

$$\mathbb{E}[Y^\top AY] = \mu^\top A\mu + \sigma^2 \mathbb{E}[Z^\top AZ].$$

Finally, by the trace trick,

$$\mathbb{E}[Z^\top AZ] = \mathbb{E}[\text{Tr}(Z^\top AZ)] = \mathbb{E}[\text{Tr}(AZZ^\top)] = \text{Tr}(A \mathbb{E}[ZZ^\top]) = \text{Tr}(A).$$

□

Lemma 5.5 (KL divergence between shifted Gaussians). *Let $\beta > 0$ and $\|v\|_2 = 1$. Let ρ and π be the densities of $\mathcal{N}(v, I_d/\beta)$ and $\mathcal{N}(0, I_d/\beta)$, respectively. Then*

$$\text{KL}(\rho \parallel \pi) = \frac{\beta}{2}.$$

Proof. For $x \in \mathbf{R}^d$, both densities share the same normalizing constant $c(\beta)$ and satisfy

$$\rho(x) = c(\beta) \exp\left(-\frac{\beta}{2}\|x - v\|_2^2\right), \quad \pi(x) = c(\beta) \exp\left(-\frac{\beta}{2}\|x\|_2^2\right).$$

Hence

$$\log \frac{\rho(x)}{\pi(x)} = -\frac{\beta}{2}(\|x - v\|_2^2 - \|x\|_2^2) = \frac{\beta}{2}(2\langle x, v \rangle - \|v\|_2^2).$$

If $X \sim \rho$ then $\mathbb{E}X = v$, and therefore

$$\text{KL}(\rho \parallel \pi) = \mathbb{E} \log \frac{\rho(X)}{\pi(X)} = \frac{\beta}{2}(2\langle \mathbb{E}X, v \rangle - \|v\|_2^2) = \frac{\beta}{2}(2\|v\|_2^2 - \|v\|_2^2) = \frac{\beta}{2}.$$

□

Lemma 5.6 (Two suprema). *Let $x \in \mathbf{R}^d$ and let $\Sigma \succeq 0$ be symmetric. Then*

$$\sup_{v \in S^{d-1}} \langle x, v \rangle = \|x\|_2, \quad \sup_{v \in S^{d-1}} v^\top \Sigma v = \lambda_{\max}(\Sigma).$$

Proof. The first identity is Cauchy–Schwarz, with equality at $v = x/\|x\|_2$ (when $x \neq 0$). For the second, diagonalize $\Sigma = U\Lambda U^\top$ with U orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. Writing $u = U^\top v$ gives $\|u\|_2 = 1$ and

$$v^\top \Sigma v = u^\top \Lambda u = \sum_{i=1}^d \lambda_i u_i^2 \leq \lambda_{\max}(\Sigma) \sum_{i=1}^d u_i^2 = \lambda_{\max}(\Sigma),$$

with equality when v is a top eigenvector. □

5.5 Proof of Theorem 4.5

Proof of Theorem 4.5. Recall the assumption of Theorem 4.4: for every $v \in S^{d-1}$ and every $\lambda \in \mathbf{R}$,

$$\mathbb{E} \exp(\lambda \langle v, X \rangle) \leq \exp\left(\frac{\lambda^2}{2} v^\top \Sigma v\right).$$

Fix parameters $\alpha > 0$ and $\beta > 0$ and a direction $v \in S^{d-1}$ (to be chosen later). Consider the function $f(X, \theta) = \alpha \langle X, \theta \rangle$, defined on $\theta \in \mathbf{R}^d$.

Let π and ρ be the densities of the Gaussians $\mathcal{N}(0, I_d/\beta)$ and $\mathcal{N}(v, I_d/\beta)$, respectively. Applying Theorem 5.3 gives that, with probability at least $1 - \delta$,

$$\mathbb{E}_{\theta \sim \rho} f(X, \theta) \leq \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{f(X, \theta)}] + \text{KL}(\rho \parallel \pi) + \log(1/\delta). \quad (14)$$

We evaluate the terms on the right-hand side. First, since $\mathbb{E}_{\theta \sim \rho} \theta = v$,

$$\mathbb{E}_{\theta \sim \rho} f(X, \theta) = \alpha \mathbb{E}_{\theta \sim \rho} \langle X, \theta \rangle = \alpha \langle X, v \rangle. \quad (15)$$

Second, using the sub-Gaussian property with direction $\theta/\|\theta\|_2$ and parameter $\lambda = \alpha\|\theta\|_2$ (the bound is trivial when $\theta = 0$),

$$\mathbb{E} e^{\alpha \langle X, \theta \rangle} = \mathbb{E} e^{\|\theta\|_2 \alpha \langle X, \theta/\|\theta\|_2 \rangle} \leq \exp\left(\frac{\alpha^2}{2} \theta^\top \Sigma \theta\right),$$

hence

$$\begin{aligned} \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{f(X, \theta)}] &= \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{\alpha \langle X, \theta \rangle}] \leq \mathbb{E}_{\theta \sim \rho} \left[\frac{\alpha^2}{2} \theta^\top \Sigma \theta \right] \\ &= \frac{\alpha^2}{2} \mathbb{E}_{\theta \sim \rho} [\theta^\top \Sigma \theta] \\ &= \frac{\alpha^2}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + v^\top \Sigma v \right) \leq \frac{\alpha^2}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + \lambda_{\max}(\Sigma) \right), \end{aligned} \quad (16)$$

where we used Theorems 5.4 and 5.6. Third, by Theorem 5.5,

$$\text{KL}(\rho \parallel \pi) = \frac{\beta}{2}. \quad (17)$$

Plugging (15)–(17) into (14), we obtain: with probability at least $1 - \delta$,

$$\alpha \langle X, v \rangle \leq \frac{\alpha^2}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + \lambda_{\max}(\Sigma) \right) + \frac{\beta}{2} + \log(1/\delta). \quad (18)$$

Divide by α and note that the right-hand side no longer depends on v after the bound $v^\top \Sigma v \leq \lambda_{\max}(\Sigma)$. Taking the supremum over $v \in S^{d-1}$ and using Theorem 5.6, we get that with probability at least $1 - \delta$,

$$\|X\|_2 \leq \frac{\alpha}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + \lambda_{\max}(\Sigma) \right) + \frac{\beta}{2\alpha} + \frac{\log(1/\delta)}{\alpha}. \quad (19)$$

Now set $\gamma = \alpha/\beta$ (so $\beta/\alpha = 1/\gamma$) to rewrite

$$\|X\|_2 \leq \frac{\gamma}{2} \text{Tr}(\Sigma) + \frac{1}{2\gamma} + \frac{\alpha}{2} \lambda_{\max}(\Sigma) + \frac{\log(1/\delta)}{\alpha}.$$

We may optimize γ and α separately. Indeed, for $a, b > 0$, the function $x \mapsto ax + b/x$ on $(0, \infty)$ is minimized at $x^* = \sqrt{b/a}$, and its minimum value is $2\sqrt{ab}$. Applying this twice gives

$$\inf_{\gamma > 0} \left\{ \frac{\gamma}{2} \text{Tr}(\Sigma) + \frac{1}{2\gamma} \right\} = \sqrt{\text{Tr}(\Sigma)}, \quad \inf_{\alpha > 0} \left\{ \frac{\alpha}{2} \lambda_{\max}(\Sigma) + \frac{\log(1/\delta)}{\alpha} \right\} = \sqrt{2 \lambda_{\max}(\Sigma) \log(1/\delta)}.$$

Substituting these choices into (19) yields

$$\|X\|_2 \leq \sqrt{\text{Tr}(\Sigma)} + \sqrt{2 \lambda_{\max}(\Sigma) \log(1/\delta)},$$

which is exactly the claim. $\square$

5.6 Gaussian concentration for Lipschitz functions

Finally, we record a classical concentration inequality for Lipschitz functions of a Gaussian vector.

Definition 5.7 (Lipschitz function). A function $f : \mathbf{R}^d \rightarrow \mathbf{R}$ is L -Lipschitz (with respect to $\|\cdot\|_2$) if

$$|f(x) - f(y)| \leq L\|x - y\|_2 \quad \text{for all } x, y \in \mathbf{R}^d.$$

Theorem 5.8 (Gaussian concentration inequality). Let $Z \sim \mathcal{N}(0, I_d)$ and let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be L -Lipschitz. Then for every $t \geq 0$,

$$\mathbb{P}(f(Z) - \mathbb{E}f(Z) \geq t) \leq \exp\left(-\frac{t^2}{2L^2}\right),$$

and consequently

$$\mathbb{P}(|f(Z) - \mathbb{E}f(Z)| \geq t) \leq 2\exp\left(-\frac{t^2}{2L^2}\right).$$

We will often apply this inequality to the Euclidean norm. Two simple auxiliary facts are helpful.

Lemma 5.9 (Existence of $\Sigma^{1/2}$). If $\Sigma \succeq 0$ is symmetric, then there exists a unique symmetric matrix $\Sigma^{1/2} \succeq 0$ such that $(\Sigma^{1/2})^2 = \Sigma$.

Proof. Diagonalize $\Sigma = U\Lambda U^\top$ with U orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_i \geq 0$. Then $\Sigma^{1/2} = U\Lambda^{1/2}U^\top$ with $\Lambda^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_d})$ is symmetric, positive semidefinite, and satisfies $(\Sigma^{1/2})^2 = \Sigma$. Uniqueness holds among symmetric positive semidefinite square roots. $\square$

Lemma 5.10 (The norm is 1-Lipschitz). For all $x, y \in \mathbf{R}^d$,

$$|\|x\|_2 - \|y\|_2| \leq \|x - y\|_2.$$

Proof. By the triangle inequality, $\|x\|_2 = \|x - y + y\|_2 \leq \|x - y\|_2 + \|y\|_2$, which gives $\|x\|_2 - \|y\|_2 \leq \|x - y\|_2$. Swapping the roles of x and y yields the reverse inequality. $\square$

Corollary 5.11 (Concentration of $\|X\|_2$ for a Gaussian with covariance Σ). Let $X \sim \mathcal{N}(0, \Sigma)$, where $\Sigma \succeq 0$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|\|X\|_2 - \mathbb{E}\|X\|_2| \leq \sqrt{2\lambda_{\max}(\Sigma) \log(2/\delta)}.$$

Proof. Let $Z \sim \mathcal{N}(0, I_d)$. Then X has the same distribution as $\Sigma^{1/2}Z$ (by Theorem 5.9). Define $f(z) = \|\Sigma^{1/2}z\|_2$. By Theorem 5.10,

$$|f(x) - f(y)| = |\|\Sigma^{1/2}x\|_2 - \|\Sigma^{1/2}y\|_2| \leq \|\Sigma^{1/2}(x - y)\|_2 \leq \sqrt{\lambda_{\max}(\Sigma)}\|x - y\|_2,$$

so f is $\sqrt{\lambda_{\max}(\Sigma)}$ -Lipschitz. Apply Theorem 5.8 and set $t = \sqrt{2\lambda_{\max}(\Sigma) \log(2/\delta)}$. $\square$

5.7 Simple proof of the Gaussian concentration inequality with a non-optimal constant

We now give a self-contained proof of a Gaussian concentration bound using a useful interpolation trick. The argument is short and robust, but it produces a slightly weaker numerical constant than the sharp one in Theorem 5.8.

Lemma 5.12 (A convex comparison lemma). *Let $\varphi : \mathbf{R} \rightarrow \mathbf{R}$ be convex and let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be differentiable. If $Z, Y \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ are independent, then*

$$\mathbb{E}[\varphi(f(Z) - \mathbb{E}f(Z))] \leq \mathbb{E}\left[\varphi\left(\frac{\pi}{2} \langle \nabla f(Z), Y \rangle\right)\right].$$

Proof. Let Z' be an independent copy of Z . Since $\mathbb{E}f(Z') = \mathbb{E}f(Z)$ and φ is convex, Jensen's inequality gives

$$\mathbb{E}[\varphi(f(Z) - \mathbb{E}f(Z))] = \mathbb{E}[\varphi(f(Z) - \mathbb{E}f(Z'))] \leq \mathbb{E}[\varphi(f(Z) - f(Z'))].$$

Fix $\theta \in [0, \pi/2]$ and define the interpolation

$$W(\theta) = Z \sin \theta + Z' \cos \theta, \quad W'(\theta) = Z \cos \theta - Z' \sin \theta.$$

Then $W(0) = Z'$ and $W(\pi/2) = Z$, hence by the fundamental theorem of calculus,

$$f(Z) - f(Z') = \int_0^{\pi/2} \frac{d}{d\theta} f(W(\theta)) d\theta = \int_0^{\pi/2} \langle \nabla f(W(\theta)), W'(\theta) \rangle d\theta,$$

where we used the chain rule.

Let $\Theta \sim \text{Unif}[0, \pi/2]$ be independent of (Z, Z') . Then

$$f(Z) - f(Z') = \frac{\pi}{2} \mathbb{E}_\Theta [\langle \nabla f(W(\Theta)), W'(\Theta) \rangle].$$

Applying Jensen's inequality once more (to φ) yields

$$\varphi(f(Z) - f(Z')) \leq \mathbb{E}_\Theta \left[\varphi\left(\frac{\pi}{2} \langle \nabla f(W(\Theta)), W'(\Theta) \rangle\right) \right].$$

Taking expectations over (Z, Z', Θ) gives

$$\mathbb{E}[\varphi(f(Z) - f(Z'))] \leq \mathbb{E}\left[\varphi\left(\frac{\pi}{2} \langle \nabla f(W(\Theta)), W'(\Theta) \rangle\right)\right].$$

Finally, for each fixed θ , the map $(Z, Z') \mapsto (W(\theta), W'(\theta))$ is an orthogonal transformation on $\mathbf{R}^{2d}$. Therefore $(W(\theta), W'(\theta))$ has the same distribution as (Z, Y) with $Z, Y \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ independent. It follows that the last expectation equals $\mathbb{E}[\varphi(\frac{\pi}{2} \langle \nabla f(Z), Y \rangle)]$, which proves the claim. $\square$

We now derive a tail bound. Assume for the moment that f is differentiable.

Proof of a Gaussian concentration bound. Let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be L -Lipschitz. Then $\|\nabla f(x)\|_2 \leq L$ for all x : indeed, for any unit vector $u \in S^{d-1}$ and any $t \in \mathbf{R}$,

$$|f(x + tu) - f(x)| \leq L|t|,$$

and letting $t \rightarrow 0$ shows $|\langle \nabla f(x), u \rangle| \leq L$; taking the supremum over u yields $\|\nabla f(x)\|_2 \leq L$.

Apply Theorem 5.12 with $\varphi(s) = e^{\lambda s}$ for $\lambda \in \mathbf{R}$:

$$\mathbb{E} \exp(\lambda(f(Z) - \mathbb{E}f(Z))) \leq \mathbb{E} \exp\left(\frac{\pi\lambda}{2} \langle \nabla f(Z), Y \rangle\right).$$

Conditioning on Z , the random variable $\langle \nabla f(Z), Y \rangle$ is Gaussian with mean 0 and variance $\|\nabla f(Z)\|_2^2$. Hence

$$\mathbb{E}\left[\exp\left(\frac{\pi\lambda}{2} \langle \nabla f(Z), Y \rangle\right) \mid Z\right] = \exp\left(\frac{1}{2} \left(\frac{\pi\lambda}{2}\right)^2 \|\nabla f(Z)\|_2^2\right) \leq \exp\left(\frac{\pi^2 \lambda^2 L^2}{8}\right).$$

Taking expectation over Z gives the MGF bound

$$\mathbb{E} \exp(\lambda(f(Z) - \mathbb{E}f(Z))) \leq \exp\left(\frac{\pi^2 \lambda^2 L^2}{8}\right), \quad \lambda \in \mathbf{R}.$$

By Chernoff's method, for any $t \geq 0$ and any $\lambda > 0$,

$$\mathbb{P}(f(Z) - \mathbb{E}f(Z) \geq t) \leq \exp\left(-\lambda t + \frac{\pi^2 \lambda^2 L^2}{8}\right).$$

Optimizing over λ gives $\lambda^* = 4t/(\pi^2 L^2)$ and therefore

$$\mathbb{P}(f(Z) - \mathbb{E}f(Z) \geq t) \leq \exp\left(-\frac{2t^2}{\pi^2 L^2}\right).$$

Applying the same bound to $-f$ and using the union bound yields

$$\mathbb{P}(|f(Z) - \mathbb{E}f(Z)| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\pi^2 L^2}\right), \quad t \geq 0.$$

Finally, if f is merely Lipschitz (not differentiable), one can apply the argument to a smooth approximation of f and pass to the limit; we omit these standard approximation details. $\square$

Remark 5.13. The proof above yields the same Gaussian t^2/L^2 scaling as Theorem 5.8, but with a worse numerical constant in the exponent. Sharper constants can be obtained, for example, via Gaussian isoperimetry or the Gaussian log-Sobolev inequality.

6 Matrices and random matrix concentration

We now turn to random matrices. The first goal is to understand how the operator norm behaves for matrices with independent entries. The second goal is to revisit sample covariance matrices and show how the same variational method used earlier for vectors extends naturally to this setting.

Throughout, if $A \in \mathbf{R}^{m \times n}$, we write its singular value decomposition in the form

$$A = \sum_{j=1}^r s_j(A) u_j v_j^\top,$$

where $r = \text{rank}(A)$, the vectors $u_1, \dots, u_r \in \mathbf{R}^m$ and $v_1, \dots, v_r \in \mathbf{R}^n$ are orthonormal, and

$$s_1(A) \geq s_2(A) \geq \dots \geq s_r(A) > 0$$

are the singular values of A . The two most common norms for us will be the operator norm

$$\|A\|_{\text{op}} = \sup_{v \in S^{n-1}} \|Av\|_2 = \sup_{u \in S^{m-1}, v \in S^{n-1}} u^\top Av$$

and the Frobenius norm

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2 \right)^{1/2}.$$

These are related to singular values by

$$\|A\|_{\text{op}} = s_1(A), \quad \|A\|_F^2 = \sum_{j=1}^r s_j(A)^2.$$

When A is square and symmetric, $\|A\|_{\text{op}}$ is the largest absolute value of an eigenvalue of A .

6.1 Euclidean nets, covering numbers, and a discretization lemma

A recurring theme in random matrix theory is that operator norms can be reduced to maxima over finite sets. The right finite sets are obtained from coverings of the Euclidean sphere.

Definition 6.1 (ε -net and covering number). Let (K, d) be a metric space and let $\varepsilon > 0$. A set $\mathcal{N} \subseteq K$ is called an ε -net of K if for every $x \in K$ there exists $y \in \mathcal{N}$ such that $d(x, y) \leq \varepsilon$. The *covering number* of K is

$$\mathcal{N}(K, d, \varepsilon) = \min\{|\mathcal{N}| : \mathcal{N} \subseteq K \text{ is an } \varepsilon\text{-net}\}.$$

Definition 6.2 (ε -separated set and packing number). Let (K, d) be a metric space and let $\varepsilon > 0$. A set $\mathcal{P} \subseteq K$ is called ε -separated if

$$d(x, y) > \varepsilon \quad \text{for all distinct } x, y \in \mathcal{P}.$$

The *packing number* of K is

$$\mathcal{P}(K, d, \varepsilon) = \max\{|\mathcal{P}| : \mathcal{P} \subseteq K \text{ is } \varepsilon\text{-separated}\}.$$

The two notions are equivalent up to harmless constants.

Lemma 6.3 (Packing and covering numbers). *For every metric space (K, d) and every $\varepsilon > 0$,*

$$\mathcal{P}(K, d, 2\varepsilon) \leq \mathcal{N}(K, d, \varepsilon) \leq \mathcal{P}(K, d, \varepsilon).$$

Proof. We begin with the right-hand inequality. Let $\mathcal{P} \subseteq K$ be a maximal ε -separated set. We claim that $\mathcal{P}$ is automatically an ε -net. Indeed, if some $x \in K$ were farther than ε from every point of $\mathcal{P}$, then $\mathcal{P} \cup \{x\}$ would still be ε -separated, contradicting maximality. Thus every maximal ε -separated set is an ε -net, and therefore

$$\mathcal{N}(K, d, \varepsilon) \leq \mathcal{P}(K, d, \varepsilon).$$

For the left-hand inequality, let $\mathcal{N}$ be an ε -net of K , and let $\mathcal{P} \subseteq K$ be 2ε -separated. Every point of $\mathcal{P}$ must lie in some ball of radius ε centered at a point of $\mathcal{N}$. Moreover, no such ball can contain two distinct points of $\mathcal{P}$, because otherwise those two points would be at distance at most 2ε , contradicting the separation property. Hence $|\mathcal{P}| \leq |\mathcal{N}|$. Taking the maximum over all 2ε -separated sets and the minimum over all ε -nets gives

$$\mathcal{P}(K, d, 2\varepsilon) \leq \mathcal{N}(K, d, \varepsilon).$$

□

We will mostly use these notions for Euclidean balls and spheres.

Lemma 6.4 (Covering number of the Euclidean ball). *Let*

$$B_2^d = \{x \in \mathbf{R}^d : \|x\|_2 \leq 1\}.$$

Then for every $\varepsilon > 0$,

$$\left(\frac{1}{\varepsilon}\right)^d \leq \mathcal{N}(B_2^d, \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d.$$

Proof. For the lower bound, let $\mathcal{N}$ be any ε -net of B_2^d . Then

$$B_2^d \subseteq \bigcup_{x \in \mathcal{N}} (x + \varepsilon B_2^d).$$

Taking volumes and using subadditivity,

$$\text{Vol}(B_2^d) \leq \sum_{x \in \mathcal{N}} \text{Vol}(x + \varepsilon B_2^d) = |\mathcal{N}| \varepsilon^d \text{Vol}(B_2^d),$$

because translations do not change volume. Therefore $|\mathcal{N}| \geq \varepsilon^{-d}$, and taking the minimum over all ε -nets proves the lower bound.

For the upper bound, let $\mathcal{P} \subseteq B_2^d$ be a maximal ε -separated set. By Theorem 6.3, it is enough to bound $|\mathcal{P}|$. The Euclidean balls

$$x + \frac{\varepsilon}{2} B_2^d, \quad x \in \mathcal{P},$$

are pairwise disjoint. Also, because $\|x\|_2 \leq 1$ for every $x \in B_2^d$,

$$x + \frac{\varepsilon}{2} B_2^d \subseteq \left(1 + \frac{\varepsilon}{2}\right) B_2^d.$$

Hence

$$|\mathcal{P}| \text{Vol}\left(\frac{\varepsilon}{2} B_2^d\right) \leq \text{Vol}\left(\left(1 + \frac{\varepsilon}{2}\right) B_2^d\right).$$

Using the scaling of volume gives

$$|\mathcal{P}| \leq \left(\frac{1 + \varepsilon/2}{\varepsilon/2}\right)^d = \left(1 + \frac{2}{\varepsilon}\right)^d.$$

Now apply Theorem 6.3. □

Remark 6.5. Since $S^{d-1} \subseteq B_2^d$, the same upper bound also applies to the sphere:

$$\mathcal{N}(S^{d-1}, \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d.$$

The next observation is the basic bridge between nets and operator norms.

Proposition 6.6 (Approximating the operator norm by a net). *Let $A \in \mathbf{R}^{m \times n}$ and let $\mathcal{M} \subseteq S^{m-1}$ and $\mathcal{N} \subseteq S^{n-1}$ be ε -nets with $\varepsilon \in (0, \frac{1}{2})$. Then*

$$\sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top A v| \leq \|A\|_{\text{op}} \leq \frac{1}{1 - 2\varepsilon} \sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top A v|.$$

Proof. The lower bound is immediate from the definition of $\|A\|_{\text{op}}$.

For the upper bound, fix $v \in S^{n-1}$. Choose $v_0 \in \mathcal{N}$ such that $\|v - v_0\|_2 \leq \varepsilon$. Then

$$\|Av\|_2 \leq \|A(v - v_0)\|_2 + \|Av_0\|_2 \leq \varepsilon \|A\|_{\text{op}} + \|Av_0\|_2.$$

Taking the supremum over $v \in S^{n-1}$ yields

$$\|A\|_{\text{op}} \leq \varepsilon \|A\|_{\text{op}} + \sup_{w \in \mathcal{N}} \|Aw\|_2,$$

hence

$$(1 - \varepsilon)\|A\|_{\text{op}} \leq \sup_{w \in \mathcal{N}} \|Aw\|_2.$$

Now fix $w \in \mathcal{N}$. For every $u \in S^{m-1}$ choose $u_0 \in \mathcal{M}$ with $\|u - u_0\|_2 \leq \varepsilon$. Then

$$|u^\top Aw| \leq |u_0^\top Aw| + |(u - u_0)^\top Aw| \leq \sup_{z \in \mathcal{M}} |z^\top Aw| + \varepsilon \|Aw\|_2.$$

Taking the supremum over $u \in S^{m-1}$ gives

$$\|Aw\|_2 \leq \sup_{z \in \mathcal{M}} |z^\top Aw| + \varepsilon \|Aw\|_2,$$

and therefore

$$(1 - \varepsilon)\|Aw\|_2 \leq \sup_{z \in \mathcal{M}} |z^\top Aw|.$$

Now take the supremum over $w \in \mathcal{N}$ to obtain

$$(1 - \varepsilon) \sup_{w \in \mathcal{N}} \|Aw\|_2 \leq \sup_{z \in \mathcal{M}, w \in \mathcal{N}} |z^\top Aw|.$$

Combining the last two estimates yields

$$(1 - \varepsilon)^2 \|A\|_{\text{op}} \leq \sup_{z \in \mathcal{M}, w \in \mathcal{N}} |z^\top Aw|.$$

Finally, since $\varepsilon \in (0, \frac{1}{2})$,

$$(1 - \varepsilon)^2 \geq 1 - 2\varepsilon,$$

which gives the claimed inequality. $\square$

6.2 Upper bounds for matrices with independent entries

We now apply the net argument to random matrices with independent sub-Gaussian entries.

Theorem 6.7 (Operator norm of a matrix with independent entries). *Let $X = (X_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$ be an $m \times n$ random matrix whose entries are independent, mean-zero random variables. Assume that*

$$\max_{1 \leq i \leq m, 1 \leq j \leq n} \|X_{ij}\|_{\psi_2} \leq K$$

for some $K > 0$. Then there exists an absolute constant $C > 0$ such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\|X\|_{\text{op}} \leq CK \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right).$$

Proof. Let $\mathcal{M} \subseteq S^{m-1}$ and $\mathcal{N} \subseteq S^{n-1}$ be $\frac{1}{4}$ -nets. By Theorem 6.4 and the remark after it, we may choose them so that

$$|\mathcal{M}| \leq 9^m, \quad |\mathcal{N}| \leq 9^n.$$

Then Theorem 6.6 gives

$$\|X\|_{\text{op}} \leq 2 \sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v|.$$

Fix a pair $(u, v) \in \mathcal{M} \times \mathcal{N}$. Since

$$u^\top X v = \sum_{i=1}^m \sum_{j=1}^n u_i v_j X_{ij},$$

this is a sum of independent mean-zero sub-Gaussian random variables. By Theorem 2.16,

$$\begin{aligned}
\|u^\top X v\|_{\psi_2}^2 &\leq C_1 \sum_{i=1}^m \sum_{j=1}^n \|u_i v_j X_{ij}\|_{\psi_2}^2 \\
&\leq C_1 K^2 \sum_{i=1}^m \sum_{j=1}^n u_i^2 v_j^2 \\
&= C_1 K^2 \left(\sum_{i=1}^m u_i^2 \right) \left(\sum_{j=1}^n v_j^2 \right) = C_1 K^2,
\end{aligned}$$

because $u \in S^{m-1}$ and $v \in S^{n-1}$. Thus there exists an absolute constant $c_1 > 0$ such that

$$\mathbb{P}(|u^\top X v| \geq s) \leq 2 \exp\left(-c_1 \frac{s^2}{K^2}\right), \quad s \geq 0.$$

Now apply the union bound over all pairs $(u, v) \in \mathcal{M} \times \mathcal{N}$:

$$\mathbb{P}\left(\sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v| \geq s\right) \leq 2 \cdot 9^{m+n} \exp\left(-c_1 \frac{s^2}{K^2}\right).$$

Choose

$$s = C_2 K \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right)$$

with C_2 large enough. Since

$$(\sqrt{m} + \sqrt{n})^2 \geq m + n,$$

the right-hand side is at most δ . Therefore, with probability at least $1 - \delta$,

$$\sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v| \leq C_2 K \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right).$$

Using the net-to-operator estimate finishes the proof. $\square$

The bound is already very informative in a number of familiar situations.

Example 6.8 (Rectangular Gaussian matrices). Let G be an $m \times n$ matrix with independent $\mathcal{N}(0, 1)$ entries. Since each entry has bounded ψ_2 norm by an absolute constant, Theorem 6.7 gives

$$\|G\|_{\text{op}} \leq C \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right)$$

with probability at least $1 - \delta$. This has the correct order in m and n .

Example 6.9 (Symmetric matrices and Wigner-type ensembles). Suppose W is an $n \times n$ symmetric random matrix such that the upper-triangular entries

$$\{W_{ij} : 1 \leq i < j \leq n\}$$

are independent mean-zero sub-Gaussian random variables, the diagonal entries are also independent mean-zero sub-Gaussian random variables, and all these variables have ψ_2 norm at most K .

To reduce to the independent-entry setting, define an upper-triangular matrix Y by

$$Y_{ij} = W_{ij} \quad (i < j), \quad Y_{ii} = \frac{W_{ii}}{2}, \quad Y_{ij} = 0 \quad (i > j).$$

Then $W = Y + Y^\top$, so

$$\|W\|_{\text{op}} \leq \|Y\|_{\text{op}} + \|Y^\top\|_{\text{op}} = 2\|Y\|_{\text{op}}.$$

Since the entries of Y are independent and $\max_{i,j} \|Y_{ij}\|_{\psi_2} \leq CK$, Theorem 6.7 yields

$$\|W\|_{\text{op}} \leq CK \left(\sqrt{n} + \sqrt{\log(1/\delta)} \right)$$

with probability at least $1 - \delta$.

In particular, for a standard Wigner matrix with Gaussian upper-triangular entries, the operator norm is of order $\sqrt{n}$. After dividing by $\sqrt{n}$, one obtains a random matrix whose norm remains of constant order.

6.3 Operator norm of the sample covariance matrix

We now return to sample covariance matrices.

Let X be a mean-zero random vector in $\mathbf{R}^d$ with covariance matrix

$$\Sigma = \mathbb{E}[XX^\top].$$

Assume that there exists a constant $L \geq 1$ such that for every $v \in \mathbf{R}^d$,

$$\|\langle X, v \rangle\|_{\psi_2} \leq L (\mathbb{E} \langle X, v \rangle^2)^{1/2} = L (v^\top \Sigma v)^{1/2}. \quad (20)$$

Let $X_1, \dots, X_n$ be independent copies of X , and define the sample covariance matrix

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top.$$

We also recall the *effective rank*

$$r(\Sigma) = \frac{\text{Tr}(\Sigma)}{\|\Sigma\|_{\text{op}}},$$

with the convention that the problem is trivial when $\Sigma = 0$.

The key point is that bilinear forms of $X_i X_i^\top - \Sigma$ are not only sub-exponential; they satisfy a sub-gamma type moment bound (for small enough λ), which is exactly what we need for a Bernstein-style argument.

Lemma 6.10 (Sub-gamma behavior of centered bilinear forms). *Assume (20). For $\theta, \nu \in \mathbf{R}^d$, define*

$$Z(\theta, \nu) = \langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu.$$

Then there exist positive constants c_L, C_L , depending only on L , such that

$$\|Z(\theta, \nu)\|_{\psi_1} \leq C_L (\theta^\top \Sigma \theta + \nu^\top \Sigma \nu),$$

and consequently

$$\log \mathbb{E} e^{\lambda Z(\theta, \nu)} \leq C_L \lambda^2 (\theta^\top \Sigma \theta + \nu^\top \Sigma \nu)^2$$

for all

$$|\lambda| \leq \frac{c_L}{\theta^\top \Sigma \theta + \nu^\top \Sigma \nu}.$$

Proof. By (20),

$$\|\langle X, \theta \rangle\|_{\psi_2} \leq L(\theta^\top \Sigma \theta)^{1/2}, \quad \|\langle X, \nu \rangle\|_{\psi_2} \leq L(\nu^\top \Sigma \nu)^{1/2}.$$

Therefore, by Theorem 3.6,

$$\|\langle X, \theta \rangle \langle X, \nu \rangle\|_{\psi_1} \leq \|\langle X, \theta \rangle\|_{\psi_2} \|\langle X, \nu \rangle\|_{\psi_2} \leq L^2(\theta^\top \Sigma \theta)^{1/2}(\nu^\top \Sigma \nu)^{1/2}.$$

For the deterministic centering term, we use $\|c\|_{\psi_1} = |c|/\log 2$ for constants c , and Cauchy–Schwarz:

$$|\theta^\top \Sigma \nu| \leq (\theta^\top \Sigma \theta)^{1/2}(\nu^\top \Sigma \nu)^{1/2} \leq \frac{1}{2}(\theta^\top \Sigma \theta + \nu^\top \Sigma \nu).$$

Combining these bounds with the triangle inequality for $\|\cdot\|_{\psi_1}$ gives

$$\|Z(\theta, \nu)\|_{\psi_1} \leq \|\langle X, \theta \rangle \langle X, \nu \rangle\|_{\psi_1} + \|\theta^\top \Sigma \nu\|_{\psi_1} \leq C_L(\theta^\top \Sigma \theta + \nu^\top \Sigma \nu)$$

for a constant C_L depending only on L .

Finally, $\mathbb{E}Z(\theta, \nu) = 0$ by definition, so the sub-exponential MGF control from Theorem 3.3(4) implies that for $|\lambda| \leq c/\|Z(\theta, \nu)\|_{\psi_1}$,

$$\log \mathbb{E}e^{\lambda Z(\theta, \nu)} \leq C \lambda^2 \|Z(\theta, \nu)\|_{\psi_1}^2.$$

Substituting the bound on $\|Z(\theta, \nu)\|_{\psi_1}$ and absorbing absolute constants into c_L, C_L yields the claim. $\square$

We are now ready for the main theorem.

Theorem 6.11 (Sample covariance in operator norm). *Assume (20), and let $\widehat{\Sigma}$ be the sample covariance matrix based on $X_1, \dots, X_n$. Then there exists a constant $C_L > 0$, depending only on L , such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq C_L \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma) + \log(1/\delta)}{n}} + \frac{r(\Sigma) + \log(1/\delta)}{n} \right).$$

Proof. If $\Sigma = 0$, then $X = 0$ almost surely, hence $\widehat{\Sigma} = 0$ and there is nothing to prove. Assume $\Sigma \neq 0$. Fix parameters $\beta > 0$ and $R > 0$ to be chosen later, and let

$$\pi = \mathcal{N}(0, \beta^{-1} I_d).$$

For each $u \in S^{d-1}$, define ρ_u as the law of $u + G$ conditioned on

$$\|\Sigma^{1/2} G\|_2 \leq R, \quad G \sim \mathcal{N}(0, \beta^{-1} I_d),$$

where $\Sigma^{1/2} \succeq 0$ is the symmetric square root of Σ . Equivalently, ρ_u has density

$$\frac{1}{p} \left(\frac{\beta}{2\pi} \right)^{d/2} \exp\left(-\frac{\beta}{2} \|\theta - u\|_2^2\right) \mathbf{1}\{\|\Sigma^{1/2}(\theta - u)\|_2 \leq R\}, \quad \theta \in \mathbf{R}^d,$$

where

$$p = \mathbb{P}(\|\Sigma^{1/2} G\|_2 \leq R).$$

Since the conditioning event is symmetric in G , we have $\mathbb{E}[G \mid \|\Sigma^{1/2} G\|_2 \leq R] = 0$, and therefore

$$\mathbb{E}_{\theta \sim \rho_u} \theta = u. \tag{21}$$

Now fix $u, v \in S^{d-1}$ and let

$$\rho_{u,v} = \rho_u \otimes \rho_v, \quad \pi_{2d} = \pi \otimes \pi.$$

Apply Theorem 5.3 on the parameter space $\mathbf{R}^d \times \mathbf{R}^d$ with

$$f((X_1, \dots, X_n), \theta, \nu) = \lambda \sum_{i=1}^n \left(\langle X_i, \theta \rangle \langle X_i, \nu \rangle - \theta^\top \Sigma \nu \right),$$

where $\lambda > 0$ will be chosen later. Using independence of $X_1, \dots, X_n$ to factorize the exponential moment, and using (21), we obtain that with probability at least $1 - \delta$, simultaneously for all $u, v \in S^{d-1}$,

$$\begin{aligned} \lambda u^\top (\widehat{\Sigma} - \Sigma) v &\leq \mathbb{E}_{(\theta, \nu) \sim \rho_{u,v}} \log \mathbb{E} \exp \left(\lambda (\langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu) \right) \\ &\quad + \frac{\text{KL}(\rho_{u,v} \| \pi_{2d}) + \log(1/\delta)}{n}, \end{aligned} \quad (22)$$

where X denotes an independent copy of X_1 .

If (θ, ν) lies in the support of $\rho_{u,v}$, then

$$\|\Sigma^{1/2}(\theta - u)\|_2 \leq R, \quad \|\Sigma^{1/2}(\nu - v)\|_2 \leq R.$$

Using $(a + b)^2 \leq 2a^2 + 2b^2$ in the Σ -inner product,

$$\theta^\top \Sigma \theta \leq 2(\theta - u)^\top \Sigma (\theta - u) + 2u^\top \Sigma u \leq 2R^2 + 2\|\Sigma\|_{\text{op}},$$

and similarly $\nu^\top \Sigma \nu \leq 2R^2 + 2\|\Sigma\|_{\text{op}}$. Choose

$$R^2 = 2\|\Sigma\|_{\text{op}}.$$

Then on the support of $\rho_{u,v}$,

$$\theta^\top \Sigma \theta + \nu^\top \Sigma \nu \leq 12\|\Sigma\|_{\text{op}}.$$

By Theorem 6.10, there exist constants $c_L, C_L > 0$ such that, for every

$$0 < \lambda \leq \frac{c_L}{\|\Sigma\|_{\text{op}}},$$

we have uniformly on the support of $\rho_{u,v}$,

$$\log \mathbb{E} \exp \left(\lambda (\langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu) \right) \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2.$$

Therefore,

$$\mathbb{E}_{(\theta, \nu) \sim \rho_{u,v}} \log \mathbb{E} \exp \left(\lambda (\langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu) \right) \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2. \quad (23)$$

We claim that

$$\text{KL}(\rho_u \| \pi) = \log(1/p) + \frac{\beta}{2}.$$

Indeed, for θ in the support of ρ_u ,

$$\log \frac{\rho_u(\theta)}{\pi(\theta)} = \log(1/p) + \frac{\beta}{2} (\|\theta\|_2^2 - \|\theta - u\|_2^2) = \log(1/p) + \frac{\beta}{2} (2 \langle \theta, u \rangle - \|u\|_2^2).$$

Taking expectation under ρ_u , using (21) and $\|u\|_2 = 1$, yields

$$\text{KL}(\rho_u \| \pi) = \log(1/p) + \frac{\beta}{2} (2 \langle \mathbb{E}_{\rho_u} \theta, u \rangle - 1) = \log(1/p) + \frac{\beta}{2}.$$

Since $\rho_{u,v}$ and π_{2d} are product measures,

$$\text{KL}(\rho_{u,v} \parallel \pi_{2d}) = \text{KL}(\rho_u \parallel \pi) + \text{KL}(\rho_v \parallel \pi) = 2\log(1/p) + \beta.$$

It remains to lower bound p . If $G \sim \mathcal{N}(0, \beta^{-1}I_d)$, then $\mathbb{E}[GG^\top] = \beta^{-1}I_d$, and therefore

$$\mathbb{E}\|\Sigma^{1/2}G\|_2^2 = \mathbb{E}\text{Tr}(\Sigma^{1/2}GG^\top\Sigma^{1/2}) = \text{Tr}(\Sigma \mathbb{E}[GG^\top]) = \frac{1}{\beta} \text{Tr}(\Sigma).$$

Choose

$$\beta = r(\Sigma) = \frac{\text{Tr}(\Sigma)}{\|\Sigma\|_{\text{op}}}.$$

Then $\mathbb{E}\|\Sigma^{1/2}G\|_2^2 = \|\Sigma\|_{\text{op}} = R^2/2$, and Markov's inequality yields

$$\mathbb{P}(\|\Sigma^{1/2}G\|_2 > R) \leq \frac{1}{2}, \quad \text{so} \quad p = \mathbb{P}(\|\Sigma^{1/2}G\|_2 \leq R) \geq \frac{1}{2}.$$

Since $\Sigma \neq 0$ implies $r(\Sigma) \geq 1$, we can bound

$$\text{KL}(\rho_{u,v} \parallel \pi_{2d}) = \beta + 2\log(1/p) \leq r(\Sigma) + 2\log 2 \leq C r(\Sigma),$$

and hence, for all $\delta \in (0, 1)$,

$$\text{KL}(\rho_{u,v} \parallel \pi_{2d}) + \log(1/\delta) \leq C(r(\Sigma) + \log(1/\delta)). \quad (24)$$

Insert (23) and (24) into (22). With probability at least $1 - \delta$, simultaneously for all $u, v \in S^{d-1}$,

$$\lambda u^\top (\widehat{\Sigma} - \Sigma)v \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2 + C_L \frac{r(\Sigma) + \log(1/\delta)}{n}, \quad 0 < \lambda \leq \frac{c_L}{\|\Sigma\|_{\text{op}}}.$$

The same inequality with $(-u, v)$ in place of (u, v) yields the corresponding lower bound, so we may write

$$\lambda |u^\top (\widehat{\Sigma} - \Sigma)v| \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2 + C_L \frac{r(\Sigma) + \log(1/\delta)}{n}.$$

Divide by λ and take the supremum over $u, v \in S^{d-1}$. Using $\|A\|_{\text{op}} = \sup_{u,v \in S^{d-1}} |u^\top A v|$, we obtain

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq C_L \left(\lambda \|\Sigma\|_{\text{op}}^2 + \frac{r(\Sigma) + \log(1/\delta)}{\lambda n} \right), \quad 0 < \lambda \leq \frac{c_L}{\|\Sigma\|_{\text{op}}}.$$

Set $a = r(\Sigma) + \log(1/\delta)$ and choose

$$\lambda = \frac{1}{\|\Sigma\|_{\text{op}}} \min \left\{ c_L, \sqrt{\frac{a}{n}} \right\}.$$

If $\sqrt{a/n} \leq c_L$, then both terms in parentheses are at most $\|\Sigma\|_{\text{op}} \sqrt{a/n}$. If $\sqrt{a/n} > c_L$, then $a/n \geq c_L^2$ and the same bound is controlled by a constant multiple of $\|\Sigma\|_{\text{op}} a/n$. In either case,

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq C_L \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{a}{n}} + \frac{a}{n} \right),$$

and recalling $a = r(\Sigma) + \log(1/\delta)$ finishes the proof. $\square$

Remark 6.12. When $n \gtrsim r(\Sigma) + \log(1/\delta)$, the linear term is dominated by the square-root term, and the theorem simplifies to

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \lesssim \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma) + \log(1/\delta)}{n}}.$$

So the earlier bound appears as a clean large-sample corollary of the more precise statement above.

Example 6.13 (The isotropic case). If $\Sigma = I_d$, then $r(\Sigma) = d$, and Theorem 6.11 becomes: for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left\| \frac{1}{n} \sum_{i=1}^n X_i X_i^\top - I_d \right\|_{\text{op}} \leq C_L \left(\sqrt{\frac{d + \log(1/\delta)}{n}} + \frac{d + \log(1/\delta)}{n} \right).$$

This is the familiar operator-norm concentration of the empirical covariance around the identity.

6.4 Matrix Bernstein Inequality

The scalar Bernstein inequality from the previous section gives sharp control for sums of independent real random variables. There is a matrix analogue with essentially the same shape, but the route to it is genuinely different. The reason is noncommutativity. For real numbers,

$$e^{\theta(x_1 + \dots + x_n)} = \prod_{i=1}^n e^{\theta x_i},$$

so independence turns the moment generating function of a sum into a product of one-dimensional MGFs. For matrices this identity fails unless the summands commute. Thus the scalar Laplace-transform method does not pass to matrices verbatim: one needs new input from matrix analysis.

Throughout this section, matrices are real and symmetric unless explicitly stated otherwise. For a symmetric matrix $A \in \mathbf{R}^{d \times d}$, we write $A \succeq 0$ if $x^\top A x \geq 0$ for all $x \in \mathbf{R}^d$, and $A \preceq B$ if $B - A \succeq 0$. We also write

$$\|A\|_{\text{op}} = \max\{\lambda_{\max}(A), \lambda_{\max}(-A)\}.$$

One of the standard modern forms of matrix Bernstein is comparatively recent. The noncommutative concentration philosophy goes back to work of Ahlswede–Winter, while the now-classical short proof through matrix Laplace transforms was clarified and popularized by Tropp in the early 2010s. The statement is as follows.

Theorem 6.14 (Matrix Bernstein inequality). *Let $X_1, \dots, X_n$ be independent, mean-zero, symmetric $d \times d$ random matrices, and assume that*

$$\|X_i\|_{\text{op}} \leq K \quad \text{almost surely for all } i.$$

Define the variance proxy

$$V = \sum_{i=1}^n \mathbb{E}[X_i^2], \quad v = \|V\|_{\text{op}}.$$

Then for every $t \geq 0$,

$$\mathbb{P}\left(\lambda_{\max}\left(\sum_{i=1}^n X_i\right) \geq t\right) \leq d \exp\left(-\frac{t^2/2}{v + Kt/3}\right).$$

Consequently,

$$\mathbb{P}\left(\left\|\sum_{i=1}^n X_i\right\|_{\text{op}} \geq t\right) \leq 2d \exp\left(-\frac{t^2/2}{v + Kt/3}\right).$$

Equivalently, if

$$x = \log(2d/\delta),$$

then with probability at least $1 - \delta$,

$$\left\|\sum_{i=1}^n X_i\right\|_{\text{op}} \leq \frac{Kx}{3} + \sqrt{\frac{K^2 x^2}{9} + 2vx} \leq \sqrt{2vx} + \frac{2Kx}{3}.$$

Compared with the scalar Bernstein inequality, three new features appear. First, the role of the variance is played by the matrix quantity $V = \sum_i \mathbb{E}[X_i^2]$, and in the final bound only its operator norm matters. Second, boundedness still enters through the same scale parameter K . Third, we pay an additional dimension factor d (or $2d$ for the two-sided operator-norm form). If all matrices commute, then they are simultaneously diagonalizable and the result reduces essentially to applying scalar Bernstein in a common eigenbasis together with a union bound over the d diagonal coordinates.

6.5 Useful facts for the proof of Theorem 6.14

The proof uses several basic facts from matrix analysis. The first group concerns the spectral functional calculus.

Proposition 6.15 (Spectral calculus and order). *Let $A \in \mathbf{R}^{d \times d}$ be symmetric, and write its spectral decomposition as*

$$A = \sum_{j=1}^d \lambda_j u_j u_j^\top,$$

where $\lambda_1, \dots, \lambda_d \in \mathbf{R}$ are the eigenvalues and $u_1, \dots, u_d$ is an orthonormal basis of eigenvectors. For a function $f : \mathbf{R} \rightarrow \mathbf{R}$, define

$$f(A) = \sum_{j=1}^d f(\lambda_j) u_j u_j^\top.$$

Then the following hold.

(a) If $f(x) \leq g(x)$ for every x in the spectrum of A , then

$$f(A) \preceq g(A).$$

In particular, if $\|A\|_{\text{op}} \leq K$ and $f(x) \leq g(x)$ for all $x \in [-K, K]$, then $f(A) \preceq g(A)$.

(b) If A is positive definite, then $\log(A)$ is well-defined by the same spectral rule:

$$\log(A) = \sum_{j=1}^d \log(\lambda_j) u_j u_j^\top.$$

It satisfies

$$e^{\log(A)} = A.$$

If H is symmetric, then e^H is positive definite and

$$\log(e^H) = H.$$

Moreover, if $0 \prec A \preceq B$, then

$$\log(A) \preceq \log(B).$$

(c) If $A \preceq B$ and f is increasing on an interval containing the spectra of both A and B , then

$$\mathrm{Tr}(f(A)) \leq \mathrm{Tr}(f(B)).$$

In particular,

$$A \preceq B \quad \implies \quad \mathrm{Tr}(e^A) \leq \mathrm{Tr}(e^B).$$

Proof. Part (a) is immediate from the spectral decomposition:

$$g(A) - f(A) = \sum_{j=1}^d (g(\lambda_j) - f(\lambda_j)) u_j u_j^\top \succeq 0.$$

Part (b) is a classical theorem in matrix analysis: the scalar logarithm is not only increasing on $(0, \infty)$, but *operator-monotone* on the positive definite cone.

For part (c), the variational characterization of eigenvalues implies that

$$\lambda_j(A) \leq \lambda_j(B), \quad j = 1, \dots, d,$$

when the eigenvalues are ordered nonincreasingly. Hence

$$\mathrm{Tr}(f(A)) = \sum_{j=1}^d f(\lambda_j(A)) \leq \sum_{j=1}^d f(\lambda_j(B)) = \mathrm{Tr}(f(B)).$$

□

Remark 6.16. Part (b) is special and worth pausing over. The matrix logarithm is defined only on the positive definite cone, and on that cone it is operator-monotone. One should *not* expect this from an arbitrary increasing scalar function. For instance, the scalar function $x \mapsto e^x$ is increasing on $\mathbf{R}$, but in general

$$A \preceq B \quad \text{does not imply} \quad e^A \preceq e^B,$$

unless A and B commute. The same warning applies to x^2 . What remains true without commutativity is the weaker trace monotonicity from part (c).

The next result is one of the first indications that $\mathrm{Tr}(e^\cdot)$ is the right noncommutative substitute for the scalar exponential.

Theorem 6.17 (Golden–Thompson inequality). *For symmetric matrices $H, K \in \mathbf{R}^{d \times d}$,*

$$\mathrm{Tr}(e^{H+K}) \leq \mathrm{Tr}(e^H e^K).$$

If H and K commute, then $e^{H+K} = e^H e^K$ and we recover equality. In general, commutativity fails, and Golden–Thompson says that after taking trace we still retain a one-sided substitute for the scalar identity. This already points in the correct direction, but it is not enough for a clean iteration over many independent summands. There is no equally simple n -term product formula that behaves well under conditioning. In the Tropp-style story of matrix concentration, Golden–Thompson explains why trace exponentials are natural, while Lieb’s theorem is the result that actually makes the iteration work.

Proposition 6.18 (Lieb’s concavity theorem). *Fix a symmetric matrix $H \in \mathbf{R}^{d \times d}$. Then the map*

$$A \longmapsto \mathrm{Tr}(e^{H+\log(A)})$$

is concave on the cone of positive definite matrices. Equivalently, for every $\alpha \in [0, 1]$ and every positive definite A, B ,

$$\mathrm{Tr}(e^{H+\log(\alpha A+(1-\alpha)B)}) \geq \alpha \mathrm{Tr}(e^{H+\log A}) + (1-\alpha) \mathrm{Tr}(e^{H+\log B}).$$

Remark 6.19. Lieb’s theorem is deep. Unlike the surrounding probabilistic argument, its proof belongs to matrix analysis and is decidedly nontrivial once $d > 1$. For matrix concentration it plays the same structural role that convexity and Jensen’s inequality play in one dimension: it is the mechanism that allows conditioning to enter the proof in a stable way.

The form we actually use is the following corollary.

Corollary 6.20 (A Jensen-type consequence of Lieb). *Let H be a deterministic symmetric matrix, and let Z be a random symmetric matrix such that $\mathbb{E}e^Z$ is finite. Then*

$$\mathbb{E} \mathrm{Tr}(e^{H+Z}) \leq \mathrm{Tr}(e^{H+\log(\mathbb{E}e^Z)}).$$

Proof. Set

$$\psi(A) = \mathrm{Tr}(e^{H+\log A}),$$

defined for positive definite A . By Theorem 6.18, the function ψ is concave. Since e^Z is positive definite almost surely, Jensen’s inequality gives

$$\mathbb{E} \psi(e^Z) \leq \psi(\mathbb{E}e^Z).$$

Finally, because $\log(e^Z) = Z$ for symmetric Z , we obtain

$$\mathbb{E} \mathrm{Tr}(e^{H+Z}) = \mathbb{E} \psi(e^Z) \leq \psi(\mathbb{E}e^Z) = \mathrm{Tr}(e^{H+\log(\mathbb{E}e^Z)}).$$

□

The next proposition is the Tropp-style “matrix Laplace transform” bound. It is the correct noncommutative replacement for the scalar identity

$$\mathbb{E}e^{\theta \sum_i X_i} = \prod_i \mathbb{E}e^{\theta X_i}.$$

Proposition 6.21 (Matrix Laplace transform method). *Let $X_1, \dots, X_n$ be independent symmetric random matrices, and set*

$$S = \sum_{i=1}^n X_i.$$

Then for every $t \in \mathbf{R}$ and every $\theta > 0$,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq e^{-\theta t} \mathrm{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E}e^{\theta X_i} \right) \right).$$

Consequently,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq \inf_{\theta > 0} \left\{ e^{-\theta t} \mathrm{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E}e^{\theta X_i} \right) \right) \right\}.$$

Proof. Fix $\theta > 0$. By Markov's inequality,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) = \mathbb{P}(e^{\theta \lambda_{\max}(S)} \geq e^{\theta t}) \leq e^{-\theta t} \mathbb{E} e^{\theta \lambda_{\max}(S)}.$$

Since the eigenvalues of $e^{\theta S}$ are $e^{\theta \lambda_j(S)}$, we have

$$e^{\theta \lambda_{\max}(S)} = \lambda_{\max}(e^{\theta S}) \leq \text{Tr}(e^{\theta S}),$$

and therefore

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq e^{-\theta t} \mathbb{E} \text{Tr}(e^{\theta S}).$$

It remains to bound $\mathbb{E} \text{Tr}(e^{\theta S})$. We condition on the last summand and apply Theorem 6.20 with

$$H = \theta \sum_{i=1}^{n-1} X_i, \quad Z = \theta X_n.$$

Because X_n is independent of $X_1, \dots, X_{n-1}$, this yields

$$\begin{aligned} \mathbb{E} \text{Tr}(e^{\theta S}) &= \mathbb{E}_{1, \dots, n-1} \mathbb{E}_n \text{Tr} \left(e^{\theta \sum_{i=1}^{n-1} X_i + \theta X_n} \right) \\ &\leq \mathbb{E}_{1, \dots, n-1} \text{Tr} \left(\exp \left(\theta \sum_{i=1}^{n-1} X_i + \log \mathbb{E} e^{\theta X_n} \right) \right). \end{aligned}$$

Now the last term has the same form as before, except that one random summand has been replaced by the deterministic matrix $\log \mathbb{E} e^{\theta X_n}$. Repeating the argument for $X_{n-1}, X_{n-2}, \dots, X_1$, we arrive at

$$\mathbb{E} \text{Tr}(e^{\theta S}) \leq \text{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E} e^{\theta X_i} \right) \right).$$

Substituting this into the previous bound proves the claim. $\square$

6.6 Proof of Theorem 6.14

The only remaining ingredient is a one-step bound on the matrix MGF of a bounded summand. This is the matrix analogue of the scalar estimate used in the proof of Bernstein's inequality.

Lemma 6.22 (A matrix MGF bound for bounded summands). *Let X be a symmetric random matrix such that $\mathbb{E}X = 0$ and $\|X\|_{\text{op}} \leq K$ almost surely. Then for every $\theta \in (0, 3/K)$,*

$$\mathbb{E} e^{\theta X} \preceq \exp(g(\theta) \mathbb{E}[X^2]), \quad g(\theta) = \frac{\theta^2/2}{1 - \theta K/3}.$$

Proof. The scalar Bernstein argument gives the inequality

$$e^{\theta x} \leq 1 + \theta x + \frac{\theta^2/2}{1 - \theta K/3} x^2, \quad x \in [-K, K], \quad \theta \in (0, 3/K).$$

Applying Theorem 6.15(a) through the spectral calculus, we obtain

$$e^{\theta X} \preceq I + \theta X + g(\theta) X^2.$$

Taking expectations and using $\mathbb{E}X = 0$,

$$\mathbb{E} e^{\theta X} \preceq I + g(\theta) \mathbb{E}[X^2].$$

Since $\mathbb{E}[X^2] \succeq 0$, we may use the scalar inequality $1 + u \leq e^u$ through the spectral calculus once more to deduce

$$I + g(\theta) \mathbb{E}[X^2] \preceq \exp(g(\theta) \mathbb{E}[X^2]).$$

This proves the claim. $\square$

Proof of Theorem 6.14. Let

$$S = \sum_{i=1}^n X_i.$$

Fix $\theta \in (0, 3/K)$. By Theorem 6.21,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq e^{-\theta t} \operatorname{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E} e^{\theta X_i} \right) \right).$$

By Theorem 6.22,

$$\mathbb{E} e^{\theta X_i} \preceq \exp(g(\theta) \mathbb{E}[X_i^2]), \quad g(\theta) = \frac{\theta^2/2}{1 - \theta K/3}.$$

Both sides are positive definite, so Theorem 6.15(b) gives

$$\log \mathbb{E} e^{\theta X_i} \preceq g(\theta) \mathbb{E}[X_i^2].$$

Summing over i and applying Theorem 6.15(c) with $f(x) = e^x$, we obtain

$$\operatorname{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E} e^{\theta X_i} \right) \right) \leq \operatorname{Tr} \left(\exp \left(g(\theta) \sum_{i=1}^n \mathbb{E}[X_i^2] \right) \right) = \operatorname{Tr}(e^{g(\theta)V}).$$

If $\mu_1, \dots, \mu_d$ are the eigenvalues of V , then $\mu_j \leq v$ for all j , so

$$\operatorname{Tr}(e^{g(\theta)V}) = \sum_{j=1}^d e^{g(\theta)\mu_j} \leq d e^{g(\theta)v}.$$

Therefore

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq d \exp(-\theta t + g(\theta)v).$$

Choosing

$$\theta = \frac{t}{v + Kt/3}$$

when $v > 0$ (and arguing by continuity if $v = 0$) gives

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq d \exp \left(-\frac{t^2/2}{v + Kt/3} \right).$$

This proves the one-sided estimate.

For the operator norm, observe that

$$\|S\|_{\text{op}} = \max\{\lambda_{\max}(S), \lambda_{\max}(-S)\}.$$

Since the matrices $-X_i$ satisfy the same assumptions and have the same variance proxy V , we may apply the one-sided bound to both S and $-S$, and then use the union bound:

$$\mathbb{P}(\|S\|_{\text{op}} \geq t) \leq 2d \exp \left(-\frac{t^2/2}{v + Kt/3} \right).$$

Finally, set $x = \log(2d/\delta)$. The inequality

$$2d \exp\left(-\frac{t^2/2}{v + Kt/3}\right) \leq \delta$$

is equivalent to

$$t^2 \geq 2x \left(v + \frac{Kt}{3}\right).$$

Solving this quadratic inequality in t gives

$$t \geq \frac{Kx}{3} + \sqrt{\frac{K^2 x^2}{9} + 2vx}.$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, we further obtain the simpler sufficient bound

$$t \geq \sqrt{2vx} + \frac{2Kx}{3}.$$

This proves the high-probability form. □

6.7 Two Useful Extensions

The matrix Bernstein theorem is already quite flexible in its symmetric form. Two standard consequences are worth recording.

Controlling $\mathbb{E}\|S\|_{\text{op}}$. Integrating the tail bound from Theorem 6.14 yields

$$\mathbb{E} \left\| \sum_{i=1}^n X_i \right\|_{\text{op}} \lesssim \sqrt{v \log(2d)} + K \log(2d).$$

Thus the same two scales appear in expectation as in the high-probability bound: a square-root variance term and a linear boundedness term. Only the dimension enters logarithmically.

Rectangular matrices and Hermitian dilation. If $A \in \mathbf{R}^{d_1 \times d_2}$ is rectangular, the standard trick is to embed it into the symmetric block matrix

$$\mathcal{H}(A) = \begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix} \in \mathbf{R}^{(d_1+d_2) \times (d_1+d_2)}.$$

Its eigenvalues are $\pm s_1(A), \dots, \pm s_r(A)$, where $s_j(A)$ are the singular values of A , so

$$\lambda_{\max}(\mathcal{H}(A)) = \|A\|_{\text{op}}.$$

Moreover,

$$\mathcal{H}(A)^2 = \begin{bmatrix} AA^\top & 0 \\ 0 & A^\top A \end{bmatrix}.$$

This reduces the rectangular case to the symmetric one.

Corollary 6.23 (Rectangular matrix Bernstein). *Let $A_1, \dots, A_n$ be independent mean-zero random matrices in $\mathbf{R}^{d_1 \times d_2}$, and assume that*

$$\|A_i\|_{\text{op}} \leq K \quad \text{almost surely for all } i.$$

Define

$$v = \max \left\{ \left\| \sum_{i=1}^n \mathbb{E}[A_i A_i^\top] \right\|_{\text{op}}, \left\| \sum_{i=1}^n \mathbb{E}[A_i^\top A_i] \right\|_{\text{op}} \right\}.$$

Then for every $t \geq 0$,

$$\mathbb{P} \left(\left\| \sum_{i=1}^n A_i \right\|_{\text{op}} \geq t \right) \leq (d_1 + d_2) \exp \left(-\frac{t^2/2}{v + Kt/3} \right).$$

Proof. Apply the one-sided part of Theorem 6.14 to the symmetric matrices $\mathcal{H}(A_i)$. They are independent, mean-zero, symmetric, and satisfy

$$\|\mathcal{H}(A_i)\|_{\text{op}} = \|A_i\|_{\text{op}} \leq K.$$

Also,

$$\sum_{i=1}^n \mathbb{E}[\mathcal{H}(A_i)^2] = \begin{bmatrix} \sum_{i=1}^n \mathbb{E}[A_i A_i^\top] & 0 \\ 0 & \sum_{i=1}^n \mathbb{E}[A_i^\top A_i] \end{bmatrix},$$

whose operator norm is exactly v . Finally,

$$\lambda_{\max} \left(\sum_{i=1}^n \mathcal{H}(A_i) \right) = \left\| \sum_{i=1}^n A_i \right\|_{\text{op}}.$$

The result follows immediately. $\square$

7 Gaussian Processes and Comparison Inequalities

We now turn to a family of results that is especially useful when the object of interest is not a single random variable, but the supremum of a random process. The guiding example is the *canonical Gaussian process*

$$X_t = \langle g, t \rangle, \quad t \in T \subseteq \mathbf{R}^d,$$

where $g \sim \mathcal{N}(0, I_d)$. Even this basic example already leads to nontrivial questions: how large can $\sup_{t \in T} X_t$ be, and how is it related to the geometry of the index set T ?

A recurrent theme is that for Gaussian processes the relevant geometry is encoded by the size of the increments. This leads to powerful comparison principles such as Slepian's lemma and the Sudakov–Fernique theorem, and from there to entropy bounds such as Sudakov minoration.

7.1 Gaussian processes and the canonical metric

Definition 7.1 (Gaussian process). Let T be an index set. A random family $(X_t)_{t \in T}$ is called a *Gaussian process* if for every finite subset

$$T_0 = \{t_1, \dots, t_m\} \subseteq T,$$

the random vector

$$(X_{t_1}, \dots, X_{t_m})$$

is multivariate Gaussian.

Definition 7.2 (Canonical metric). Let $(X_t)_{t \in T}$ be a Gaussian process. Its *canonical metric* is

$$d_X(s, t) = (\mathbb{E}(X_t - X_s)^2)^{1/2}, \quad s, t \in T.$$

Strictly speaking, d_X is in general only a pseudometric: it may happen that $d_X(s, t) = 0$ for distinct points $s \neq t$. In all entropy statements below, this causes no difficulty; equivalently, one may identify points at distance zero and work on the resulting metric quotient.

For many questions involving $\sup_{t \in T} X_t$, the canonical metric is the right quantity to look at. Indeed, adding a common Gaussian variable to all coordinates does not change the increments at all, and it also does not change the supremum except by a random shift. Thus the process is largely controlled by its increment structure rather than by its pointwise variances alone.

7.2 Slepian and Sudakov–Fernique inequalities

In many applications, one wants to compare the random quantities

$$\sup_{t \in T} X_t \quad \text{and} \quad \sup_{t \in T} Y_t$$

for two Gaussian processes $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$.

Remark 7.3 (Making T finite). To avoid measurability issues, one may think of

$$\mathbb{E} \sup_{t \in T} X_t$$

as shorthand for

$$\sup_{T_0 \subseteq T, |T_0| < \infty} \mathbb{E} \max_{t \in T_0} X_t.$$

Thus it is enough to prove the comparison inequalities below for finite index sets T ; the general separable case then follows by approximation. Thus, in the comparison theorems below, we will state the results for finite index sets.

Theorem 7.4 (Slepian inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean-zero Gaussian processes. Assume that for all $s, t \in T$,*

$$\mathbb{E} X_t^2 = \mathbb{E} Y_t^2 \quad \text{and} \quad \mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2.$$

Then for every $\tau \in \mathbf{R}$,

$$\mathbb{P} \left\{ \sup_{t \in T} X_t \geq \tau \right\} \leq \mathbb{P} \left\{ \sup_{t \in T} Y_t \geq \tau \right\}.$$

Consequently,

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t.$$

We prove Slepian inequality by Gaussian interpolation. The first step is Stein's identity in dimension one.

Lemma 7.5 (Stein's identity in dimension one). *Let $X \sim N(0, 1)$. Then for every differentiable function $f : \mathbf{R} \rightarrow \mathbf{R}$,*

$$\mathbb{E}[Xf(X)] = \mathbb{E}f'(X),$$

assuming both expectations exist and are finite.

Proof. Let

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

be the standard Gaussian density. Then

$$\mathbb{E}[Xf(X)] = \int_{\mathbf{R}} xf(x)\varphi(x) dx.$$

Since $\varphi'(x) = -x\varphi(x)$, integration by parts gives

$$\int_{\mathbf{R}} xf(x)\varphi(x) dx = - \int_{\mathbf{R}} f(x)\varphi'(x) dx = \int_{\mathbf{R}} f'(x)\varphi(x) dx = \mathbb{E}f'(X),$$

where the boundary term vanishes because φ decays exponentially. □

Lemma 7.6 (Multivariate Stein identity). *Let $X \sim N(0, \Sigma)$ in $\mathbf{R}^n$. Then for every differentiable function $f : \mathbf{R}^n \rightarrow \mathbf{R}$,*

$$\mathbb{E}[Xf(X)] = \Sigma \mathbb{E}\nabla f(X),$$

assuming all expectations exist and are finite. Equivalently, for every $i = 1, \dots, n$,

$$\mathbb{E}[X_i f(X)] = \sum_{j=1}^n \Sigma_{ij} \mathbb{E}[\partial_j f(X)].$$

Proof. Write $X = AZ$, where $Z \sim N(0, I_n)$ and $AA^\top = \Sigma$. Then

$$X_i = \sum_{k=1}^n A_{ik} Z_k,$$

so

$$\mathbb{E}[X_i f(X)] = \sum_{k=1}^n A_{ik} \mathbb{E}[Z_k f(AZ)].$$

Conditioning on all coordinates of Z except Z_k , and applying Theorem 7.5 to the k -th coordinate, we obtain

$$\mathbb{E}[Z_k f(AZ)] = \mathbb{E}[\partial_k(f(AZ))].$$

By the chain rule,

$$\partial_k(f(AZ)) = \sum_{j=1}^n A_{jk} \partial_j f(X).$$

Therefore

$$\mathbb{E}[X_i f(X)] = \sum_{k=1}^n A_{ik} \sum_{j=1}^n A_{jk} \mathbb{E}[\partial_j f(X)] = \sum_{j=1}^n (AA^\top)_{ij} \mathbb{E}[\partial_j f(X)] = \sum_{j=1}^n \Sigma_{ij} \mathbb{E}[\partial_j f(X)].$$

□

Lemma 7.7 (Gaussian interpolation). *Consider two independent Gaussian random vectors*

$$X \sim N(0, \Sigma^X), \quad Y \sim N(0, \Sigma^Y).$$

Define the interpolation Gaussian vector

$$Z(u) := \sqrt{u} X + \sqrt{1-u} Y, \quad u \in [0, 1].$$

Then for every twice differentiable function $f : \mathbf{R}^n \rightarrow \mathbf{R}$,

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i,j=1}^n (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E} [\partial_{ij} f(Z(u))],$$

assuming all expectations exist and are finite.

Proof. Using the chain rule,

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \sum_{i=1}^n \mathbb{E} \left[\partial_i f(Z(u)) \frac{dZ_i(u)}{du} \right].$$

Since

$$\frac{dZ_i(u)}{du} = \frac{X_i}{2\sqrt{u}} - \frac{Y_i}{2\sqrt{1-u}},$$

we obtain

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i=1}^n \mathbb{E} \left[\partial_i f(Z(u)) \left(\frac{X_i}{\sqrt{u}} - \frac{Y_i}{\sqrt{1-u}} \right) \right].$$

Let us compute the part containing X_i . Condition on Y , and define

$$g_i(x) := \partial_i f(\sqrt{u} x + \sqrt{1-u} Y).$$

Then

$$\frac{1}{\sqrt{u}} \mathbb{E} [X_i \partial_i f(Z(u)) \mid Y] = \frac{1}{\sqrt{u}} \mathbb{E} [X_i g_i(X) \mid Y].$$

Applying Theorem 7.6 to the Gaussian vector X , we get

$$\frac{1}{\sqrt{u}} \mathbb{E} [X_i g_i(X) \mid Y] = \frac{1}{\sqrt{u}} \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E} [\partial_j g_i(X) \mid Y].$$

By the chain rule,

$$\partial_j g_i(x) = \sqrt{u} \partial_{ij} f(\sqrt{u} x + \sqrt{1-u} Y).$$

Therefore

$$\frac{1}{\sqrt{u}} \mathbb{E} [X_i \partial_i f(Z(u)) \mid Y] = \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E} [\partial_{ij} f(Z(u)) \mid Y].$$

Taking expectation in Y , we obtain

$$\frac{1}{\sqrt{u}} \mathbb{E} [X_i \partial_i f(Z(u))] = \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E} [\partial_{ij} f(Z(u))].$$

Similarly,

$$\frac{1}{\sqrt{1-u}} \mathbb{E} [Y_i \partial_i f(Z(u))] = \sum_{j=1}^n \Sigma_{ij}^Y \mathbb{E} [\partial_{ij} f(Z(u))].$$

Substituting these identities into the derivative formula above proves the result. $\square$

Lemma 7.8 (Slepian inequality, functional form). *Consider two mean-zero Gaussian random vectors X and Y in $\mathbf{R}^n$. Assume that for all $i, j = 1, \dots, n$,*

$$\mathbb{E}X_i^2 = \mathbb{E}Y_i^2 \quad \text{and} \quad \mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2.$$

Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be twice differentiable and satisfy

$$\partial_{ij}f(x) \geq 0 \quad \text{for all } x \in \mathbf{R}^n \text{ and all } i \neq j.$$

Then

$$\mathbb{E}f(X) \geq \mathbb{E}f(Y),$$

assuming both expectations exist and are finite.

Proof. Let Σ^X and Σ^Y be the covariance matrices of X and Y . Since

$$\mathbb{E}(X_i - X_j)^2 = \Sigma_{ii}^X + \Sigma_{jj}^X - 2\Sigma_{ij}^X,$$

and similarly for Y , the assumptions imply that

$$\Sigma_{ii}^X = \Sigma_{ii}^Y \quad \text{and} \quad \Sigma_{ij}^X \geq \Sigma_{ij}^Y \quad \text{for all } i \neq j.$$

Since the conclusion depends only on the marginal laws of X and Y , we may take them independent. Applying Theorem 7.7 to

$$Z(u) = \sqrt{u} X + \sqrt{1-u} Y,$$

we get

$$\frac{d}{du} \mathbb{E}f(Z(u)) = \frac{1}{2} \sum_{i,j=1}^n (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E}[\partial_{ij}f(Z(u))].$$

The diagonal terms vanish because $\Sigma_{ii}^X = \Sigma_{ii}^Y$, so

$$\frac{d}{du} \mathbb{E}f(Z(u)) = \frac{1}{2} \sum_{i \neq j} (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E}[\partial_{ij}f(Z(u))].$$

Each term in this sum is nonnegative, hence

$$\frac{d}{du} \mathbb{E}f(Z(u)) \geq 0.$$

Therefore $u \mapsto \mathbb{E}f(Z(u))$ is increasing, and so

$$\mathbb{E}f(X) = \mathbb{E}f(Z(1)) \geq \mathbb{E}f(Z(0)) = \mathbb{E}f(Y).$$

□

Proof of Theorem 7.4. By Theorem 7.3, it is enough to consider finite T . Relabeling if needed, write

$$T = \{1, \dots, n\}, \quad X = (X_1, \dots, X_n), \quad Y = (Y_1, \dots, Y_n).$$

Fix $\tau \in \mathbf{R}$. For $\varepsilon > 0$, choose a twice differentiable, non-increasing function

$$h_\varepsilon : \mathbf{R} \rightarrow [0, 1]$$

such that

$$h_\varepsilon(x) = 1 \quad \text{for } x \leq \tau - \varepsilon, \quad h_\varepsilon(x) = 0 \quad \text{for } x \geq \tau.$$

Then $h_\varepsilon(x) \rightarrow \mathbf{1}_{(-\infty, \tau)}(x)$ pointwise as $\varepsilon \downarrow 0$.

Define

$$f_\varepsilon(x) := \prod_{i=1}^n h_\varepsilon(x_i), \quad x = (x_1, \dots, x_n) \in \mathbf{R}^n.$$

Then $f_\varepsilon(x) \rightarrow \mathbf{1}_{\{\max_{i \leq n} x_i < \tau\}}$ pointwise as $\varepsilon \downarrow 0$. Moreover, for $i \neq j$,

$$\partial_{ij} f_\varepsilon(x) = h'_\varepsilon(x_i) h'_\varepsilon(x_j) \prod_{k \notin \{i, j\}} h_\varepsilon(x_k) \geq 0,$$

because $h'_\varepsilon \leq 0$ and $0 \leq h_\varepsilon \leq 1$. Thus Theorem 7.8 gives

$$\mathbb{E} f_\varepsilon(X) \geq \mathbb{E} f_\varepsilon(Y).$$

Letting $\varepsilon \downarrow 0$ and using dominated convergence, we obtain

$$\mathbb{P} \left\{ \max_{i \leq n} X_i < \tau \right\} \geq \mathbb{P} \left\{ \max_{i \leq n} Y_i < \tau \right\}.$$

Equivalently,

$$\mathbb{P} \left\{ \max_{i \leq n} X_i \geq \tau \right\} \leq \mathbb{P} \left\{ \max_{i \leq n} Y_i \geq \tau \right\}.$$

This proves the tail comparison. The expectation comparison follows immediately. $\square$

Theorem 7.9 (Sudakov–Fernique). *Let $X = (X_t)_{t \in T}$ and $Y = (Y_t)_{t \in T}$ be centered Gaussian processes. Assume that*

$$\mathbb{E}[(X_t - X_s)^2] \leq \mathbb{E}[(Y_t - Y_s)^2] \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \max_{t \in T} X_t \leq \mathbb{E} \max_{t \in T} Y_t.$$

Remark 7.10. We will use Theorem 7.9 repeatedly, but we will not prove it here. Compared with Theorem 7.4, the point is that one no longer needs the equality of pointwise variances. Only the increment structure matters.

7.3 Packing, covering, and Sudakov minoration

We now connect Gaussian suprema to the metric entropy of the index set. Recall the definition. Let T be a subset of a metric space (E, d) , and let $\varepsilon > 0$. The covering number $N(T, d, \varepsilon)$ is the smallest number of closed d -balls of radius ε in E whose union covers T .

An ε -packing is a subset $P \subseteq T$ such that

$$d(s, t) > \varepsilon \quad \text{for all distinct } s, t \in P.$$

Also recall that we have

$$N(T, d, \varepsilon) \leq M(T, d, \varepsilon).$$

Theorem 7.11 (Sudakov minoration). *Let $(X_t)_{t \in T}$ be a centered separable Gaussian process, and let*

$$d_X(s, t) = (\mathbb{E}(X_t - X_s)^2)^{1/2}$$

be its canonical metric. Then for every $\varepsilon > 0$,

$$\varepsilon \sqrt{\log N(T, d_X, \varepsilon)} \leq c \mathbb{E} \sup_{t \in T} X_t,$$

where $c > 0$ is an absolute constant.

Proof. Fix $\varepsilon > 0$. If $N(T, d_X, \varepsilon) = \infty$, then $M(T, d_X, \varepsilon) = \infty$ as well. Hence for every integer $m \geq 1$, there exists an ε -packing

$$P_{\varepsilon, m} = \{t_1, \dots, t_m\} \subseteq T.$$

Applying the argument below to $P_{\varepsilon, m}$ shows that

$$\mathbb{E} \sup_{t \in T} X_t \geq c \varepsilon \sqrt{\log m}$$

for all m , and therefore $\mathbb{E} \sup_{t \in T} X_t = +\infty$. In this case the desired inequality is trivial. We may therefore assume that $N(T, d_X, \varepsilon) < \infty$.

Since $N(T, d_X, \varepsilon) \leq M(T, d_X, \varepsilon)$, we can choose a finite ε -packing

$$P_\varepsilon = \{t_1, \dots, t_m\} \subseteq T$$

with

$$m \geq N(T, d_X, \varepsilon).$$

Of course,

$$\mathbb{E} \sup_{t \in T} X_t \geq \mathbb{E} \max_{1 \leq j \leq m} X_{t_j}.$$

Now let $g_1, \dots, g_m \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, independent of everything else, and define a new Gaussian process

$$Y_{t_j} = \frac{\varepsilon}{\sqrt{2}} g_j, \quad j = 1, \dots, m.$$

For distinct i, j ,

$$\mathbb{E}(Y_{t_i} - Y_{t_j})^2 = \frac{\varepsilon^2}{2} \mathbb{E}(g_i - g_j)^2 = \varepsilon^2.$$

On the other hand, because P_ε is an ε -packing,

$$\mathbb{E}(X_{t_i} - X_{t_j})^2 = d_X(t_i, t_j)^2 > \varepsilon^2.$$

Therefore

$$\mathbb{E}(Y_{t_i} - Y_{t_j})^2 \leq \mathbb{E}(X_{t_i} - X_{t_j})^2 \quad \text{for all } i, j,$$

and Theorem 7.9 yields

$$\mathbb{E} \max_{1 \leq j \leq m} Y_{t_j} \leq \mathbb{E} \max_{1 \leq j \leq m} X_{t_j} \leq \mathbb{E} \sup_{t \in T} X_t.$$

Finally,

$$\mathbb{E} \max_{1 \leq j \leq m} Y_{t_j} = \frac{\varepsilon}{\sqrt{2}} \mathbb{E} \max_{1 \leq j \leq m} g_j.$$

A standard estimate for the maximum of m independent standard Gaussians gives

$$\mathbb{E} \max_{1 \leq j \leq m} g_j \geq c_0 \sqrt{\log m}$$

for an absolute constant $c_0 > 0$. Hence

$$\mathbb{E} \sup_{t \in T} X_t \geq \frac{c_0}{\sqrt{2}} \varepsilon \sqrt{\log m} \geq \frac{c_0}{\sqrt{2}} \varepsilon \sqrt{\log N(T, d_X, \varepsilon)}.$$

This proves the theorem. □

7.4 Canonical Gaussian process: covering-number examples

A particularly important example is the canonical Gaussian process on a set $T \subseteq \mathbf{R}^d$:

$$X_t = \langle g, t \rangle, \quad t \in T, \quad g \sim \mathcal{N}(0, I_d).$$

Its canonical metric is exactly the Euclidean metric. Indeed,

$$d_X(s, t)^2 = \mathbb{E}(\langle g, t - s \rangle)^2 = \|t - s\|_2^2.$$

Thus Theorem 7.11 translates bounds on Gaussian suprema into covering-number bounds in Euclidean space.

Example 7.12 (An ellipsoid). Let

$$T = \Sigma^{1/2} B_2^d = \{\Sigma^{1/2} u : \|u\|_2 \leq 1\},$$

where $\Sigma \succeq 0$. Then

$$\sup_{t \in T} \langle g, t \rangle = \sup_{\|u\|_2 \leq 1} \langle \Sigma^{1/2} g, u \rangle = \|\Sigma^{1/2} g\|_2.$$

Therefore

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle = \mathbb{E} \|\Sigma^{1/2} g\|_2 \leq \sqrt{\mathbb{E} \|\Sigma^{1/2} g\|_2^2} = \sqrt{\text{Tr}(\Sigma)},$$

where we used Jensen's inequality. Applying Theorem 7.11, we obtain

$$\log N(\Sigma^{1/2} B_2^d, \|\cdot\|_2, \varepsilon) \leq \frac{c \text{Tr}(\Sigma)}{\varepsilon^2}.$$

Example 7.13 (The ℓ_1 ball). Let

$$T = B_1^d = \{x \in \mathbf{R}^d : \|x\|_1 \leq 1\}.$$

Then

$$\sup_{t \in B_1^d} \langle g, t \rangle = \|g\|_\infty,$$

so

$$\mathbb{E} \sup_{t \in B_1^d} \langle g, t \rangle = \mathbb{E} \|g\|_\infty.$$

By Theorem 3.25 applied to the $2d$ centered Gaussian variables

$$g_1, \dots, g_d, -g_1, \dots, -g_d,$$

we get

$$\mathbb{E}\|g\|_\infty = \mathbb{E} \max_{1 \leq i \leq d} |g_i| \leq \sqrt{2 \log(2d)}.$$

Therefore Theorem 7.11 yields

$$\log N(B_1^d, \|\cdot\|_2, \varepsilon) \leq \frac{c \log(2d)}{\varepsilon^2}.$$

This estimate is very effective when ε is not too small. For smaller ε , a volumetric argument gives a complementary bound. Indeed, the inclusions

$$B_1^d \subseteq B_2^d \subseteq \sqrt{d} B_1^d$$

imply

$$B_1^d + \frac{\varepsilon}{2} B_2^d \subseteq \left(1 + \frac{\varepsilon \sqrt{d}}{2}\right) B_1^d.$$

Hence

$$\begin{aligned} N(B_1^d, \|\cdot\|_2, \varepsilon) &\leq \frac{\text{Vol}(B_1^d + \frac{\varepsilon}{2} B_2^d)}{\text{Vol}((\varepsilon/2) B_2^d)} \\ &\leq \frac{\text{Vol}\left(\left(1 + \frac{\varepsilon \sqrt{d}}{2}\right) B_1^d\right)}{\text{Vol}((\varepsilon/2) B_2^d)} \\ &= \left(1 + \frac{\varepsilon \sqrt{d}}{2}\right)^d \left(\frac{2}{\varepsilon}\right)^d \frac{\text{Vol}(B_1^d)}{\text{Vol}(B_2^d)}. \end{aligned}$$

Using the standard bound

$$\frac{\text{Vol}(B_1^d)}{\text{Vol}(B_2^d)} \leq \left(\frac{c}{\sqrt{d}}\right)^d,$$

we arrive at

$$N(B_1^d, \|\cdot\|_2, \varepsilon) \leq \left(c \left(1 + \frac{2}{\varepsilon \sqrt{d}}\right)\right)^d.$$

Equivalently,

$$\log N(B_1^d, \|\cdot\|_2, \varepsilon) \leq d \log \left(c \left(1 + \frac{2}{\varepsilon \sqrt{d}}\right)\right).$$

Combining the two estimates, we get

$$\log N(B_1^d, \|\cdot\|_2, \varepsilon) \leq \min \left\{ d \log \left(c \left(1 + \frac{2}{\varepsilon \sqrt{d}}\right)\right), \frac{c \log(2d)}{\varepsilon^2} \right\}.$$

In particular, when $\varepsilon \asymp d^{-1/2}$, the volumetric bound is of order d , whereas the Sudakov bound is of order $d \log d$. Thus the volumetric argument is better in the very small-scale regime, while Sudakov minoration becomes more informative at larger scales.

7.5 Dual Sudakov minoration

Sudakov minoration bounds the entropy of a set in terms of the Gaussian supremum over that same set. There is a dual statement in which one covers the Euclidean ball by translates of a polar body.

Definition 7.14 (Polar set). Let $T \subseteq \mathbf{R}^d$ be convex and symmetric, meaning $T = -T$. Its *polar* is

$$T^\circ = \left\{ y \in \mathbf{R}^d : \sup_{x \in T} \langle x, y \rangle \leq 1 \right\}.$$

A few standard examples are useful to keep in mind:

- $(B_2^d)^\circ = B_2^d$;
- if $1 \leq p \leq \infty$ and q is the conjugate exponent, then

$$(B_p^d)^\circ = B_q^d;$$

- if T is an ellipsoid with semi-axes $a_1, \dots, a_d > 0$, then T° is an ellipsoid with semi-axes $a_1^{-1}, \dots, a_d^{-1}$.

We will also use the set-based covering-number notation

$$N(A, K),$$

which denotes the minimum number of translates of K needed to cover A . Thus

$$N(T, \|\cdot\|_2, \varepsilon) = N(T, \varepsilon B_2^d).$$

Theorem 7.15 (Dual Sudakov minoration). *Let $T \subseteq \mathbf{R}^d$ be a symmetric convex body. Then for every $\varepsilon > 0$,*

$$\varepsilon \sqrt{\log N(B_2^d, \varepsilon T^\circ)} \leq c \mathbb{E} \sup_{t \in T} \langle g, t \rangle,$$

where $g \sim \mathcal{N}(0, I_d)$ and $c > 0$ is an absolute constant.

We will not prove Theorem 7.15 here. The important point is that it gives entropy bounds for coverings of the Euclidean ball by a body dual to the one defining the Gaussian supremum.

Example 7.16 (A Euclidean-ball covering estimate). Let

$$T = \Sigma^{1/2} B_2^d, \quad \Sigma \succ 0.$$

By Theorem 7.12,

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle \leq \sqrt{\text{Tr}(\Sigma)}.$$

On the other hand,

$$T^\circ = \left\{ y \in \mathbf{R}^d : \sup_{\|u\|_2 \leq 1} \langle \Sigma^{1/2} u, y \rangle \leq 1 \right\} = \left\{ y \in \mathbf{R}^d : \|\Sigma^{1/2} y\|_2 \leq 1 \right\}.$$

Thus covering B_2^d by translates of εT° is the same as covering B_2^d by balls of radius ε for the metric

$$d_\Sigma(s, t) = \|\Sigma^{1/2}(s - t)\|_2.$$

Indeed,

$$s \in t + \varepsilon T^\circ \iff \|\Sigma^{1/2}(s - t)\|_2 \leq \varepsilon.$$

Therefore Theorem 7.15 yields

$$\log N(B_2^d, d_\Sigma, \varepsilon) \leq \frac{c \text{Tr}(\Sigma)}{\varepsilon^2}.$$

7.6 Gaussian width

The quantity that appears repeatedly in the previous discussion deserves its own name.

Definition 7.17 (Gaussian width). Let $T \subseteq \mathbf{R}^d$ be nonempty, and let $g \sim \mathcal{N}(0, I_d)$. The *Gaussian width* of T is

$$W(T) = \mathbb{E} \sup_{t \in T} \langle g, t \rangle \in [0, \infty].$$

Geometrically, $W(T)$ is the expected support function of T in a random Gaussian direction. It is a very natural size parameter in high-dimensional probability: it appears in comparison theorems, in entropy estimates, in random matrix bounds, and later in learning-theoretic applications.

Proposition 7.18 (Basic properties of Gaussian width). *Let $T, K \subseteq \mathbf{R}^d$ be nonempty sets, let Q be an orthogonal matrix, and let $y \in \mathbf{R}^d$. Then:*

1. $W(T) < \infty$ if and only if T is bounded;
2. $W(QT + y) = W(T)$;
3. $W(T + K) = W(T) + W(K)$, where

$$T + K = \{t + k : t \in T, k \in K\};$$

4. for every $\alpha \in \mathbf{R}$,

$$W(\alpha T) = |\alpha| W(T);$$

- 5.

$$W(T) = \frac{1}{2} W(T - T) = \frac{1}{2} \mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle;$$

6. if T is finite, then

$$W(T) \leq \sqrt{2 \log |T|} \operatorname{diam}(T),$$

where

$$\operatorname{diam}(T) = \sup_{x, y \in T} \|x - y\|_2.$$

Proof. (1) If T is bounded, then

$$\sup_{t \in T} \langle g, t \rangle \leq \|g\|_2 \sup_{t \in T} \|t\|_2,$$

so $W(T) < \infty$.

Conversely, suppose T is unbounded. Choose $t_k \in T$ with $\|t_k\|_2 \rightarrow \infty$, and write $u_k = t_k / \|t_k\|_2$. By compactness of the unit sphere, some subsequence u_{k_ℓ} converges to a unit vector u . On the event $\langle g, u \rangle > 0$, which has probability $1/2$, we have $\langle g, u_{k_\ell} \rangle > \langle g, u \rangle / 2$ for all large ℓ , and hence

$$\langle g, t_{k_\ell} \rangle = \|t_{k_\ell}\|_2 \langle g, u_{k_\ell} \rangle \rightarrow \infty.$$

Therefore $\sup_{t \in T} \langle g, t \rangle = +\infty$ with positive probability, so $W(T) = +\infty$.

(2) We have

$$W(QT + y) = \mathbb{E} \sup_{t \in T} \langle g, Qt + y \rangle = \mathbb{E} \sup_{t \in T} (\langle Q^\top g, t \rangle + \langle g, y \rangle).$$

Since $Q^\top g \sim \mathcal{N}(0, I_d)$ and $\mathbb{E} \langle g, y \rangle = 0$, this gives $W(QT + y) = W(T)$.

(3) For every fixed g ,

$$\sup_{z \in T+K} \langle g, z \rangle = \sup_{t \in T, k \in K} \langle g, t+k \rangle = \sup_{t \in T} \langle g, t \rangle + \sup_{k \in K} \langle g, k \rangle.$$

Taking expectations proves the claim.

(4) If $\alpha \geq 0$, then

$$W(\alpha T) = \alpha W(T).$$

If $\alpha < 0$, then

$$W(\alpha T) = |\alpha| \mathbb{E} \sup_{t \in T} \langle -g, t \rangle = |\alpha| W(T),$$

since $-g$ has the same law as g .

(5) For every fixed g ,

$$\sup_{x, y \in T} \langle g, x-y \rangle = \sup_{x \in T} \langle g, x \rangle + \sup_{y \in T} \langle -g, y \rangle.$$

Taking expectations and using the symmetry of g gives

$$W(T - T) = 2W(T).$$

(6) Fix $t_0 \in T$. Then

$$W(T) = \mathbb{E} \max_{t \in T} \langle g, t \rangle = \mathbb{E} \max_{t \in T} \langle g, t - t_0 \rangle,$$

because $\mathbb{E} \langle g, t_0 \rangle = 0$. For each $t \in T$, the random variable $\langle g, t - t_0 \rangle$ is centered Gaussian with variance

$$\|t - t_0\|_2^2 \leq \text{diam}(T)^2.$$

Applying Theorem 3.25 to the family $\{\langle g, t - t_0 \rangle : t \in T\}$, we obtain

$$W(T) \leq \sqrt{2 \log |T|} \text{diam}(T).$$

□

Example 7.19 (Basic Gaussian width computations). Let $g \sim \mathcal{N}(0, I_d)$. Then:

1. For the Euclidean ball,

$$W(B_2^d) = \mathbb{E} \sup_{\|t\|_2 \leq 1} \langle g, t \rangle = \mathbb{E} \|g\|_2 \leq \sqrt{\mathbb{E} \|g\|_2^2} = \sqrt{d}.$$

2. For an ellipsoid,

$$W(\Sigma^{1/2} B_2^d) = \mathbb{E} \sup_{\|u\|_2 \leq 1} \langle \Sigma^{1/2} g, u \rangle = \mathbb{E} \|\Sigma^{1/2} g\|_2 \leq \sqrt{\text{Tr}(\Sigma)}.$$

3. For the ℓ_1 ball,

$$W(B_1^d) = \mathbb{E} \|g\|_\infty \leq \sqrt{2 \log(2d)}.$$

4. For the ℓ_∞ ball,

$$W(B_\infty^d) = \mathbb{E} \sup_{\|t\|_\infty \leq 1} \langle g, t \rangle = \mathbb{E} \|g\|_1 = \sum_{i=1}^d \mathbb{E} |g_i| = d \sqrt{\frac{2}{\pi}}.$$

7.7 A useful consequence: Gaussian contraction

We end with a very useful corollary of Theorem 7.9. It says that applying coordinatewise 1-Lipschitz maps can only decrease Gaussian width.

Corollary 7.20 (Gaussian contraction inequality). *Let $T \subseteq \mathbf{R}^d$, let $g = (g_1, \dots, g_d) \sim \mathcal{N}(0, I_d)$, and let*

$$\phi_1, \dots, \phi_d : \mathbf{R} \rightarrow \mathbf{R}$$

be 1-Lipschitz functions. Then

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^d g_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^d g_i t_i = W(T).$$

Proof. Define two centered Gaussian processes on T by

$$X_t = \sum_{i=1}^d g_i \phi_i(t_i), \quad Y_t = \sum_{i=1}^d g_i t_i.$$

For any $s, t \in T$,

$$\mathbb{E}(X_t - X_s)^2 = \sum_{i=1}^d (\phi_i(t_i) - \phi_i(s_i))^2 \leq \sum_{i=1}^d (t_i - s_i)^2 = \mathbb{E}(Y_t - Y_s)^2,$$

since each ϕ_i is 1-Lipschitz. By the Sudakov–Fernique theorem,

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t.$$

The right-hand side equals $\mathbb{E} \sup_{t \in T} \langle g, t \rangle = W(T)$ by definition, which proves the claim. $\square$

8 Generic chaining

We now return to Gaussian processes from a more structural point of view. In the previous section we saw that the size of

$$\mathbb{E} \sup_{t \in T} X_t$$

is governed by the geometry of the index set through the increment metric. Sudakov minoration gave a lower bound in terms of covering numbers. The next question is the upper bound.

A natural first answer is Dudley’s entropy integral. It is already a multiscale argument, and in that sense it is already a chaining bound. But from the conceptual point of view, the more fundamental object is Talagrand’s γ_2 -functional. So we will organize the material in the following order: first we prove the generic chaining upper bound in terms of γ_2 , and only then we recover Dudley’s entropy integral as a corollary.

Historically, Dudley’s entropy bound was one of the first systematic upper bounds of this type. Fernique then introduced the majorizing-measure point of view, and Talagrand proved that the resulting geometric quantity gives the correct order of $\mathbb{E} \sup_{t \in T} X_t$ for Gaussian processes. This majorizing measure theorem is one of the landmarks of the modern theory of Gaussian processes.

Throughout this section, the upper bounds will only use the increment condition

$$\|X_t - X_s\|_{\psi_2} \leq L d(s, t), \quad s, t \in T,$$

for some metric d on T and some constant $L > 0$. So exact Gaussianity is not needed for the upper half of the theory. The genuinely Gaussian step is the matching lower bound.

A canonical example to keep in mind is

$$X_t = \langle g, t \rangle, \quad t \in T \subseteq \mathbf{R}^d,$$

where $g \sim \mathcal{N}(0, I_d)$. Then

$$X_t - X_s = \langle g, t - s \rangle \sim \mathcal{N}(0, \|t - s\|_2^2),$$

and therefore

$$\|X_t - X_s\|_{\psi_2} \lesssim \|t - s\|_2.$$

So the canonical Gaussian process has sub-Gaussian increments with respect to the Euclidean metric.

8.1 Admissible sequences and the γ_2 functional

The generic chaining functional is built from multiscale approximations whose sizes are prescribed in advance.

Definition 8.1 (Admissible sequence). Let T be a nonempty set. A sequence of subsets

$$T_0, T_1, T_2, \dots \subseteq T$$

is called *admissible* if

$$|T_0| = 1 \quad \text{and} \quad |T_k| \leq 2^{2^k} \quad \text{for all } k \geq 1.$$

Definition 8.2 (Talagrand's γ_2 functional). Let (T, d) be a metric space. We define

$$\gamma_2(T, d) = \inf_{(T_k) \text{ admissible}} \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k),$$

where

$$d(t, T_k) = \inf_{s \in T_k} d(t, s).$$

Two simple facts follow immediately from the definition. First, for every $c \geq 0$,

$$\gamma_2(T, cd) = c \gamma_2(T, d).$$

Second, if $d_1 \leq d_2$ pointwise on $T \times T$, then

$$\gamma_2(T, d_1) \leq \gamma_2(T, d_2).$$

The particular cardinality bound 2^{2^k} is not accidental. At scale k , the proof will compare points in T_k with points in T_{k+1} . The number of such pairs is at most

$$|T_k| |T_{k+1}| \leq 2^{2^k} \cdot 2^{2^{k+1}} = 2^{3 \cdot 2^k},$$

so its logarithm is of order 2^k . After taking a square root, the natural probabilistic price is exactly $2^{k/2}$. This is why the geometry of γ_2 is perfectly matched to the sub-Gaussian union-bound computation.

8.2 The generic chaining upper bound

We now prove the main upper bound. As promised, the proof uses only sub-Gaussian increments.

Theorem 8.3 (Generic chaining upper bound). *Let $(X_t)_{t \in T}$ be a centered process such that*

$$\|X_t - X_s\|_{\psi_2} \leq L d(s, t) \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL \gamma_2(T, d),$$

where $C > 0$ is an absolute constant.

Proof. We prove the result first when T is finite.

Choose an admissible sequence $(T_k)_{k \geq 0}$. Since T is finite, by enlarging the sequence if necessary we may assume that

$$T_k = T \quad \text{for all sufficiently large } k.$$

Let $T_0 = \{t_0\}$. For each $t \in T$ and each $k \geq 0$, choose a point $\pi_k(t) \in T_k$ such that

$$d(t, \pi_k(t)) = d(t, T_k).$$

Because $T_k = T$ for all large k , we have $\pi_k(t) = t$ eventually. Therefore, for every $t \in T$,

$$X_t - X_{t_0} = \sum_{k=0}^{\infty} (X_{\pi_{k+1}(t)} - X_{\pi_k(t)}),$$

with only finitely many nonzero terms.

Now fix $u \geq 1$. Consider the event

$$\Omega_u = \bigcap_{k=0}^{\infty} \bigcap_{a \in T_k, b \in T_{k+1}} \left\{ |X_b - X_a| \leq C_1 L u 2^{k/2} d(a, b) \right\},$$

where $C_1 > 0$ is a constant to be chosen.

For a fixed k , $a \in T_k$, and $b \in T_{k+1}$, the increment $X_b - X_a$ is centered and sub-Gaussian, with

$$\|X_b - X_a\|_{\psi_2} \leq L d(a, b).$$

Hence, by Theorem 2.13(1),

$$\mathbb{P}\left(|X_b - X_a| > C_1 L u 2^{k/2} d(a, b)\right) \leq 2 \exp(-c C_1^2 u^2 2^k)$$

for an absolute constant $c > 0$.

On the other hand,

$$|T_k| |T_{k+1}| \leq 2^{2^k} \cdot 2^{2^{k+1}} \leq 2^{2^{k+2}}.$$

Therefore, by the union bound,

$$\mathbb{P}(\Omega_u^c) \leq 2 \sum_{k=0}^{\infty} 2^{2^{k+2}} \exp(-c C_1^2 u^2 2^k).$$

Choose C_1 so large that

$$cC_1^2 \geq 4 \log 2 + 1.$$

Then for every $u \geq 1$,

$$2^{2^{k+2}} \exp(-cC_1^2 u^2 2^k) \leq \exp(-u^2 2^k),$$

and consequently

$$\mathbb{P}(\Omega_u^c) \leq 2 \sum_{k=0}^{\infty} e^{-u^2 2^k} \leq 4e^{-u^2}.$$

Now work on the event Ω_u . For every $t \in T$,

$$\begin{aligned} X_t - X_{t_0} &= \sum_{k=0}^{\infty} (X_{\pi_{k+1}(t)} - X_{\pi_k(t)}) \\ &\leq \sum_{k=0}^{\infty} |X_{\pi_{k+1}(t)} - X_{\pi_k(t)}| \\ &\leq C_1 L u \sum_{k=0}^{\infty} 2^{k/2} d(\pi_{k+1}(t), \pi_k(t)). \end{aligned}$$

By the triangle inequality,

$$d(\pi_{k+1}(t), \pi_k(t)) \leq d(t, \pi_{k+1}(t)) + d(t, \pi_k(t)) = d(t, T_{k+1}) + d(t, T_k).$$

Hence

$$\begin{aligned} \sum_{k=0}^{\infty} 2^{k/2} d(\pi_{k+1}(t), \pi_k(t)) &\leq \sum_{k=0}^{\infty} 2^{k/2} (d(t, T_k) + d(t, T_{k+1})) \\ &\leq 2 \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k). \end{aligned}$$

Therefore, on Ω_u ,

$$\sup_{t \in T} (X_t - X_{t_0}) \leq 2C_1 L u \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k).$$

Set

$$A = \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k).$$

We have shown that

$$\mathbb{P}\left(\sup_{t \in T} (X_t - X_{t_0}) > 2C_1 L u A\right) \leq 4e^{-u^2}, \quad u \geq 1.$$

Since $t = t_0$ gives $X_t - X_{t_0} = 0$, the random variable

$$Z = \sup_{t \in T} (X_t - X_{t_0})$$

is nonnegative. Using the tail-integral formula,

$$\begin{aligned} \mathbb{E}Z &= \int_0^{\infty} \mathbb{P}(Z > s) ds \\ &\leq 2C_1 L A + \int_{2C_1 L A}^{\infty} \mathbb{P}(Z > s) ds. \end{aligned}$$

With the substitution $s = 2C_1 LA u$, the tail estimate gives

$$\mathbb{E}Z \leq 2C_1 LA + 2C_1 LA \int_1^\infty 4e^{-u^2} du \leq C_2 LA$$

for an absolute constant $C_2 > 0$.

Finally,

$$\mathbb{E} \sup_{t \in T} X_t = \mathbb{E} X_{t_0} + \mathbb{E} \sup_{t \in T} (X_t - X_{t_0}) = \mathbb{E} Z,$$

because $\mathbb{E} X_{t_0} = 0$. Thus

$$\mathbb{E} \sup_{t \in T} X_t \leq C_2 LA.$$

Since this holds for every admissible sequence, taking the infimum over all admissible sequences yields

$$\mathbb{E} \sup_{t \in T} X_t \leq C_2 L \gamma_2(T, d).$$

For separable processes, one passes to the limit over increasing finite subsets in the standard way. $\square$

Remark 8.4. At this stage, the main upper bound is already proved. Nothing beyond sub-Gaussian increment control was used. This is one of the reasons generic chaining is so robust: the upper half of the theory is not specifically Gaussian.

8.3 Dudley's entropy integral as a corollary

We now compare γ_2 with covering numbers. This is the step that turns the abstract chaining functional into the familiar entropy integral.

Proposition 8.5 (Comparing γ_2 with entropy). *Let (T, d) be a bounded metric space, and write*

$$D = \sup_{s, t \in T} d(s, t).$$

Then

$$\gamma_2(T, d) \leq C \int_0^D \sqrt{\log N(T, d, \varepsilon)} d\varepsilon,$$

where $C > 0$ is an absolute constant.

Proof. If $N(T, d, \varepsilon) = \infty$ for some $\varepsilon > 0$, then the right-hand side is $+\infty$, and there is nothing to prove. So we may assume that the relevant covering numbers are finite.

For each $k \geq 0$, define

$$\varepsilon_k = \inf \{ \varepsilon > 0 : N(T, d, \varepsilon) \leq 2^{2^k} \}.$$

Since $N(T, d, D) = 1$, the quantity ε_k is well-defined, and $\varepsilon_k \downarrow 0$.

Choose $T_k \subseteq T$ to be an ε_k -net with

$$|T_k| \leq 2^{2^k}.$$

Then (T_k) is admissible, and therefore

$$\gamma_2(T, d) \leq \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k) \leq \sum_{k=0}^{\infty} 2^{k/2} \varepsilon_k.$$

It remains to compare the last sum with the integral. By the definition of ε_k , whenever $\varepsilon < \varepsilon_k$,

$$N(T, d, \varepsilon) > 2^{2^k}.$$

Hence for every $\varepsilon \in [\varepsilon_{k+1}, \varepsilon_k)$,

$$\sqrt{\log N(T, d, \varepsilon)} \geq \sqrt{\log(2^{2^k})} = \sqrt{\log 2} 2^{k/2}.$$

Therefore

$$\int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log N(T, d, \varepsilon)} d\varepsilon \geq \sqrt{\log 2} 2^{k/2} (\varepsilon_k - \varepsilon_{k+1}).$$

Summing over $k \geq 0$, we obtain

$$\int_0^D \sqrt{\log N(T, d, \varepsilon)} d\varepsilon \geq c \sum_{k=0}^{\infty} 2^{k/2} (\varepsilon_k - \varepsilon_{k+1})$$

for an absolute constant $c > 0$.

Now observe that

$$\sum_{k=0}^{\infty} 2^{k/2} (\varepsilon_k - \varepsilon_{k+1}) = \varepsilon_0 + \sum_{k=1}^{\infty} (2^{k/2} - 2^{(k-1)/2}) \varepsilon_k.$$

Since

$$2^{k/2} - 2^{(k-1)/2} = (1 - 2^{-1/2}) 2^{k/2},$$

the right-hand side is bounded below by a constant multiple of

$$\sum_{k=0}^{\infty} 2^{k/2} \varepsilon_k.$$

Combining the previous estimates gives

$$\sum_{k=0}^{\infty} 2^{k/2} \varepsilon_k \leq C \int_0^D \sqrt{\log N(T, d, \varepsilon)} d\varepsilon,$$

and hence

$$\gamma_2(T, d) \leq C \int_0^D \sqrt{\log N(T, d, \varepsilon)} d\varepsilon.$$

□

Corollary 8.6 (Dudley's entropy integral). *Let $(X_t)_{t \in T}$ be a centered process such that*

$$\|X_t - X_s\|_{\psi_2} \leq L d(s, t) \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL \int_0^D \sqrt{\log N(T, d, \varepsilon)} d\varepsilon, \quad D = \sup_{s, t \in T} d(s, t),$$

where $C > 0$ is an absolute constant.

Proof. Combine Theorem 8.3 with Theorem 8.5. □

Remark 8.7. In this order of presentation, Dudley's entropy integral appears as a clean corollary of generic chaining rather than as the starting point. This is conceptually useful: the geometric object of real interest is γ_2 , while the entropy integral is a convenient upper estimate for it.

8.4 Talagrand's majorizing measure theorem

The upper bound from Theorem 8.3 is already quite general. What makes the Gaussian case special is that there is also a matching lower bound.

Theorem 8.8 (Talagrand's majorizing measure theorem). *Let $(X_t)_{t \in T}$ be a centered separable Gaussian process, and let*

$$d_X(s, t) = (\mathbb{E}(X_t - X_s)^2)^{1/2}$$

be its canonical metric. Then there exist absolute constants $c, C > 0$ such that

$$c \gamma_2(T, d_X) \leq \mathbb{E} \sup_{t \in T} X_t \leq C \gamma_2(T, d_X).$$

Remark 8.9. The upper bound is exactly the generic chaining estimate we have just proved. The lower bound is the deep part. Fernique introduced the majorizing-measure viewpoint, and Talagrand proved that it gives the correct answer up to absolute constants. This theorem is one of the central structural results in the theory of Gaussian processes.

It is also useful to keep in mind how it fits with the entropy bounds from the previous section. The upper half contains Dudley's bound after the comparison in Theorem 8.5. The lower half contains Sudakov minoration as a rough entropy shadow. So γ_2 sits between the two classical entropy bounds and refines both of them.

8.5 A comparison principle

The majorizing measure theorem immediately yields a very useful comparison theorem. It says that once a process has sub-Gaussian increments dominated by those of a Gaussian process, its supremum is controlled by the corresponding Gaussian supremum.

Corollary 8.10 (Talagrand's comparison theorem). *Let T be an index set. Let $(X_t)_{t \in T}$ be a centered process, and let $(Y_t)_{t \in T}$ be a centered Gaussian process. Assume that*

$$\|X_t - X_s\|_{\psi_2} \leq L (\mathbb{E}(Y_t - Y_s)^2)^{1/2} \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL \mathbb{E} \sup_{t \in T} Y_t,$$

where $C > 0$ is an absolute constant.

Proof. Define

$$d_Y(s, t) = (\mathbb{E}(Y_t - Y_s)^2)^{1/2}.$$

By Theorem 8.3,

$$\mathbb{E} \sup_{t \in T} X_t \leq C_1 L \gamma_2(T, d_Y).$$

Since Y is Gaussian with canonical metric d_Y , Theorem 8.8 gives

$$\gamma_2(T, d_Y) \leq C_2 \mathbb{E} \sup_{t \in T} Y_t.$$

Combining the two inequalities proves the claim. □

A particularly important special case is the canonical Gaussian process.

Corollary 8.11 (Sub-Gaussian increments and Gaussian width). *Let $T \subseteq \mathbf{R}^d$, and let $(X_t)_{t \in T}$ be a centered process such that*

$$\|X_t - X_s\|_{\psi_2} \leq L \|t - s\|_2 \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL W(T),$$

where $W(T)$ is the Gaussian width from Theorem 7.17.

Proof. Apply Theorem 8.10 with

$$Y_t = \langle g, t \rangle, \quad g \sim \mathcal{N}(0, I_d).$$

Then

$$(\mathbb{E}(Y_t - Y_s)^2)^{1/2} = \|t - s\|_2,$$

and

$$\mathbb{E} \sup_{t \in T} Y_t = W(T).$$

□

8.6 Examples

We conclude with two examples. The first is a general finite-set estimate. The second shows, in a concrete way, why γ_2 can be much smaller than Dudley's entropy integral.

Proposition 8.12 (Finite sets). *Let T be a finite metric space. Then*

$$\gamma_2(T, d) \leq C \sqrt{\log |T|} \sup_{s, t \in T} d(s, t),$$

where $C > 0$ is an absolute constant.

Proof. Write

$$T = \{t_1, \dots, t_m\}, \quad D = \sup_{s, t \in T} d(s, t).$$

For each $k \geq 0$, define

$$T_k = \{t_1, \dots, t_{m \wedge 2^{2^k}}\}.$$

Then (T_k) is admissible. If $2^{2^k} \geq m$, then $T_k = T$, so $d(t, T_k) = 0$ for all $t \in T$. Otherwise $d(t, T_k) \leq D$ for all $t \in T$. Therefore

$$\gamma_2(T, d) \leq D \sum_{k: 2^{2^k} < m} 2^{k/2}.$$

If k_0 is the largest index with $2^{2^{k_0}} < m$, then $2^{k_0} \asymp \log m$, and hence

$$\sum_{k=0}^{k_0} 2^{k/2} \leq C 2^{k_0/2} \leq C \sqrt{\log m}.$$

This proves the claim. □

Example 8.13 (A set for which Dudley is loose). For each $d \geq 1$, define

$$T_d = \left\{ \frac{e_i}{\sqrt{1 + \log i}} : 1 \leq i \leq d \right\} \subseteq \mathbf{R}^d.$$

Consider the canonical Gaussian process

$$X_t = \langle g, t \rangle, \quad g \sim \mathcal{N}(0, I_d).$$

Then

$$\sup_{t \in T_d} X_t = \max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}}.$$

We first show that the Gaussian supremum is uniformly bounded in d . For $u > 0$,

$$\mathbb{P} \left(\max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} \geq u \right) \leq \sum_{i=1}^d \mathbb{P} \left(g_i \geq u \sqrt{1 + \log i} \right).$$

Using the Gaussian Chernoff bound,

$$\mathbb{P}(g_i \geq a) \leq e^{-a^2/2}, \quad a \geq 0,$$

we get

$$\mathbb{P} \left(\max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} \geq u \right) \leq e^{-u^2/2} \sum_{i=1}^d i^{-u^2/2}.$$

For $u \geq 3$, the series on the right is bounded by an absolute constant, uniformly in d . Hence

$$\mathbb{P} \left(\max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} \geq u \right) \leq C e^{-u^2/2}, \quad u \geq 3.$$

Integrating the tail, we conclude that

$$\sup_{d \geq 1} \mathbb{E} \max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} < \infty.$$

Since $X_t = \langle g, t \rangle$ is Gaussian, Theorem 8.8 therefore implies

$$\sup_{d \geq 1} \gamma_2(T_d, \|\cdot\|_2) < \infty.$$

Now let us look at Dudley's entropy integral. Fix $\varepsilon \in (0, \frac{1}{2}]$, and set

$$m = \left\lfloor \exp \left(\frac{1}{4\varepsilon^2} \right) \right\rfloor.$$

Assume $m \leq d$. If $i \neq j$ and $1 \leq i, j \leq m$, then

$$\left\| \frac{e_i}{\sqrt{1 + \log i}} - \frac{e_j}{\sqrt{1 + \log j}} \right\|_2^2 = \frac{1}{1 + \log i} + \frac{1}{1 + \log j} \geq \frac{2}{1 + \log m}.$$

Since $\varepsilon \leq \frac{1}{2}$, we have $1 \leq \frac{1}{4\varepsilon^2}$, so

$$1 + \log m \leq 1 + \frac{1}{4\varepsilon^2} \leq \frac{1}{2\varepsilon^2}.$$

Therefore

$$\left\| \frac{e_i}{\sqrt{1 + \log i}} - \frac{e_j}{\sqrt{1 + \log j}} \right\|_2^2 \geq 4\varepsilon^2,$$

which means that the first m points form a 2ε -packing. By Theorem 6.3,

$$N(T_d, \|\cdot\|_2, \varepsilon) \geq m.$$

Hence

$$\log N(T_d, \|\cdot\|_2, \varepsilon) \geq \frac{c}{\varepsilon^2}$$

whenever $m \leq d$, that is, whenever

$$\varepsilon \geq \frac{c'}{\sqrt{\log d}}.$$

Consequently, for all sufficiently large d ,

$$\int_0^{1/2} \sqrt{\log N(T_d, \|\cdot\|_2, \varepsilon)} \, d\varepsilon \geq c \int_{c'/\sqrt{\log d}}^{1/2} \frac{d\varepsilon}{\varepsilon} \geq c'' \log \log d.$$

So Dudley's entropy integral diverges with d , while $\gamma_2(T_d, \|\cdot\|_2)$ remains uniformly bounded.

This is a very instructive example. The entropy is large at many scales, but different points are responsible for different scales. Dudley's bound pays the worst possible price at every scale. The γ_2 -functional is more flexible, because the supremum stays inside the multiscale sum. That is exactly the feature that makes generic chaining sharper.

Remark 8.14. The moral is that Dudley's entropy integral is an extremely useful first estimate, but it is not the end of the story. For Gaussian processes, Talagrand's theorem says that the true multiscale geometry is captured by γ_2 , and it does so up to absolute constants.

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